



Euclid's Garden

A Journey Through Geometry
from First Wonder to the Olympiad

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First Complete Edition • Prologue to Olympiad

Euclid's Garden — *A Journey Through Geometry, from First Wonder to the Olympiad.*

This is the **First Complete Edition**: the whole journey, from the first wonder to the olympiad. It contains the Prologue, all five Parts (Chapters 1–30), an Epilogue, and four Appendices. An interactive companion app, with draggable versions of the key figures, is in development.

Text and original diagrams prepared with $\text{\LaTeX}$ and `tkz-euclide`.

Companion-app figures (in development) built with JSXGraph.

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A Letter to the Reader

Welcome.

If you have ever drawn a star without lifting your pen, folded a paper exactly in half, or wondered why a ball is round, then you have already done geometry. You simply did not know it had a name.

This book has one ambition: to make you *fall in love* with geometry, and then to walk beside you, patiently and without ever skipping a step, all the way to the doorstep of the International Mathematical Olympiad. That is a long road. But it is a beautiful one, and you will not walk it alone.

I have tried to write the book I wish someone had handed me when I was starting. So a few promises:

- **We begin at zero.** No prior geometry is assumed. If you can count and hold a pencil, you can begin Chapter 1.
- **Wonder comes before rigour.** Every chapter opens with a question worth caring about. You will feel the mystery before we ever write the word “theorem.”
- **You will see everything.** Geometry is the most visual of all mathematics. There is a diagram on nearly every page, and for the most important ones a living, draggable version is planned for the companion app (currently in development).
- **Nothing is wasted.** Every idea you learn will be used again. You will always know *why* you are learning something.

How to read this book

Read with a pencil and paper beside you. Geometry is not a spectator sport. When you reach a diagram, redraw it. When you reach a “Worked Example,” cover the solution and try first.

Watch for these signposts along the way:

★ Wonder

A puzzle or story that opens a chapter and plants a question.

Aha!

The key insight of a section — the moment it “clicks.”

→ Try it live

A pointer to the companion app (in development), where you will be able to drag the figure and watch the idea move.

🔍 Worked Example

A complete solution with the *thinking* shown, not just the answer.

Mathematician's Margin

A short story about the human beings behind the mathematics.

✗ Common Trap

A classic mistake — and how to step around it.

★ Olympiad Spark

A real competition-style problem to end the chapter. You will be ready for it.

The exercises

At the end of each chapter the problems climb a gentle staircase: Warm-up to settle the ideas, Practice to build fluency, Challenge to stretch you, and an Olympiad Spark to show you the summit. Do not rush. A single problem, truly understood, is worth a hundred skimmed.

Turn the page. The garden is waiting.

— *The Author*

Prologue: Why Geometry?

★ WONDER

Four thousand years ago, on the banks of the Nile, surveyors called *rope-stretchers* carried a loop of rope knotted into twelve equal parts. By pulling it taut into a triangle with sides of 3, 4, and 5 knots, they could create a *perfect right angle* — anywhere, anytime, with nothing but a rope. How did a knotted cord come to hold a law of the universe?

The rope-stretchers did not know *why* the trick worked. They only knew that it did. Twenty centuries later, a Greek would prove why — and that act of proving, of moving from “it seems to work” to “it *must* work, always, forever,” is the moment mathematics was born. His name was Pythagoras, and we will meet his theorem properly in Chapter 9. But the deeper lesson begins now.

Geometry is thinking you can see

Most of mathematics hides inside symbols. Geometry does not. It lives in pictures — in triangles and circles, in folds and reflections, in shadows and stars. This makes it the friendliest doorway into serious mathematics, because your eyes can lead where your logic has not yet learned to walk.

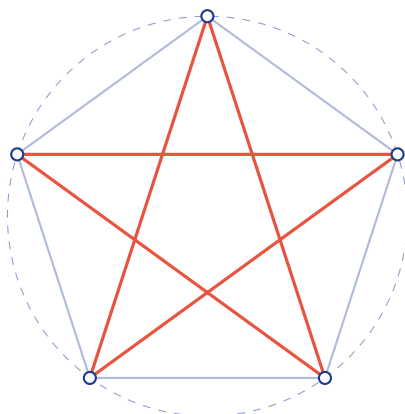
But geometry is also the strictest of teachers, and here is its first secret:

AHA!

A picture can suggest the truth, but only a proof can guarantee it. Your eyes are a brilliant guide and an occasional liar. Geometry trains you to tell the difference — to know not just *that* something is true, but *why* it cannot be otherwise.

Consider the five-pointed star below. Draw it in one stroke and you have traced a *pentagram*, a shape the ancient Greeks believed held the secret of beauty. Hidden inside it is the *golden ratio*, the same proportion that appears in sunflower seeds, nautilus shells, and the spiral of galaxies. You cannot see the golden ratio by looking. But by the end of this book, you will be able to *prove* it

is there.



The pentagram. Five lines, infinite consequences.

What is a proof, really?

A proof is a story so carefully told that no one — not the cleverest skeptic, not you at your most doubtful — can find a gap in it. It begins from things we agree are obviously true (we will call them *axioms* and *definitions*), and it advances one undeniable step at a time until the conclusion is forced upon us.

That is the great trade of mathematics. We give up the right to say “it looks true,” and in return we gain something no other human pursuit can offer: *certainty*. A theorem proved by Euclid in 300 BCE is exactly as true today, and will be exactly as true in a million years. Empires fall; the angle sum of a triangle does not.

MATHEMATICIAN'S MARGIN

The word *geometry* comes from the Greek *geō* (earth) and *metron* (measure): literally, “earth-measuring,” after those Egyptian surveyors. Around 300 BCE, Euclid of Alexandria gathered the scattered knowledge of his age into thirteen books called the *Elements* — the most successful textbook in history, studied continuously for over two thousand years.

The journey ahead

This book is a single road, and it climbs. Here is the map:

- **Part I — First Steps.** Points, lines, angles, parallels, triangles, and the art of proof itself. We build the alphabet.

- **Part II — The Triangle's Secrets.** Similarity, the Pythagorean theorem, the hidden centres of every triangle.
- **Part III — The Circle.** The most perfect shape, and the surprising theorems it hides.
- **Part IV — The Toolbox.** Angle chasing, trigonometry, coordinates, vectors, transformations — the techniques that turn a student into a solver.
- **Part V — Olympiad Geometry.** The famous configurations, the great strategies, and a gallery of real olympiad problems solved from scratch.

You do not need to be a genius to walk this road. You need curiosity, a pencil, and the willingness to be patient with yourself. The summit — a hard, beautiful olympiad problem yielding to your own hand — is closer than you think.

★ OLYMPIAD SPARK

Here is a problem you cannot yet solve. Keep it in mind; you will return to it as a master.

A square is folded so that one corner lands exactly on the midpoint of a non-adjacent side. Prove that the three triangles formed along the bottom edge have sides in the ratio 3 : 4 : 5.

The rope-stretchers' triangle, hidden inside a folded square. Geometry is full of such echoes. Let us learn to hear them.

PART I

First Steps

1

Points, Lines, and the Art of Definition

★ WONDER

Try to define the word “point” to a friend, using only simpler words. Then define *those* words. Keep going. You will find, surprisingly quickly, that you run out of simpler words entirely. Where does a chain of definitions *begin*?

Every dictionary is secretly a circle. Look up “line” and it sends you to “straight”; look up “straight” and it sends you back to “line.” For ordinary language this is harmless. For mathematics, where we demand certainty, it is a crisis — because if our definitions chase each other in circles, our proofs rest on nothing.

The Greeks found the only possible escape, and it is the foundation of everything that follows.

1.1 The terms we agree not to define

We choose a tiny handful of words and agree, openly, *not* to define them. Instead we describe how they behave. These are the **primitive** (or undefined) terms. In plane geometry there are just three.

Definition 1.1 (The primitives). We take as undefined the notions of **point**, **line**, and **plane**. We do not say what they *are*; we only agree on the rules they obey.

Picture them this way, while remembering these pictures are crutches, not definitions:

- A **point** marks a position. It has no size — no length, no width, no thickness. Our dots are merely visible stand-ins for something infinitely small. We name points with capital letters: A , B , P .
- A **line** is straight, has no thickness, and extends forever in both directions. We name a line by two of its points with a bar, $\overleftrightarrow{AB}$, or by a single lowercase letter, ℓ .
- A **plane** is a perfectly flat surface extending forever in all directions — the endless sheet of paper on which our drawings live.

AHA!

We do not build mathematics by defining everything. We build it by defining *almost* everything in terms of a few honest, undefined starting points — and then never sneaking in a hidden assumption again. Knowing where to *stop* defining is the first act of mathematical maturity.

From these three primitives, every other object is *built* by definition. Here are the first and most important.

Definition 1.2 (Segment, ray, and their relatives). Let A and B be distinct points.

- The **segment** $\overline{AB}$ is the point A , the point B , and all points between them. Its two *endpoints* are A and B .
- The **ray** $\overrightarrow{AB}$ is the point A together with all points of $\overleftrightarrow{AB}$ on the same side as B . The point A is its single endpoint, or *origin*.
- The **length** of $\overline{AB}$, written AB , is the distance between A and B .

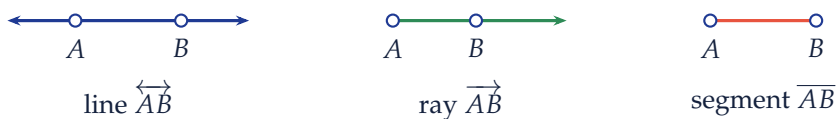


Figure 1.1 — The same two points A, B give a line (infinite), a ray (half-infinite), and a segment (finite). The arrowheads mean “continues forever.”

→ TRY IT LIVE Drag A and B in the “Primitives” demo and watch the line, ray, and segment update together.

1.2 The rules of the game: a first axiom

An **axiom** (or *postulate*) is a statement we accept as true without proof — a rule of the game, agreed at the start. Everything we ever prove will trace back to axioms. Our very first one captures something you have always known:

Theorem 1.3 (The Two-Point Axiom). Through any two distinct points there passes *exactly one* line.

Read it twice, because it says two things. “At least one”: you can always draw a line through two points. “At most one”: you can never draw two *different* straight lines through the same pair. Together: the line through two points is *unique*. This is why “ $\overleftrightarrow{AB}$ ” is allowed to name a single, definite object.

Definition 1.4 (Collinear and concurrent). Points are **collinear** if a single line passes through all of them. Lines are **concurrent** if a single point lies on all of them.

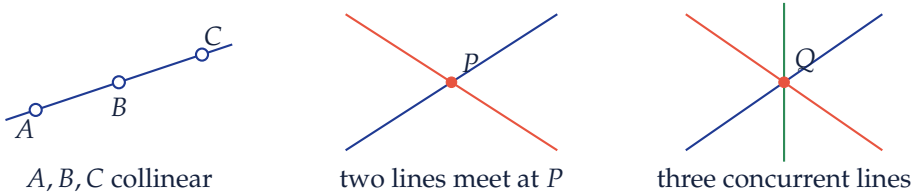


Figure 1.2 — Collinear points share a line; concurrent lines share a point. The two ideas are mirror images of each other, a symmetry we will exploit again and again.

X COMMON TRAP

“Two lines always meet at a point.” Not so! Two lines in a plane either meet at exactly one point or never meet at all. Lines that never meet have their own name — *parallel* — and an entire chapter (Chapter 3) devoted to the strange and wonderful rules they obey.

1.3 Distance, betweenness, and the midpoint

A line is more than a set of points: along it we can *measure*. The everyday fact that “the whole equals the sum of its parts” becomes, on a line, a working axiom we will lean on constantly.

Theorem 1.5 (Segment Addition). If a point B lies *between* A and C on a line, then $AB + BC = AC$. Conversely, if $AB + BC = AC$, then B lies between A and C .

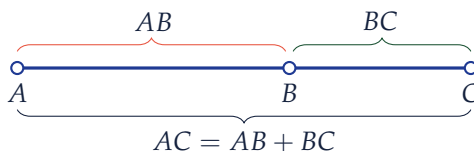


Figure 1.3 — Betweenness made arithmetic: the part AB plus the part BC rebuilds the whole AC .

A point that splits a segment into two *equal* halves is its **midpoint**: M is the midpoint of $\overline{AB}$ when $AM = MB = \frac{1}{2}AB$.

 **Worked Example.** Point B lies between A and C , with $AB = 7$ and $AC = 12$. Find BC .

By Segment Addition, $AB + BC = AC$, so $BC = AC - AB = 12 - 7 = 5$. The picture is doing exactly what arithmetic says: the whole (12) minus the part you have (7) leaves the rest (5). ■

 **Worked Example.** M is the midpoint of $\overline{AB}$, with $AM = 3x - 1$ and $MB = x + 5$. Find AB .

A midpoint makes the two halves equal, so $AM = MB$:

$$3x - 1 = x + 5 \Rightarrow 2x = 6 \Rightarrow x = 3.$$

Then $AM = 3(3) - 1 = 8$, and since AB is *two* halves, $AB = 2 \cdot AM = 16$. (Check: $MB = x + 5 = 8$ too, as it must.) Geometry has quietly turned into a one-line equation — a foreshadowing of Chapter 21. ■

1.4 Why this much care?

It may feel excessive to fuss over the word “point.” But this care is the price of certainty, and it pays for itself almost immediately. Because our terms are pinned down, the next time we write “let ℓ be the line through A and B ,” there is no ambiguity, no hidden assumption, no crack for an error to slip through. We have built bedrock. Now we can build upward.

 **Worked Example.** How many lines are determined by 5 points, if no three of them are collinear?

Each line is fixed by choosing 2 of the points (the Two-Point Axiom), and since no three points lie on one line, different pairs give different lines. The number of ways to choose 2 points from 5 is

$$\binom{5}{2} = \frac{5 \cdot 4}{2} = 10.$$

So there are exactly 10 lines. The phrase “no three collinear” is doing real work: without it, three points on one line would be counted as three pairs but yield only *one* line, and the answer would drop.

→ YOUR TURN

At a party of 9 people, everyone shakes hands exactly once with everyone else. How many handshakes take place?

Set-up (you finish it): a handshake is a choice of 2 people from 9 — the very same count as the number of segments joining 9 points with no three collinear. So evaluate $\binom{9}{2} = \frac{9 \cdot 8}{2}$.

Answer: $\binom{9}{2} = 36$ handshakes.

We spent a whole chapter learning where to stop asking “but what *is* that?”, and on that small patch of bedrock everything else in this book will stand.

KEY IDEAS & FORMULAS

- **Primitives** (undefined): *point*, *line*, *plane* — described by behaviour, not defined.
- **Built objects:** segment $\overline{AB}$ (finite), ray $\overrightarrow{AB}$ (one endpoint, a direction), line $\overleftrightarrow{AB}$ (infinite); AB denotes length.
- **Two-Point Axiom:** exactly one line through two distinct points.
- **Collinear** points share a line; **concurrent** lines share a point.
- **Segment Addition:** B between $A, C \Rightarrow AB + BC = AC$; the **midpoint** gives $AM = MB = \frac{1}{2}AB$.
- **Counting:** n points, no three collinear, determine $\binom{n}{2} = \frac{n(n-1)}{2}$ lines.

Exercises

Warm-up.

1. Name three real-world objects that behave roughly like (a) a point, (b) a line, (c) a plane. In each case, say how the real object *fails* to be the ideal one.
2. How many segments have both endpoints among the points A, B, C, D ? List them.
3. True or false, with a reason: "A ray $\overrightarrow{AB}$ is the same as the ray $\overrightarrow{BA}$."
4. B lies between A and C with $AB = 5$ and $BC = 8$. Find AC .
5. M is the midpoint of $\overline{PQ}$ and $PQ = 18$. Find PM .
6. Three points have $AB = 4$, $BC = 6$. Are they collinear (with B between) if $AC = 10$? If instead $AC = 9$?

Practice.

7. Six points lie on a circle, so that no three are collinear. How many lines do they determine? How many segments?
8. Three distinct lines are drawn in a plane. What are *all* the possible numbers of intersection points? Draw a picture for each case.
9. Points A, B, C, D are collinear in that order, with $AB = 3$, $AC = 8$, and $AD = 11$. Find BC , CD , and BD .
10. M is the midpoint of $\overline{AB}$, with $AM = 2x + 1$ and $MB = 4x - 7$. Find x and AB .
11. Nine points lie in a plane, no three collinear. How many lines do they determine? How many triangles can be formed with vertices among them?

Challenge.

12. n points lie in a plane with no three collinear. Find a formula, in terms of n , for the number of lines they determine. Justify it.
13. In a plane there are n lines, no two parallel and no three concurrent. Find a formula for the number of intersection points, and explain why both conditions are needed.
14. A convex polygon has n sides. Show that it has exactly $\frac{n(n-3)}{2}$ diagonals, and check your formula on a pentagon and a hexagon.

★ OLYMPIAD SPARK

(Combinatorial geometry.) Ten lines are drawn in the plane. Among them, exactly four pass through a single common point, and apart from that coincidence no two are parallel and no three are concurrent. Into how many regions do these ten lines divide the plane?

Hint for later: a new line adds one region for itself, plus one more for each new point at which it crosses the previous lines. We will develop this counting idea fully; for now, see how far the picture takes you.

2

Angles and How to Measure Them

★ WONDER

The hour hand of a clock creeps from 12 all the way around back to 12. The minute hand does the same in one hour; the second hand, in one minute. All three return to where they began — yet we feel that the second hand has “turned more.” What exactly is the thing that is being measured when something turns?

That “thing” is **angle**: the measure of turning, or equivalently, of the opening between two directions. It is the single most important quantity in all of elementary geometry. Master angles and you hold the key to triangles, circles, and — as we will see in Part IV — to half of all olympiad problems.

2.1 The anatomy of an angle

Definition 2.1 (Angle). An **angle** is the figure formed by two rays sharing a common endpoint. The shared endpoint is the **vertex**; the two rays are the **sides**. The angle with vertex V and sides through A and B is written $\angle AVB$ (or $\angle BVA$, or simply $\angle V$ when there is no ambiguity).

The middle letter always names the vertex — a small convention with a big payoff in crowded diagrams.

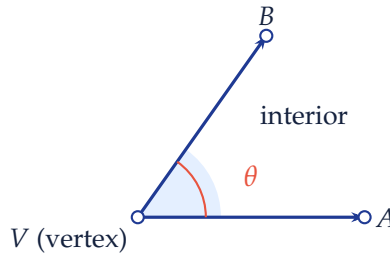


Figure 2.1 — The angle $\angle AVB$, with measure θ . The shaded region is its *interior*.

2.2 Measuring the turn: degrees

To measure an angle we compare it to a full turn. Slice one complete revolution into 360 equal parts and call each part one **degree**, written 1° . Then a full turn is 360° , a half-turn is 180° , and a quarter-turn is 90° .

MATHEMATICIAN'S MARGIN

Why 360, and not a rounder number like 100? The choice descends from the Babylonians, who used a base-60 number system and reckoned a year at about 360 days. The number 360 is also gloriously divisible — by 2, 3, 4, 5, 6, 8, 9, 10, 12 and more — so most natural fractions of a turn come out whole. We keep it out of a four-thousand-year-old habit, and because it is genuinely convenient.

Angles earn names by their size. Commit these to memory; we will use them constantly.

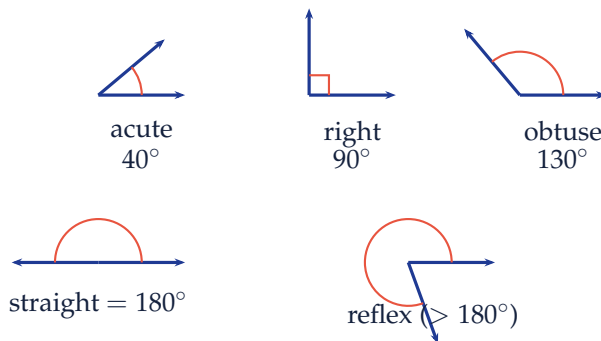


Figure 2.2 — The family of angles: *acute* ($< 90^\circ$), *right* ($= 90^\circ$, marked with a small square), *obtuse* (between 90° and 180°), *straight* ($= 180^\circ$), and *reflex* ($> 180^\circ$).

The right angle is special enough to deserve its own mark: a small square at the vertex, never an arc. When two lines meet at a right angle we say they are **perpendicular**, written $\ell \perp m$.

2.3 Angles that add: a working axiom

If a ray $\overrightarrow{VC}$ lies in the interior of $\angle AVB$, then it splits the big angle into two smaller ones that must account for the whole. This is the **Angle Addition Postulate**:

$$\angle AVB = \angle AVC + \angle CVB.$$

Obvious? Yes. Useful? Endlessly — it is the engine of every “angle chase” you will ever run, and the reason the simple act of *drawing one extra ray* so often cracks a problem open.

A ray that splits an angle into two *equal* parts is its **bisector**. Two other relationships appear so often they are named:

Definition 2.2 (Complementary and supplementary). Two angles are **complementary** if their measures sum to 90° , and **supplementary** if they sum to 180° . When two angles are supplementary *and* share a side, forming a straight line, they are a **linear pair**.

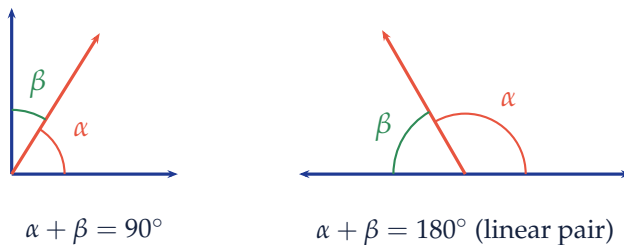


Figure 2.3 — Complementary angles complete a right angle; a linear pair completes a straight line.

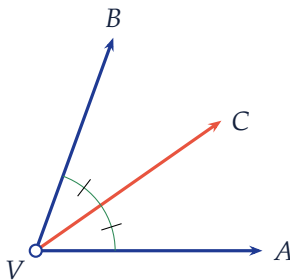


Figure 2.4 — A bisector $\overrightarrow{VC}$ splits $\angle AVB$ into two equal parts (matching marks):
 $\angle AVC = \angle CVB = \frac{1}{2}\angle AVB$.

 **Worked Example.** Ray $\overrightarrow{VC}$ bisects $\angle AVB$, with $\angle AVC = (2x + 5)^\circ$ and $\angle CVB = (3x - 10)^\circ$. Find $\angle AVB$.

A bisector makes the two halves equal, so $\angle AVC = \angle CVB$:

$$2x + 5 = 3x - 10 \Rightarrow x = 15.$$

Then $\angle AVC = 2(15) + 5 = 35^\circ$, and by Angle Addition $\angle AVB = 2 \cdot 35^\circ = 70^\circ$. (Check: $\angle CVB = 3(15) - 10 = 35^\circ$ too.) ■

 **Worked Example.** Find the angle that is exactly twice its own complement.

Let the angle be x . Its complement is $90^\circ - x$, and we want $x = 2(90^\circ - x)$:

$$x = 180^\circ - 2x \Rightarrow 3x = 180^\circ \Rightarrow x = 60^\circ.$$

So the angle is 60° and its complement is 30° — and indeed 60 is twice 30. Naming a relationship (“complementary”) turned a sentence of English into a one-line equation. ■

2.4 Your first theorem: vertical angles

When two lines cross, they make four angles. The two that sit “across” from each other — sharing only the vertex — are called **vertical angles**. Your eye insists they are equal. Let us prove it, and feel for the first time the difference between seeing and *knowing*.

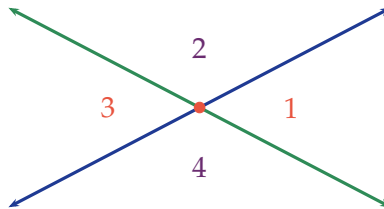


Figure 2.5 — Two crossing lines. Angles 1 and 3 are vertical; so are 2 and 4.

Theorem 2.3 (Vertical Angle Theorem). If two lines intersect, then vertical angles are equal. In the figure, $\angle 1 = \angle 3$.

Proof. Angles 1 and 2 form a linear pair, so $\angle 1 + \angle 2 = 180^\circ$. Angles 2 and 3 also form a linear pair, so $\angle 2 + \angle 3 = 180^\circ$. Setting these equal,

$$\angle 1 + \angle 2 = \angle 2 + \angle 3.$$

Subtract $\angle 2$ from both sides: $\angle 1 = \angle 3$. The same argument gives $\angle 2 = \angle 4$. ■

AHA!

Look at what just happened. We did not measure a single angle. We never appealed to the picture. We used only the meaning of “linear pair” and ordinary arithmetic — and out came a fact that holds for *every* pair of crossing lines that ever has been or will be drawn. That is the power you are here to learn.

→ TRY IT LIVE In the “Crossing Lines” demo, rotate one line and watch the vertical angles change together, always staying equal.

 **Worked Example.** Two lines cross. One of the four angles measures 35° . Find the other three.

Its vertical partner is also 35° . Each of the remaining two forms a linear pair with the 35° angle, so each is $180^\circ - 35^\circ = 145^\circ$. The four angles are $35^\circ, 145^\circ, 35^\circ, 145^\circ$, and indeed they go around the point summing to 360° .

 **Worked Example. Back to the clock.** What is the angle between the hour and minute hands at exactly 4:00?

The twelve hour-marks split the full 360° into twelve equal steps of 30° each. At 4:00 the minute hand points at 12 and the hour hand at 4, a gap of four steps:

$$4 \times 30^\circ = 120^\circ.$$

So the hands make 120° . The Wonder’s vague feeling of “how much turning” is now a number you can compute from nothing but a fraction of a full revolution. (Beware the moving hour hand at times like 4:30 — see the exercises.) ■

→ YOUR TURN

Two angles are supplementary, and one of them is exactly four times the other. Find both angles.

Set-up (you finish it): call the smaller angle x ; the larger is $4x$, and “supplementary” means $x + 4x = 180^\circ$. Solve for x , then double-check the two add to 180° .

Answer: $x = 36^\circ$ and $4x = 144^\circ$ (and $36 + 144 = 180$).

What began as the second hand’s vague feeling of “more turning” is now a quantity you can measure, add, and reason with. And you proved your very first theorem along the way.

KEY IDEAS & FORMULAS

- **Angle:** two rays from a vertex; written $\angle AVB$ (middle letter = vertex). Measured in **degrees**, a full turn = 360° .
- **Names:** acute $< 90^\circ$, right = 90° (perpendicular, $\perp$), obtuse between 90° and 180° , straight = 180° , reflex $> 180^\circ$.
- **Angle Addition:** an interior ray gives $\angle AVB = \angle AVC + \angle CVB$; a **bisector** halves an angle.
- **Complementary** sum to 90° ; **supplementary** sum to 180° ; a **linear pair** is supplementary and straight.
- **Vertical angles** (from two crossing lines) are equal — proved from two linear pairs.

Exercises

Warm-up.

1. Classify each angle: 17° , 90° , 179° , 245° , 90.0001° .
2. Find the complement and the supplement of 31° .
3. An angle equals its own supplement. What is it?
4. Find the complement and the supplement of 62° .
5. An angle equals half of its own supplement. Find it.
6. Two lines cross, and one of the four angles is 110° . Find the other three.

Practice.

7. An angle is 24° more than its complement. Find both angles.

8. Ray $\overrightarrow{VC}$ bisects $\angle AVB$, and $\angle AVB = 76^\circ$. A further ray bisects $\angle AVC$. What angle does it make with $\overrightarrow{VA}$?
9. Three lines pass through a single point, making six angles. If one of them is 50° , must any of the others equal 50° ? Explain.
10. In $\angle AVB = 90^\circ$, an interior ray $\overrightarrow{VC}$ makes $\angle AVC$ twice $\angle CVB$. Find both angles.
11. Find the angle between the clock hands at 2:00 and at 5:00.
12. Two supplementary angles are in the ratio 2 : 7. Find both.

Challenge.

13. The hands of a clock at 3:00 make a 90° angle. What is the exact angle between them at 3:30? (Careful — the hour hand moves too.)
14. Two supplementary angles are in the ratio 4 : 5. A line bisects each. Find the angle between the two bisectors.
15. Between 3:00 and 4:00, at what times (to the nearest second) are the clock hands exactly perpendicular? (*There are two; one of them is obvious.*)

★ OLYMPIAD SPARK

(Angle chase, warm-up level.) Two lines cross at P . A third line through P bisects one of the four angles. Prove that it also bisects the vertical angle opposite to it — and that the two bisectors of a pair of vertical angles always form a single straight line.

3

Parallel Lines and the Transversal

★ WONDER

Railway tracks run side by side, never meeting, all the way to the horizon — and yet in a photograph they appear to converge to a single point. Are the rails “really” parallel, or do our eyes know something our rulers do not? And if two lines truly never meet, the angles around them are no longer free; they fall into a strict and beautiful lockstep.

Parallel lines are where geometry first becomes genuinely surprising. Two lines that never touch, no matter how far you follow them, turn out to impose strict, beautiful rules on every angle in their neighbourhood. Those rules are the heartbeat of the angle chase — the technique that will carry you to the olympiad.

3.1 What “parallel” means

Definition 3.1 (Parallel lines). Two lines in the same plane are **parallel** if they never meet, however far they are extended. We write $\ell \parallel m$.

A line that cuts across two (or more) other lines is called a **transversal**. When a transversal crosses two lines it creates eight angles — four at each crossing — and these eight angles come in families with names worth learning.

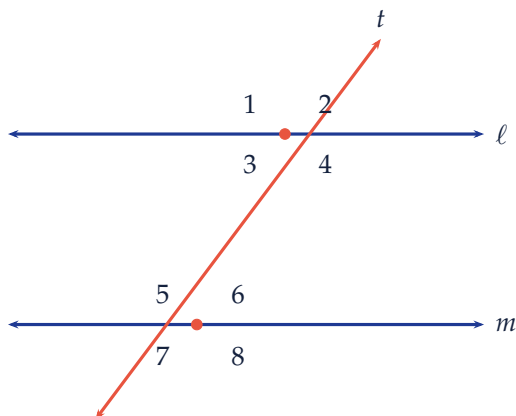


Figure 3.1 — A transversal t crossing two lines $\ell \parallel m$, producing eight angles.

The vocabulary, all referring to Figure 3.1:

- **Corresponding angles** sit in matching corners at the two crossings: (1, 5), (2, 6), (3, 7), (4, 8). (Think of the letter F.)
- **Alternate interior angles** lie between the two lines, on opposite sides of t : (3, 6) and (4, 5). (Think of the letter Z.)
- **Co-interior** (or *same-side interior*) **angles** lie between the lines, on the *same* side of t : (3, 5) and (4, 6). (Think of the letter C.)

3.2 The Parallel Postulate, in working form

Here is the axiom that makes everything go. It is the modern, practical face of Euclid's famous fifth postulate.

Theorem 3.2 (Corresponding Angles Axiom). A transversal cuts two lines so that a pair of corresponding angles are *equal* **if and only if** the two lines are parallel.

The phrase “if and only if” means the street runs both ways: parallel lines *force* corresponding angles to be equal, and equal corresponding angles *guarantee* the lines are parallel. The second direction is how we will later *prove* that two lines are parallel — by catching a pair of equal angles.

From this single axiom, the other two families follow as theorems. Watch how little work it takes.

Theorem 3.3 (Alternate Interior Angles). If $\ell \parallel m$, then alternate interior angles are equal: $\angle 3 = \angle 6$.

Proof. Since $\ell \parallel m$, corresponding angles are equal, so $\angle 3 = \angle 7$. But $\angle 7$ and $\angle 6$ are vertical angles, so $\angle 7 = \angle 6$ (Theorem 2.3). Chaining these, $\angle 3 = \angle 7 = \angle 6$. ■

Theorem 3.4 (Co-interior Angles). If $\ell \parallel m$, then co-interior angles are supplementary: $\angle 3 + \angle 5 = 180^\circ$.

Proof. Angles 3 and 6 are alternate interior, so $\angle 3 = \angle 6$. Angles 6 and 5 form a linear pair, so $\angle 6 + \angle 5 = 180^\circ$. Substituting, $\angle 3 + \angle 5 = 180^\circ$. ■

AHA!

Notice the pattern of the last two proofs: we never started from scratch. We borrowed a fact we had already secured (corresponding angles, vertical angles, linear pairs) and took one more step. This is how *all* of geometry is built — each theorem standing on the shoulders of the ones before. You are watching a tower rise.

 **Worked Example.** Lines $\ell \parallel m$ are cut by a transversal, and one of the eight angles measures 70° . Find the angle that is (a) *corresponding* to it, (b) *alternate interior* to it, (c) *co-interior* with it.

Corresponding angles are equal across parallels, so (a) is 70° . Alternate interior angles are equal, so (b) is 70° . Co-interior angles are supplementary, so (c) is $180^\circ - 70^\circ = 110^\circ$. Two of the three relationships preserve the angle; only the co-interior one flips it to its supplement — a distinction worth burning into memory. ■

 **Worked Example.** A transversal crosses two lines, making a pair of *co-interior* angles measuring $(2x + 20)^\circ$ and $(3x + 10)^\circ$. For what value of x are the two lines parallel?

Run the Corresponding Angles Axiom in reverse: the lines are parallel exactly when the angle relationships hold. Co-interior angles across

parallels are supplementary, so we *require*

$$(2x + 20) + (3x + 10) = 180 \Rightarrow 5x + 30 = 180 \Rightarrow x = 30.$$

At $x = 30$ the angles are 80° and 100° (summing to 180°), so the lines are parallel. This is the everyday way parallelism is *proved*: catch an angle relationship and quote the converse. ■

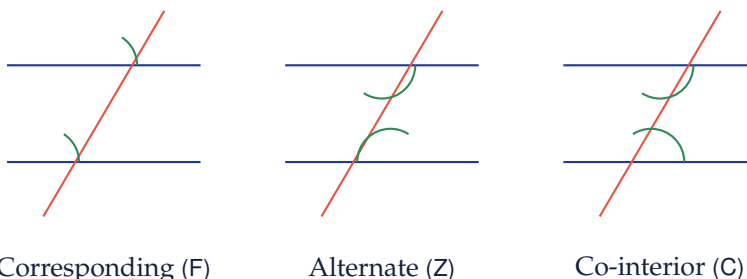


Figure 3.2 — The three families, highlighted. F angles are equal, Z angles are equal, C angles are supplementary.

→ TRY IT LIVE Open “Parallel Explorer,” slide the transversal, and watch all eight angles update — with the F-, Z-, and C-pairs highlighted.

✗ COMMON TRAP

The neat equalities hold *only when the two lines are parallel*. If ℓ and m are not parallel, corresponding angles are simply unequal and carry no special relationship. Always confirm the parallel marks (the little arrowheads $\rightarrow\rightarrow$ on the lines) before you use these rules.

3.3 A trick worth its weight in gold: the auxiliary parallel

Here is a problem that looks impossible until one extra line makes it trivial — your first taste of the single most powerful move in olympiad geometry: *adding a helpful line*.

Worked Example. Two horizontal lines are parallel. A point P lies between them. A segment runs from the upper line down to P , making an angle of a with the upper line; another runs from P down to the lower

line, making an angle of b with the lower line, on the *opposite* side. Show that the angle at P (the bend) equals $a + b$.

The idea: draw a third line through P parallel to the two given lines. It splits the bend angle at P into two pieces. The upper piece and a are alternate interior angles between two parallels, so the upper piece $= a$. The lower piece and b are likewise alternate interior, so the lower piece $= b$. By the Angle Addition Postulate, the whole bend is the sum of its pieces:

$$\angle P = a + b. \quad \blacksquare$$

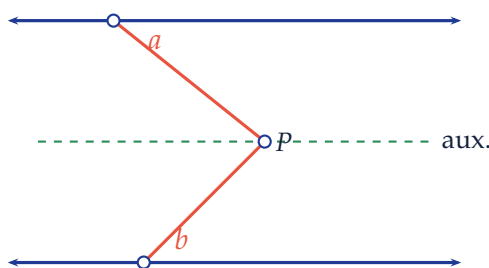


Figure 3.3 — The “saw-tooth.” One auxiliary parallel through P turns a mystery into $a + b$.

AHA!

Keep this manoeuvre close. When a configuration involves parallel lines and a stubborn angle, the question to ask is almost always: “*what if I drew one more line parallel to those?*” Half of beginning olympiad geometry is knowing which line to add.

→ YOUR TURN

Lines $\ell \parallel m$ are cut by a transversal, and a pair of *alternate interior* angles measures $(2x + 15)^\circ$ and $(4x - 25)^\circ$. Find x and the size of each angle.

Set-up (you finish it): alternate interior angles across parallels are *equal*, so set $2x + 15 = 4x - 25$, solve for x , then substitute back to get the angle.

Answer: $x = 20$, and each angle is $2(20) + 15 = 55^\circ$.

KEY IDEAS & FORMULAS

- **Parallel** ($\ell \parallel m$): lines in a plane that never meet. A **transversal** crosses them, making eight angles.
- **Corresponding** (F) angles equal *iff* the lines are parallel — the working Parallel Postulate, true both ways.
- **Alternate interior** (Z) angles equal; **co-interior** (C) angles supplementary (sum 180°).
- Each relationship has a **converse** used to *prove* two lines parallel — catch the angle relationship, quote the converse.
- **Auxiliary parallel**: through a troublesome point, draw a line parallel to the given ones; alternate-interior angles then split the mystery angle into nameable pieces.

Exercises**Warm-up.**

1. In Figure 3.1, $\angle 1 = 110^\circ$. Find all eight angles, naming the rule you use for each.
2. Lines $\ell \parallel m$ are cut by a transversal. One co-interior angle is 73° . Find its partner.
3. Name the type of each pair in Figure 3.1: $(2, 6)$, $(4, 5)$, $(3, 5)$, $(1, 8)$.
4. Lines $\ell \parallel m$ are cut by a transversal and one angle is 125° . Find the angle corresponding to it, the one alternate-interior to it, and the one co-interior with it.
5. A transversal makes a co-interior pair measuring $(4x)^\circ$ and $(5x)^\circ$ across two parallel lines. Find x .

Practice.

6. Two angles formed by a transversal across parallel lines are $(3x + 10)^\circ$ and $(5x - 30)^\circ$ and are corresponding. Find x .
7. Prove the converse of the Alternate Interior Angles theorem: if a transversal makes a pair of alternate interior angles equal, the two lines are parallel.
8. A “saw-tooth” path bends three times between two parallel lines, making angles a , then b (the other way), then c . Generalise the worked example to relate the bend angles.
9. A transversal crosses two lines, making alternate interior angles $(3x + 5)^\circ$ and $(x + 45)^\circ$. Find the value of x that forces the two lines to be parallel,

and give the resulting angle.

Challenge.

10. In Figure 3.3, suppose instead the second segment makes angle b on the *same* side as a . What is $\angle P$ now? Draw it and prove your answer.
11. Two mirrors meet at angle θ . A light ray bounces off one, then the other (angle in = angle out at each). Show the total turning of the ray is 2θ , independent of where it strikes. (*This is why a corner reflector sends light straight back.*)

★ OLYMPIAD SPARK

(Classic olympiad lemma.) Points A and B lie on one of two parallel lines and C on the other. Show that as C slides along its line, the area of triangle ABC never changes. Then use this “equal-area slide” to prove that the median of a triangle divides it into two triangles of equal area.

4

Triangles and the 180° Miracle

★ WONDER

Draw *any* triangle you like — long and thin, squat and wide, perfectly even. Tear off its three corners and lay them side by side, tip to tip. Every time, without exception, they fit together into a perfectly straight line. A skinny triangle, a fat triangle, a triangle the size of a galaxy: always a straight line. *Why* should three angles that look nothing alike always add up to the very same total?

The triangle is the most important shape in all of geometry. It is the simplest figure that encloses space, the unit from which every polygon can be built, and the stage on which nearly every theorem you will ever prove first appears. And it guards a remarkable secret, the one your torn corners just revealed.

4.1 The triangle, defined

Definition 4.1 (Triangle). A **triangle** is the figure formed by three points, not all collinear, joined in pairs by segments. The three points are its **vertices**, the three segments its **sides**, and the angles at the vertices its **interior angles**. The triangle with vertices A, B, C is written $\triangle ABC$.

The condition “not all collinear” matters: three points on one line give no triangle at all, just a flattened segment. With that settled, we can state and prove the miracle.

4.2 The angle-sum theorem

Theorem 4.2 (Angle Sum of a Triangle). The three interior angles of any triangle sum to 180° .

Before the proof, savour the strategy, because it is a perfect example of the auxiliary-line idea from Chapter 3. The angles of the triangle are scattered at three different vertices — hard to compare. We will use one cleverly drawn parallel line to *transport* all three angles to a single point, where they visibly line up.

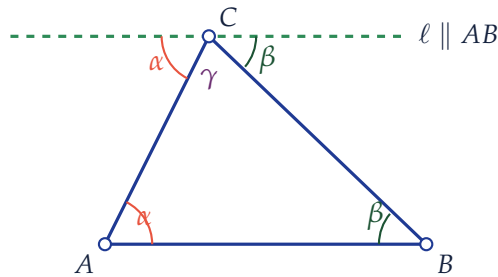


Figure 4.1 — Through C draw ℓ parallel to AB . The angles α and β reappear at C as alternate interior angles, flanking γ along a straight line.

Proof. Let the interior angles at A, B, C be α, β, γ . Through the vertex C , draw the line ℓ parallel to the opposite side AB (possible, and unique, by the Parallel Postulate).

The side CA is a transversal cutting the parallels ℓ and AB . Hence the angle it makes with ℓ on the left equals α , as *alternate interior angles* (Theorem 3.3). Likewise CB is a transversal, and the angle it makes with ℓ on the right equals β .

Now look along the line ℓ at the vertex C . The three angles there — the transported α , the original γ , and the transported β — together fill the straight line ℓ . Angles on a straight line sum to 180° , so

$$\alpha + \gamma + \beta = 180^\circ.$$

That is exactly the claim. ■

AHA!

The triangle's three angles were never really independent. The parallel line revealed that they are simply the three pieces of a single straight angle, broken apart and carried to different corners. The “miracle” was a straight line in disguise all along — just as we suspected back in Chapter 2.

→ TRY IT LIVE In “Angle-Sum Live,” drag any vertex of the triangle. The three angle measures change wildly, but their sum stays frozen at 180° .

4.3 An immediate harvest

One theorem, proved well, pays dividends at once. Here is a corollary we get almost for free, and which is itself a workhorse of angle chasing.

Theorem 4.3 (Exterior Angle Theorem). If one side of a triangle is extended, the **exterior angle** so formed equals the sum of the two *remote* (non-adjacent) interior angles.

Proof. Extend side AB beyond B to a point D . The exterior angle $\angle CBD$ and the interior angle $\beta = \angle CBA$ form a linear pair, so $\angle CBD = 180^\circ - \beta$. But by the angle-sum theorem $\alpha + \gamma = 180^\circ - \beta$ as well. Therefore $\angle CBD = \alpha + \gamma$, the sum of the two remote interior angles. ■

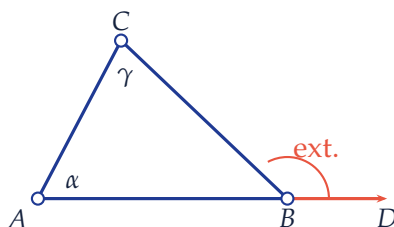


Figure 4.2 — The exterior angle at B equals $\alpha + \gamma$, the two remote interior angles.

Worked Example. An exterior angle of a triangle measures 130° , and the two remote interior angles are *equal*. Find all three interior angles. By the Exterior Angle Theorem, the exterior angle equals the sum of the two remote interior angles. With those two equal to x each,

$$2x = 130^\circ \Rightarrow x = 65^\circ.$$

So two interior angles are 65° each, and the third (adjacent to the exterior angle) is $180^\circ - 130^\circ = 50^\circ$. (Check: $65 + 65 + 50 = 180$.) Notice we never needed the angle-sum theorem directly — the exterior angle did the work in one step. ■

4.4 A family portrait: classifying triangles

Triangles are sorted in two independent ways — by their angles and by their sides.

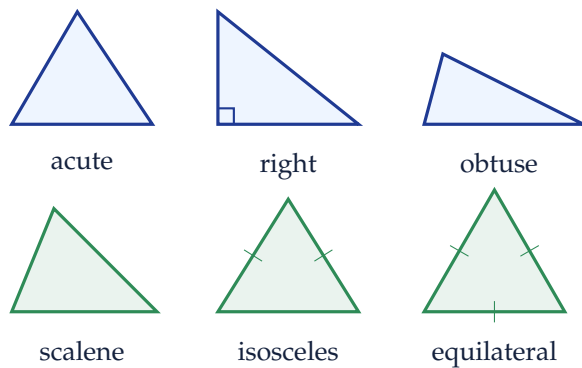


Figure 4.3 — Top, by largest angle: *acute* (all $< 90^\circ$), *right* (one $= 90^\circ$), *obtuse* (one $> 90^\circ$). Bottom, by equal sides (tick marks): *scalene* (none equal), *isosceles* (two equal), *equilateral* (all three equal).

These two classifications interact in ways worth noticing: a triangle can be *right* *and* *isosceles*, but — as the angle-sum theorem quietly guarantees — never *obtuse* and *equilateral*. (Why? An equilateral triangle's angles are all equal, hence each is $180^\circ/3 = 60^\circ$, all acute.)

Worked Example. In $\triangle ABC$, $\angle A = 2\angle B$ and $\angle C = \angle B + 30^\circ$. Find all three angles.

Let $\angle B = x$. Then $\angle A = 2x$ and $\angle C = x + 30^\circ$. By the angle-sum theorem,

$$2x + x + (x + 30^\circ) = 180^\circ \Rightarrow 4x = 150^\circ \Rightarrow x = 37.5^\circ.$$

So $\angle B = 37.5^\circ$, $\angle A = 75^\circ$, $\angle C = 67.5^\circ$. (Check: they sum to 180° . ✓)

 **Worked Example.** An isosceles triangle has its apex angle equal to 40° . Find the two base angles.

In an isosceles triangle the two base angles are equal (Chapter 7); call each β . The three angles sum to 180° :

$$40^\circ + 2\beta = 180^\circ \Rightarrow 2\beta = 140^\circ \Rightarrow \beta = 70^\circ.$$

Each base angle is 70° . The angle-sum theorem, plus the single fact that two angles are equal, pins the whole triangle down. ■

→ YOUR TURN

The three angles of a triangle are in the ratio 3 : 4 : 5. Find all three, and classify the triangle by its angles.

Set-up (you finish it): the parts total $3 + 4 + 5 = 12$, and they share 180° , so one part is $\frac{180^\circ}{12}$. Multiply that by 3, 4, and 5; then check whether the largest is below, equal to, or above 90° .

Answer: one part = 15° , so the angles are $45^\circ, 60^\circ, 75^\circ$ — all below 90° , an *acute* triangle.

KEY IDEAS & FORMULAS

- **Triangle:** three non-collinear points joined in pairs; vertices, sides, interior angles.
- **Angle Sum:** the three interior angles total 180° — proved by a parallel through one vertex (an auxiliary line transporting the angles onto a straight line).
- **Exterior Angle:** an exterior angle equals the sum of the two remote interior angles.
- **Classification — by angles:** acute (all $< 90^\circ$), right (one $= 90^\circ$), obtuse (one $> 90^\circ$). **By sides:** scalene, isosceles (two equal), equilateral (all equal, each 60°).
- **Polygons:** a convex n -gon has interior angle sum $(n - 2) \cdot 180^\circ$; its exterior angles always total 360° .

Exercises

Warm-up.

1. Two angles of a triangle are 48° and 67° . Find the third.
2. Can a triangle have two right angles? Two obtuse angles? Explain using the angle-sum theorem.
3. An exterior angle of a triangle is 115° and one remote interior angle is 40° . Find the other two interior angles.
4. An isosceles triangle has a base angle of 50° . Find its apex angle.
5. An exterior angle of a triangle is 105° , and its two remote interior angles are equal. Find those two angles.

Practice.

6. The angles of a triangle are in the ratio $2 : 3 : 4$. Find them.
7. Prove that each angle of an equilateral triangle is 60° .
8. In $\triangle ABC$, the bisector of $\angle A$ meets BC at D . If $\angle B = 50^\circ$ and $\angle C = 70^\circ$, find $\angle ADB$. (Revisit the Olympiad Spark from the style preview — you can now do it.)
9. The angles of a triangle are in the ratio $1 : 2 : 3$. Find all three, and say what kind of triangle it is.

Challenge.

10. Prove that the sum of the interior angles of any (convex) quadrilateral is 360° . (*Hint: one diagonal.*)
11. Generalise: prove that the interior angles of a convex polygon with n sides sum to $(n - 2) \cdot 180^\circ$.
12. Prove that the *exterior* angles of any convex polygon, one at each vertex, sum to exactly 360° — no matter how many sides it has.

★ OLYMPIAD SPARK

(A star is born — olympiad classic.) Five points sit on a circle and are joined in order $1 \rightarrow 3 \rightarrow 5 \rightarrow 2 \rightarrow 4 \rightarrow 1$ to draw a five-pointed star (the pentagram from the Prologue). Prove that the sum of the five angles at the star's points is exactly 180° — regardless of where the five points are placed on the circle.

Everything you need is now in your hands: the angle-sum theorem and the exterior angle theorem. Find the right triangles hiding in the star.

5

Congruence: When Are Two Things the Same?

★ WONDER

A factory must stamp out thousands of identical metal triangles. To check a finished piece, an inspector is not allowed to lay it on top of the master copy. She may only *measure* it. What is the *fewest* measurements she needs to be *certain* the new triangle is a perfect copy — not merely close, but exactly the same?

The answer — remarkably — is *three*, and the story of *which* three is the theory of congruence. It is the first place geometry pays you a dividend: a short list of checks that guarantee two figures are identical, so you never have to slide one onto the other.

5.1 What congruence means

Definition 5.1 (Congruent figures). Two figures are **congruent** if one can be moved onto the other — by sliding, turning, and if necessary flipping — so that they coincide exactly. For triangles this means all three pairs of corresponding sides are equal *and* all three pairs of corresponding angles are equal. We write $\triangle ABC \cong \triangle DEF$.

The order of letters is a promise, not a decoration. Writing $\triangle ABC \cong \triangle DEF$ asserts six equalities at once: $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, so $AB = DE$, $BC = EF$, $CA = FD$ and $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$. Always list vertices in

corresponding order.

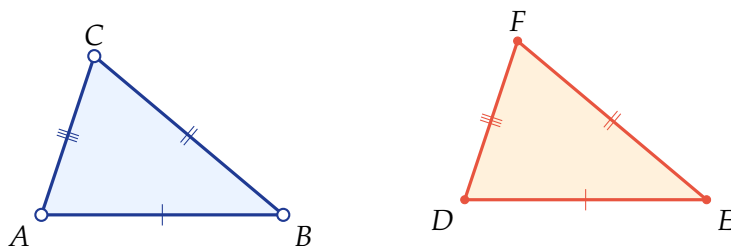


Figure 5.1 — $\triangle ABC \cong \triangle DEF$. Matching tick marks show equal sides; equal angles sit at corresponding vertices.

5.2 The shortcuts: congruence criteria

Checking all six equalities would defeat the purpose. The great discovery is that the right *three* already force the other three. Each of the following is taken as a basic criterion (provable from Euclid's axioms; we will use them as trustworthy tools).

Theorem 5.2 (The congruence criteria). Two triangles are congruent if any one of the following holds:

- SSS** three pairs of sides are equal;
- SAS** two pairs of sides and the *included* angle (the angle *between* them) are equal;
- ASA** two pairs of angles and the *included* side are equal;
- AAS** two pairs of angles and a *non*-included side are equal;
- RHS** in right triangles, the hypotenuse and one other side are equal.

The word *included* is the whole game in SAS and ASA. The angle in SAS must be the one squeezed between the two named sides; the side in ASA must be the one joining the two named angles.

AHA!

There is no “SSA” and no “AAA.” Two sides and a *non*-included angle need not fix a triangle, and three equal angles fix only the *shape*, not the *size* (that is similarity, the subject of Chapter 8). Knowing which letter-combinations *fail* is as important as knowing which succeed.

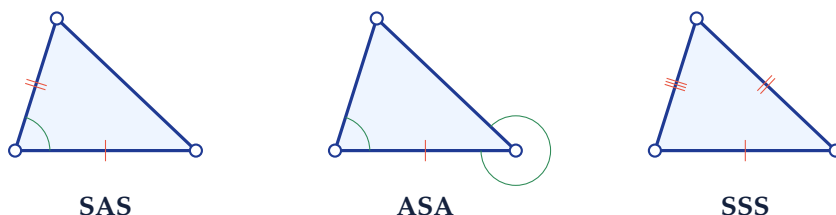


Figure 5.2 — The marked data in SAS (included angle, green), ASA (included side), and SSS. In each, the three marked facts force the entire triangle.

Worked Example. The inspector's three measurements. A finished triangle is checked by measuring one side, 6 cm long, together with the two angles *at its ends*, 50° and 60° . Is the triangle now completely determined? Yes. The measured side is the one *included* between the two measured angles, so this is exactly **ASA** — and ASA forces the whole triangle. (As a bonus, the third angle is pinned too: $180^\circ - 50^\circ - 60^\circ = 70^\circ$.) This answers the chapter's Wonder: three well-chosen measurements — a side and its two end-angles — certify a perfect copy, with no need to lay one triangle on another. ■

5.3 Why SSA can fail

It is worth *seeing* the failure, so the rule sticks. Fix an angle at A and a side AB . From B swing a segment of fixed length. If that length is right, it can reach the far ray at *two* different points C_1 and C_2 — giving two genuinely different triangles that share the same “side-side-angle.”

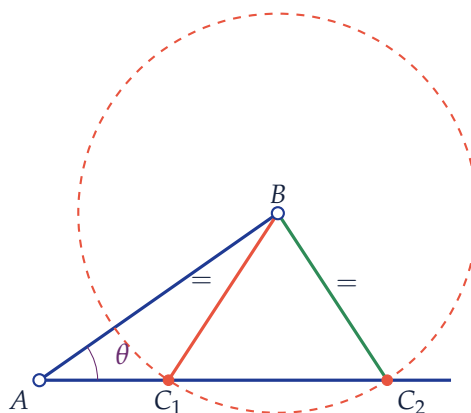


Figure 5.3 — The ambiguous case. Same angle at A , same AB , same length $BC_1 = BC_2$ — yet $\triangle ABC_1 \not\cong \triangle ABC_2$. “SSA” is no criterion.

5.4 Congruence as a proof engine

The real power of congruence is indirect: to prove two *segments* or two *angles* are equal, find two triangles containing them, prove the triangles congruent, then read off the equality as “corresponding parts.” This last step is used so often it has an abbreviation: **CPCTC** — *corresponding parts of congruent triangles are congruent*.

 **Worked Example.** C is the midpoint of both AB and DE (the segments cross at C). Prove $AD = BE$.

Look at $\triangle ACD$ and $\triangle BCE$. We have $AC = BC$ (midpoint of AB) and $DC = EC$ (midpoint of DE). The angles $\angle ACD$ and $\angle BCE$ are vertical angles, hence equal. That is two sides and the *included* angle: by **SAS**, $\triangle ACD \cong \triangle BCE$. Therefore $AD = BE$ by CPCTC. ■

COMMON TRAP

CPCTC may be used *only after* congruence is already established. Writing “ $AD = BE$ because the triangles look congruent” proves nothing. First earn the congruence with SSS/SAS/ASA/AAS/RHS; *then* harvest the corresponding parts.

 **Worked Example.** In $\triangle PQR$, $PQ = PR$, and PS bisects $\angle P$ with S on QR . Prove that $QS = SR$ and that $PS \perp QR$.

Compare $\triangle PQS$ and $\triangle PRS$. We have $PQ = PR$ (given), $\angle QPS = \angle RPS$ (the bisector), and $PS = PS$ (shared). That is two sides and the *included* angle: by **SAS**, $\triangle PQS \cong \triangle PRS$. Now harvest corresponding parts (CPCTC): $QS = SR$, so S is the midpoint of QR ; and $\angle PSQ = \angle PSR$. But those two angles form a linear pair along QR , so each is $\frac{180^\circ}{2} = 90^\circ$ — hence $PS \perp QR$. One congruence delivered *both* a length equality and a perpendicularity at once. ■

YOUR TURN

In quadrilateral $ABCD$ the diagonal AC is drawn, and $AB = AD$ while $CB = CD$. Which criterion proves $\triangle ABC \cong \triangle ADC$, and what does it

tell you about $\angle B$ and $\angle D$?

Set-up (you finish it): the two triangles share the side AC ; together with $AB = AD$ and $CB = CD$ that is three pairs of equal sides. Name the criterion, then read off the corresponding angles at B and D by CPCTC.

Answer: **SSS**, so $\triangle ABC \cong \triangle ADC$ and therefore $\angle B = \angle D$.

Congruence is the first real dividend geometry pays: three well-chosen measurements, and you *know* two figures are twins. From now on, whenever you must prove two things equal, you will go hunting for the triangles that hold them.

KEY IDEAS & FORMULAS

- **Congruent** ($\triangle ABC \cong \triangle DEF$): identical after sliding/turning/flipping; the letter order names the correspondence (six equalities).
- **Criteria:** **SSS**, **SAS** (included angle), **ASA** (included side), **AAS**, **RHS** (right triangles). Each fixes the whole triangle from three facts.
- **No SSA, no AAA:** a non-included angle can give two triangles; equal angles fix shape, not size (that is similarity, Chapter 8).
- **CPCTC:** once triangles are proved congruent, all corresponding parts are equal — the engine for proving segments or angles equal.
- **Strategy:** to prove two lengths or angles equal, find two triangles containing them and prove those congruent first.

Exercises

Warm-up.

1. If $\triangle ABC \cong \triangle PQR$, list all six pairs of equal parts.
2. Name the criterion (SSS/SAS/ASA/AAS/RHS) that applies to each: (a) three equal sides; (b) two equal angles and the side between them; (c) right angle, equal hypotenuse, one equal leg.
3. In $\triangle ABC \cong \triangle DEF$, $AB = 5$, $\angle B = 40^\circ$. State DE and $\angle E$.
4. Explain in one sentence why “AAA” does not prove congruence.
5. Two triangles share three pairs of equal angles and one pair of equal sides. Are they congruent? Which criterion?
6. Sketch a counterexample showing two non-congruent triangles with the

same “SSA” data.

Practice.

7. In Figure 5.1’s style, $AB = DE$, $\angle A = \angle D$, $\angle B = \angle E$. Prove $\triangle ABC \cong \triangle DEF$ and name the criterion.
8. $AB = AC$ and M is the midpoint of BC . Prove $\triangle ABM \cong \triangle ACM$, and deduce $AM \perp BC$.
9. In quadrilateral $ABCD$, $AB = CD$ and $AB \parallel CD$. Prove $\triangle ABC \cong \triangle CDA$ and hence that $ABCD$ is a parallelogram.
10. Two circles of equal radius centred at A and B meet at P and Q . Prove $\triangle APB \cong \triangle AQB$.
11. $\overrightarrow{BD}$ bisects $\angle ABC$, and $\angle BAD = \angle BCD = 90^\circ$. Prove $AD = CD$.
12. Point O is equidistant from the endpoints of segment PQ and so is point R . Prove that line OR is the perpendicular bisector of PQ .
13. In $\triangle ABC$, D lies on BC with $BD = DC$ and $AD \perp BC$. Prove $AB = AC$.

Challenge.

14. (Kite.) $ABCD$ has $AB = AD$ and $CB = CD$. Prove that diagonal AC bisects both $\angle A$ and $\angle C$, and that $AC \perp BD$.
15. $\triangle ABC$ is equilateral. Points X, Y, Z lie on BC, CA, AB respectively with $BX = CY = AZ$. Prove that $\triangle XYZ$ is equilateral.
16. In $\triangle ABC$, the bisector of $\angle A$ meets BC at D , and $DB = DC$. Prove that $\triangle ABC$ is isosceles.
17. Squares $ABPQ$ and $ACRS$ are drawn outward on sides AB and AC of $\triangle ABC$. Prove that $QC = BS$, and in fact that $QC \perp BS$. (*Hint: a 90° rotation about A sends $Q \rightarrow B$ and $C \rightarrow S$, carrying segment QC onto BS .*)
18. Prove the converse direction of the kite problem (Problem 14): if AC bisects $\angle A$ and $\angle C$ in quadrilateral $ABCD$, then $AB = AD$ and $CB = CD$.

★ OLYMPIAD SPARK

(A competition classic.) Let $ABCD$ be a square. Points E and F lie on sides BC and CD respectively such that $\angle EAF = 45^\circ$. Prove that $EF = BE + DF$. (*Hint: rotate $\triangle ABE$ by 90° about A so that AB lands on AD ; congruence does the rest. Rotation + congruence is a signature olympiad move.*)

(Second spark.) In $\triangle ABC$, the medians from B and C are equal in length. Prove that $AB = AC$ (an isosceles triangle). *Try to find the congruent*

triangles; this is harder than it looks.

6

Compass and Straightedge

★ WONDER

You are given only two tools: a *straightedge* (an unmarked ruler — it can draw a line through two points, but cannot measure) and a *compass* (it can draw a circle of any radius about any centre). No protractor, no numbers. With these alone, the Greeks built almost all of geometry. Can you, with just these two tools, cut any angle *exactly* in half — with no measuring at all?

You can, and you will, in this chapter. Constructions are not a quaint old game. They are *proof made physical*: every legal construction corresponds to a theorem about what *must* exist. And the discipline of using only these two idealised tools is what gives geometry its bedrock certainty.

6.1 The rules

Exactly two moves are permitted:

- **Straightedge:** given two existing points, draw the (infinite) line through them.
- **Compass:** given a centre point and a radius (the distance between two existing points), draw a circle.

New points may be created only where drawn lines and circles *cross*. That is the whole game — and it is astonishingly powerful.

6.2 Construction 1: the perpendicular bisector

To construct the perpendicular bisector of a segment AB.

Open the compass wider than half of AB . Draw an arc centred at A , then an arc of the *same* radius centred at B . They meet at two points, P above and Q below. The line PQ is the perpendicular bisector.

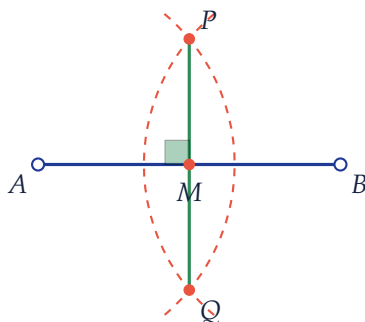


Figure 6.1 — Equal arcs from A and B cross at P and Q . Line PQ bisects AB at right angles.

Why it works. Both P and Q are equidistant from A and from B (equal compass radius). The set of all points equidistant from A and B is the perpendicular bisector — so P and Q both lie on it, and two points determine the line. (You proved exactly this fact in Exercise 5.12.)

6.3 Construction 2: bisecting an angle

To bisect a given angle with vertex A .

Draw an arc centred at A , cutting the two sides at P and Q . Now draw two equal arcs centred at P and at Q ; they meet at R . The ray AR bisects the angle.

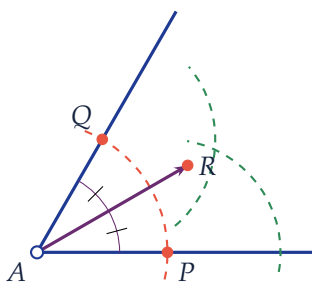


Figure 6.2 — Equal arcs from P and Q meet at R ; ray AR splits the angle into two equal halves (matching ticks).

Why it works. By construction $AP = AQ$ (one arc) and $PR = QR$ (equal arcs), and AR is shared. So $\triangle APR \cong \triangle AQR$ by **SSS**, and therefore $\angle PAR = \angle QAR$ by CPCTC. Congruence, doing exactly the job it was built for.

6.4 Construction 3: the equilateral triangle (Euclid, Proposition I.1)

The very first proposition of Euclid's *Elements* builds an equilateral triangle on a given segment AB . Draw a circle centred at A through B , and another centred at B through A . Where they meet is the apex C .

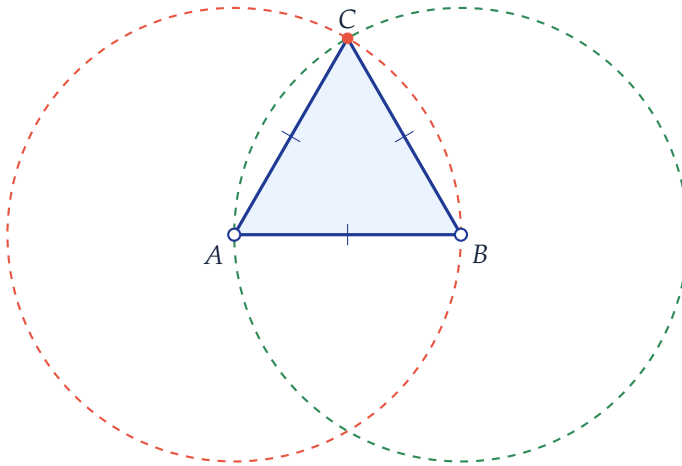


Figure 6.3 — Two circles of radius AB meet at C . Then $CA = AB = BC$: an equilateral triangle, built from nothing but two circles.

Why it works. $CA = AB$ because both are radii of the first circle; $CB = AB$ because both are radii of the second. Hence all three sides equal AB . It is hard to overstate how much rests on this little figure: it is, quite literally, theorem number one.

6.5 Building specific angles and lengths

The three basic moves combine into a surprising reach. Two short examples show the pattern — and answer the chapter's Wonder.

 **Worked Example.** Construct a 45° angle with straightedge and compass alone.

Steps. First erect a *right* angle: take a line, and at a point on it construct the perpendicular (Construction 1's idea, applied at the point). That gives 90° . Now *bisect* that right angle by Construction 2.

Why it works. The perpendicular is exactly 90° (the perpendicular-bisector construction produces a right angle), and bisection halves it: $\frac{90^\circ}{2} = 45^\circ$.

No protractor ever touched the page — two certified moves, stacked. ■

 **Worked Example.** From a segment of length 1, construct a segment of length $\sqrt{2}$.

Construct a right angle at a point O , and with the compass mark off length 1 along each arm, reaching A and B . Join A to B . By the Pythagorean theorem (Chapter 9), the hypotenuse satisfies $AB^2 = 1^2 + 1^2 = 2$, so $AB = \sqrt{2}$.

The wow: an *irrational* length — one with no exact decimal — is built *exactly*, in three strokes, from a unit and a right angle. Repeating the trick on the new hypotenuse builds $\sqrt{3}$, then $\sqrt{4} = 2$, then $\sqrt{5}, \dots$ — the elegant spiral of Theodorus. ■

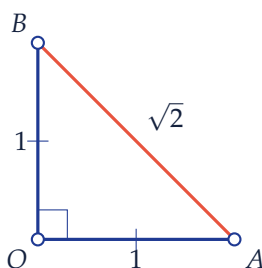


Figure 6.4 — A right angle and two unit legs build a hypotenuse of length $\sqrt{2}$, exactly — the first step of the spiral of Theodorus.

AHA!

Notice the rhythm of every construction: *do*, then *prove*. The compass and straightedge produce a figure; congruence (or the equidistance idea) certifies it is the figure we wanted. A construction without its justification is a magic trick; with it, it is mathematics.

MATHEMATICIAN'S MARGIN

For over two thousand years, three construction puzzles resisted every attempt: *trisecting* an arbitrary angle, *doubling the cube*, and *squaring the circle*. Only in the 1800s, using algebra and the new idea of *constructible numbers*, were all three proven **impossible** with straightedge and compass alone. Some of the deepest theorems are about what *cannot* be done.

Two tools, no numbers, and yet every figure you build arrives with a proof that it must be exactly what you claim. Construction is geometry rolling up its sleeves.

→ YOUR TURN

Outline how to construct a 30° angle with straightedge and compass.

Set-up (you finish it): you already know how to build a 60° angle for free — it is a corner of the equilateral triangle of Construction 3. What single further move turns 60° into 30° ?

Answer: build a 60° angle (one vertex of an equilateral triangle), then *bisect* it (Construction 2): $\frac{60^\circ}{2} = 30^\circ$.

KEY IDEAS & FORMULAS

- **Two moves only:** straightedge (line through two points) and compass (circle from a centre and a radius); new points appear only where drawn lines/circles cross.
- **Core constructions:** perpendicular bisector and midpoint; angle bisector; equilateral triangle (hence a 60° angle); perpendiculars; copying an angle; a parallel through a point.
- **Built angles:** $60^\circ, 30^\circ, 15^\circ$ (equilateral, then bisect); $90^\circ, 45^\circ$ (perpendicular, then bisect).
- **Every construction = do, then prove:** the figure is certified by congruence (often **SSS**) or the equidistance idea.
- **Provably impossible** with these tools: trisecting a general angle, doubling the cube, squaring the circle.

Exercises

Warm-up.

1. List the only two moves permitted in classical construction, and say where new points are allowed to appear.
2. In the perpendicular-bisector construction, why must the compass be opened to *more* than half of AB ?
3. In Construction 2, which congruence criterion proves the angle is bisected?
4. True or false: a straightedge may be used to measure a length. Explain.
5. Why are CA , AB , and BC all equal in Figure 6.3?

Practice. (For each, describe the steps, then justify why it works.)

6. Construct the midpoint of a given segment.
7. Construct a line through a given point P perpendicular to a given line ℓ , when P is *not* on ℓ .
8. Construct a line through a point P that lies *on* ℓ , perpendicular to ℓ .
9. Copy a given angle onto a given ray.
10. Construct a 60° angle. Then bisect it to obtain 30° , and again for 15° .
11. Construct a 90° angle, then a 45° angle, using only the tools.
12. Given a line ℓ and a point P off it, construct the line through P *parallel* to ℓ . (Hint: copy an angle, or drop two perpendiculars.)

Challenge.

13. Construct a square given one of its sides. Justify each step.
14. Divide a given segment into *four* equal parts using only compass and straightedge.
15. Construct the three angle bisectors of a triangle and observe that they meet at one point (the *incentre*). You will prove this concurrency in Chapter 10.
16. Given a circle but *not* its centre, construct the centre. (Hint: two chords and their perpendicular bisectors.)
17. Construct a triangle given the lengths of its three sides a, b, c (with a drawn). When is this impossible?

★ OLYMPIAD SPARK

(Constructibility.) Starting from a segment of length 1, show how to construct segments of length $\sqrt{2}$, then $\sqrt{3}$, and in general $\sqrt{n}$ for any positive integer n . (Hint: right triangles, and the theorem of the next-but-two chapter. The elegant “spiral of Theodorus” stacks these square roots one on the next.)

(Second spark.) Prove that a 20° angle *cannot* be constructed with straightedge and compass, given that a 60° angle can. (This is the famous impossibility of trisecting the 60° angle — state clearly what you would need and why the tools cannot deliver it; a full proof needs algebra you will meet later.)

PART II

The Triangle's Secrets

7

Isosceles, Equilateral, and the Bridge of Asses

★ WONDER

Fold a sheet of paper in half and cut a triangle whose base sits on the fold line. Unfold it: the two halves match perfectly. Some triangles have this hidden mirror; most do not. What does that single line of symmetry secretly force to be true about the triangle's *angles*?

A triangle with two equal sides is called **isosceles** (from the Greek for “equal legs”). That one equality — two sides the same — ripples outward into a whole family of consequences, and gives us a theorem so central, and historically so feared by struggling students, that it earned a nickname: the *Bridge of Asses*.

7.1 The Isosceles Triangle Theorem

Theorem 7.1 (Base angles of an isosceles triangle). If two sides of a triangle are equal, then the angles opposite them are equal. That is, in $\triangle ABC$ with $AB = AC$, we have $\angle B = \angle C$.

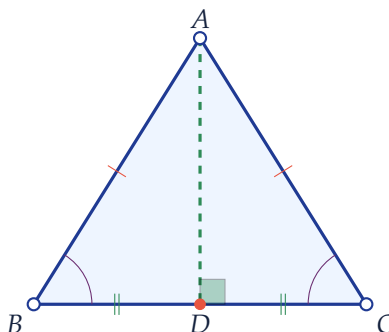


Figure 7.1 — $AB = AC$ forces $\angle B = \angle C$. The line AD to the base's midpoint is simultaneously median, altitude, and angle bisector.

Proof. Let D be the midpoint of BC and draw AD . Compare $\triangle ABD$ and $\triangle ACD$:

$$AB = AC \text{ (given),} \quad BD = DC \text{ (midpoint),} \quad AD = AD \text{ (shared).}$$

By **SSS**, $\triangle ABD \cong \triangle ACD$. Reading off corresponding parts (CPCTC): $\angle B = \angle C$. As a bonus, $\angle ADB = \angle ADC$; since they form a linear pair they are each 90° , so $AD \perp BC$ — and $\angle BAD = \angle CAD$, so AD bisects the apex too. ■

AHA!

One stroke — the line AD — did four jobs at once. In an isosceles triangle the *median*, the *altitude*, the *angle bisector* from the apex, and the *perpendicular bisector* of the base are all the *same line*: the axis of symmetry. Whenever you meet an isosceles triangle, look for this line. It is almost always the key.

The converse is also true, and just as useful for *proving* a triangle isosceles.

Theorem 7.2 (Converse). If two angles of a triangle are equal, then the sides opposite them are equal. So $\angle B = \angle C$ implies $AB = AC$.

Proof. Drop the bisector of $\angle A$, meeting BC at D . Then $\angle BAD = \angle CAD$, $\angle B = \angle C$ (given), and AD is shared, so $\triangle ABD \cong \triangle ACD$ by **AAS**. Hence $AB = AC$ by CPCTC. ■

MATHEMATICIAN'S MARGIN

This theorem is *Pons Asinorum*, the “Bridge of Asses” — Proposition I.5 of Euclid. The figure Euclid drew, with the triangle’s sides extended below the base, resembles a footbridge; and medieval students who could not “cross” it (master the proof) were teased as the asses who balked at the bridge. Take heart: with congruence in hand, you crossed it in three lines.

7.2 The equilateral triangle

An **equilateral** triangle has all three sides equal. Applying the base-angle theorem to each pair of equal sides forces all three angles equal — and three equal angles summing to 180° must each be 60° .

Corollary 7.3. A triangle is equilateral if and only if it is equiangular, in which case every angle is 60° .

 **Worked Example.** In $\triangle ABC$, $AB = AC$ and $\angle A = 40^\circ$. Find $\angle B$ and $\angle C$.

The base angles are equal: $\angle B = \angle C$. Since the three sum to 180° ,

$$40^\circ + 2\angle B = 180^\circ \Rightarrow \angle B = \angle C = 70^\circ.$$

 **Worked Example. Using the converse.** In $\triangle ABC$, $\angle A = 50^\circ$ and $\angle B = 65^\circ$. Show the triangle is isosceles, and name the equal sides.

First find the third angle: $\angle C = 180^\circ - 50^\circ - 65^\circ = 65^\circ$. So $\angle B = \angle C$, and by the *converse* of the base-angle theorem the sides opposite them are equal. The side opposite $\angle B$ is AC and the side opposite $\angle C$ is AB , so $AB = AC$ — the triangle is isosceles with apex A . (Spotting two equal angles is the standard way to *prove* a triangle isosceles.) ■

 **Worked Example.** In $\triangle ABC$, $AB = AC$, and the *exterior* angle at B measures 110° . Find $\angle A$.

The exterior angle at B and the interior $\angle B$ form a linear pair, so $\angle B = 180^\circ - 110^\circ = 70^\circ$. Since $AB = AC$, the base angles are equal: $\angle C =$

$\angle B = 70^\circ$. Then $\angle A = 180^\circ - 70^\circ - 70^\circ = 40^\circ$. Two chapters' tools — the linear pair from Chapter 2 and the base-angle theorem here — combine in three short lines. ■

One equality between two sides, and a single fold of symmetry hands you equal angles, a right angle, a bisector, and a median all at once. Few lines in geometry work as hard as the one down the middle of an isosceles triangle.

→ YOUR TURN

An isosceles triangle has an apex angle of 36° . Find its two (equal) base angles.

Set-up (you finish it): the base angles are equal; call each β . The three angles sum to 180° , so $36^\circ + 2\beta = 180^\circ$. Solve for β .

Answer: $\beta = \frac{180^\circ - 36^\circ}{2} = 72^\circ$. (These 72° base angles are exactly the ones hiding in a regular pentagon.)

KEY IDEAS & FORMULAS

- **Isosceles:** two equal sides. **Base-angle theorem:** $AB = AC \Rightarrow \angle B = \angle C$; the **converse** ($\angle B = \angle C \Rightarrow AB = AC$) is the usual way to *prove* a triangle isosceles.
- **Axis of symmetry:** from the apex, the median, altitude, angle bisector, and perpendicular bisector of the base are one and the same line — the first thing to draw.
- **Equilateral** $\iff$ **equiangular**, every angle 60° .
- **Strategy:** meet an isosceles triangle, look for its axis of symmetry; meet two equal angles, suspect equal sides.

Exercises

Warm-up.

1. In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = 100^\circ$. Find $\angle B$ and $\angle C$.
2. In isosceles $\triangle ABC$ with $AB = AC$, $\angle B = 65^\circ$. Find $\angle A$.
3. Each base angle of an isosceles triangle is 50° . Find the apex angle.
4. Is every equilateral triangle isosceles? Is every isosceles triangle equilateral? Explain.

5. Name the four lines that coincide along the axis of symmetry of an isosceles triangle.
6. An isosceles triangle has a 90° apex angle. Find its base angles.

Practice.

7. In $\triangle ABC$, $\angle B = \angle C = 72^\circ$. A point D on AB satisfies $CD = CB$. Find all angles of $\triangle BCD$.
8. The apex angle of an isosceles triangle is 4 times a base angle. Find all three angles.
9. $\triangle ABC$ is isosceles with $AB = AC$. The bisector of $\angle B$ meets AC at D . If $\angle A = 40^\circ$, find $\angle BDC$.
10. Prove: if the bisector of the apex angle of a triangle is also the median to the opposite side, the triangle is isosceles.
11. In equilateral $\triangle ABC$, P lies on BC . Prove $\angle BAP + \angle CAP = 60^\circ$ and find $\angle APB + \angle APC$.
12. Two isosceles triangles ABC ($AB = AC$) and DBC ($DB = DC$) sit on the same base BC , with A, D on the same side. Prove that line AD is the perpendicular bisector of BC .
13. In $\triangle ABC$, $AB = AC$. Points E on AB and F on AC satisfy $AE = AF$. Prove $EF \parallel BC$.

Challenge.

14. (The 80–80–20 triangle, a famous teaser.) In $\triangle ABC$, $AB = AC$ and $\angle A = 20^\circ$. Point D on AC and E on AB satisfy $\angle DBC = 60^\circ$ and $\angle ECB = 50^\circ$. Find $\angle BDE$. (Start by finding every angle you can.)
15. Equilateral $\triangle ABC$. Equilateral triangles BCP , CAQ , ABR are erected outward. Prove $AP = BQ = CR$.
16. In $\triangle ABC$, prove that $\angle A = 2\angle B$ if and only if $a^2 = b(b + c)$. (Draw the bisector of $\angle A$ to D on BC ; show $\triangle ABD$ is isosceles and $\triangle CAD \sim \triangle CBA$.)
17. Inside square $ABCD$, an equilateral triangle ABP is drawn (with P inside). Find $\angle PCD$.
18. Prove that the three medians of an equilateral triangle are equal in length, and conversely that a triangle with three equal medians is equilateral.

★ OLYMPIAD SPARK

(Olympiad classic.) In $\triangle ABC$, $AB = AC$. A point P inside the triangle satisfies $\angle PBC = \angle PCA$ and $\angle PCB = \angle PBA$. Prove that AP bisects

$\angle BAC$. (*Symmetry is begging to be used.*)

(Second spark.) Let $\triangle ABC$ be isosceles with $AB = AC$, and let M be the midpoint of BC . A line through M meets AB at P and line AC at Q . Prove that $\frac{1}{BP} - \frac{1}{CQ}$ stays constant as the line rotates about M . (Save this one until after Chapter 8.)

8

Similarity and the Power of Ratio

★ WONDER

On a sunny afternoon a metre stick casts a shadow 1.5 metres long. At the same moment, a tall tree casts a shadow 24 metres long. Without climbing, without any special instrument, you can state the tree's height *exactly*. How can a shadow on the ground measure a height in the air?

The secret is **similarity**: figures that have the *same shape* but not necessarily the same size. Where congruence asks “are these identical?”, similarity asks the subtler and more powerful question “are these scaled copies?” — and the answer unlocks heights, distances, and a vast share of all olympiad geometry.

8.1 Same shape, any size

The intuition is everyday: a photograph and its enlargement show the *same* picture at different scales. Every angle is preserved, and every length is multiplied by one fixed number. Triangles are the place to pin this down precisely, because a triangle's shape is completely controlled by its angles.

Definition 8.1 (Similar triangles). Two triangles are **similar**, written $\triangle ABC \sim \triangle DEF$, if their corresponding angles are equal and their corresponding sides are in proportion:

$$\angle A = \angle D, \angle B = \angle E, \angle C = \angle F \quad \text{and} \quad \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = k.$$

The common ratio k is the **scale factor**.

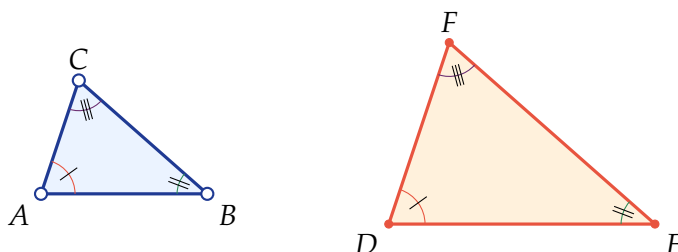


Figure 8.1 — $\triangle ABC \sim \triangle DEF$. Equal angles (matching arcs), sides scaled by a common factor k .

8.2 Three criteria — and the one you will use most

For *congruence* we needed three facts (SSS, SAS, ASA, ...). Similarity is cheaper, because we no longer insist the sizes match — only the shape. There are three standard tests.

Theorem 8.2 (Similarity criteria). Two triangles are similar if any one of the following holds.

AA Two angles of one equal two angles of the other.

SAS An angle of one equals an angle of the other, and the two sides enclosing those angles are in proportion.

SSS All three pairs of corresponding sides are in proportion.

Why two angles are enough. Fix two angles of a triangle; the third is then forced by the 180° sum, so the triangle is completely determined *up to size*. Stretch it uniformly and the angles never change — you simply get a larger copy. That is exactly what **AA** captures: equal angles compel proportional sides. SAS and SSS are the back-ups for when you are handed sides rather than angles.

Where you will use it. **AA** is the workhorse of the whole book. Almost every time you “find equal angles” — from parallel lines, vertical angles, a shared angle, a cyclic quadrilateral (Part III) — a pair of similar triangles is waiting, and with it a proportion that cracks the problem. Train your eye to ask, at every figure: *which two triangles here are similar?*

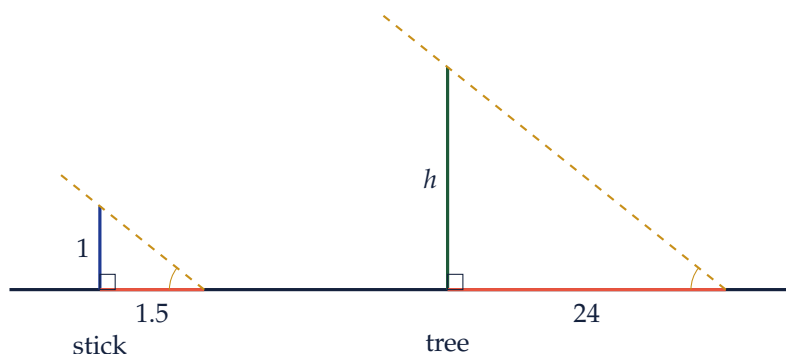


Figure 8.2 — Same sun, same instant: the stick and the tree make similar right triangles (equal shadow-angle, both upright), so heights are proportional to shadows.

 **Worked Example.** Solve the shadow problem: a 1 m stick casts 1.5 m; a tree casts 24 m.

The Sun's rays strike both at the same angle, and both stick and tree stand vertically, so the two “object-shadow” right triangles share two equal angles: by **AA** they are similar. Corresponding sides are proportional:

$$\frac{\text{tree height}}{\text{stick height}} = \frac{\text{tree shadow}}{\text{stick shadow}} \Rightarrow \frac{h}{1} = \frac{24}{1.5}.$$

So $h = 16$ m. The shadow on the ground measured the height in the air. ■

 **Worked Example.** **An angle you are simply handed.** In $\triangle ABC$, a point D lies on AB and E on AC so that $\angle ADE = \angle ACB$. Given $AD = 3$, $AC = 6$, $AB = 8$, find AE .

Compare $\triangle ADE$ and $\triangle ACB$. They share the angle at A , and we are given $\angle ADE = \angle ACB$; by **AA** they are similar. The subtlety is the *correspondence*: matching equal angles, $A \leftrightarrow A$, $D \leftrightarrow C$, $E \leftrightarrow B$, so

$$\frac{AD}{AC} = \frac{AE}{AB} \Rightarrow \frac{3}{6} = \frac{AE}{8} \Rightarrow AE = 4.$$

Notice we did *not* write $AD/AB = AE/AC$ — that would pair the wrong vertices. Equal angles tell you *which* sides correspond; get that order right and the proportion is automatic. ■

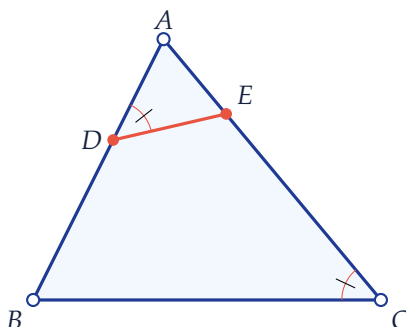


Figure 8.3 — The handed angle $\angle ADE = \angle ACB$ (matching marks) makes $\triangle ADE \sim \triangle ACB$, with $D \leftrightarrow C$ and $E \leftrightarrow B$.

X COMMON TRAP

Similarity statements are *ordered*: $\triangle ABC \sim \triangle DEF$ means $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, and the proportion must pair sides accordingly. Writing the letters in the wrong order is the most common similarity mistake. And beware: just as **SSA** fails to prove congruence, “two sides proportional plus a *non-included* angle equal” does *not* prove similarity. The equal angle in SAS must sit *between* the two proportional sides.

8.3 Thales’ Basic Proportionality Theorem

The single most useful consequence of similarity appears the moment a line is drawn *parallel* to one side of a triangle: it slices the other two sides into matching proportions. Intuitively, a parallel cut makes a small triangle at the apex that is a scaled copy of the whole, so its sides keep the same ratios.

Theorem 8.3 (Basic Proportionality / Thales). A line parallel to one side of a triangle, meeting the other two sides, divides them in the same ratio. In $\triangle ABC$ with D on AB , E on AC and $DE \parallel BC$,

$$\frac{AD}{DB} = \frac{AE}{EC}.$$

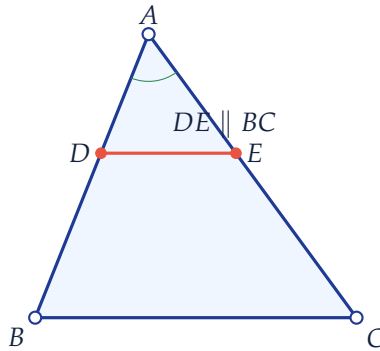


Figure 8.4 — A line $DE \parallel BC$ cuts AB and AC proportionally. Here $\triangle ADE \sim \triangle ABC$.

Sketch. Because $DE \parallel BC$, the corresponding angles give $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$; the angle at A is shared. By **AA**, $\triangle ADE \sim \triangle ABC$, so $AD/AB = AE/AC$. A little algebra rearranges this to $AD/DB = AE/EC$. ■

Worked Example. Whole side or part? In $\triangle ABC$, D lies on AB and E on AC with $DE \parallel BC$. Given $AD = 4$, $AB = 10$, and $DE = 6$, find BC .

Since $DE \parallel BC$ we have $\triangle ADE \sim \triangle ABC$, so the ratio of *corresponding* sides is constant. Here DE corresponds to BC , and AD corresponds to the *whole* side AB — not to DB . Hence

$$\frac{AD}{AB} = \frac{DE}{BC} \Rightarrow \frac{4}{10} = \frac{6}{BC} \Rightarrow BC = 15.$$

The trap is to divide by $DB = 6$ instead of by $AB = 10$: the side DE lives in the *big* triangle ABC , so it must be compared with the big triangle's full side. When in doubt, name the two similar triangles first, then read corresponding sides straight off the similarity statement. ■

The *converse* is equally prized: if a line divides two sides of a triangle proportionally, it is parallel to the third side — a clean test for parallelism.

Corollary 8.4 (Midsegment Theorem). The segment joining the midpoints of two sides of a triangle is parallel to the third side and exactly half its length.

8.4 The geometric mean in a right triangle

Drop the altitude from the right angle of a right triangle to the hypotenuse, and similarity hands you one of the prettiest length relations in geometry — and a second route to Pythagoras.

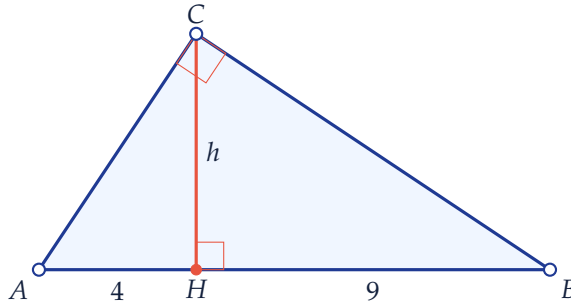


Figure 8.5 — The altitude CH to the hypotenuse splits a right triangle into two smaller ones, each similar to the whole: $\triangle ACH \sim \triangle ABC \sim \triangle CBH$.

Worked Example. In right $\triangle ABC$ (right angle at C), the altitude from C meets the hypotenuse AB at H , with $AH = 4$ and $HB = 9$. Find the altitude CH .

In $\triangle ACH$ and $\triangle CBH$: each has a right angle (at H), and $\angle HAC = \angle HCB$ (both are the complement of $\angle ACH$). By **AA**, $\triangle ACH \sim \triangle CBH$. Matching the right angles and these equal angles, H -side to H -side,

$$\frac{AH}{CH} = \frac{CH}{HB} \Rightarrow CH^2 = AH \cdot HB = 4 \cdot 9 = 36 \Rightarrow CH = 6.$$

The altitude is the **geometric mean** of the two pieces it cuts: $CH = \sqrt{AH \cdot HB}$. The same pair of similar triangles gives $CA^2 = AH \cdot AB$ and $CB^2 = HB \cdot AB$ — and adding those two recovers $CA^2 + CB^2 = AB^2$, Pythagoras itself (we meet this again, head-on, in Chapter 9). ■

→ YOUR TURN

A right triangle has its altitude to the hypotenuse equal to 6, and that altitude meets the hypotenuse so that one of the two pieces has length 4. Find the other piece and the full hypotenuse.

Set-up (you finish it): the altitude is the geometric mean of the two pieces, so $6^2 = 4 \cdot (\text{other piece})$. Solve for the other piece, then add the two

pieces for the hypotenuse.

Answer: other piece = 9, hypotenuse = $4 + 9 = 13$.

AHA!

Similarity converts *shape information* (parallels, equal angles) into *number information* (equal ratios). That bridge — from geometry to algebra and back — is the single most productive idea in the whole subject. Almost every length you cannot see directly, you can reach through a proportion.

MATHEMATICIAN'S MARGIN

A nice fact for later: if two triangles are similar with scale factor k , their areas are in the ratio k^2 . Double every length and the area quadruples; triple the lengths and area grows nine-fold. Lengths scale by k , areas by k^2 , and (in three dimensions) volumes by k^3 .

Congruence asked whether two figures are identical. Similarity asks the deeper question of whether they are scaled echoes, and that single shift lets a shadow on the ground weigh a tree in the air.

KEY IDEAS & FORMULAS

- **Similar** $\triangle ABC \sim \triangle DEF$: corresponding angles equal, corresponding sides in one ratio k (the scale factor).
- **Criteria**: **AA** (two equal angles) — the workhorse; **SAS** (equal included angle + the two sides proportional); **SSS** (all three sides proportional).
- **Basic Proportionality (Thales)**: $DE \parallel BC \Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$; converse tests parallelism. **Midsegment**: joins two midpoints, $\parallel$ third side and half its length.
- **Right-triangle altitude**: $CH = \sqrt{AH \cdot HB}$, and each $\text{leg}^2 = (\text{its hypotenuse-segment}) \times (\text{hypotenuse})$.
- **Scaling**: lengths $\times k$, areas $\times k^2$, volumes $\times k^3$.

Exercises

Warm-up.

1. $\triangle ABC \sim \triangle DEF$ with $AB = 4$, $DE = 6$. Find the scale factor from ABC to DEF .
2. In the same pair, $BC = 5$. Find EF .
3. Two triangles have angles $(50^\circ, 60^\circ, 70^\circ)$ and $(60^\circ, 70^\circ, 50^\circ)$. Are they similar? Why?
4. A 2 m pole casts a 3 m shadow. A nearby building casts a 30 m shadow. How tall is the building?
5. In $\triangle ABC$, D, E are midpoints of AB, AC . If $BC = 14$, find DE .
6. State the AA criterion in your own words.
7. Name the three similarity criteria, and for each give the minimum information it needs.
8. Triangles have sides $(3, 4, 5)$ and $(6, 8, 10)$. Are they similar? By which criterion, and what is the scale factor?
9. $\triangle PQR \sim \triangle XYZ$ with scale factor $3 : 2$ (from PQR to XYZ). If $PQ = 9$, find XY .
10. In a right triangle the altitude to the hypotenuse divides it into segments 2 and 8. Find the altitude.

Practice.

11. In $\triangle ABC$, $DE \parallel BC$ with D on AB , E on AC . If $AD = 3$, $DB = 5$, $AE = 4.2$, find EC .
12. With $DE \parallel BC$: $AD = x$, $DB = x + 2$, $AE = 3$, $EC = 6$. Find x .
13. A line parallel to BC cuts off a small triangle at A with $AD/AB = 2/5$. Find the ratio of the area of $\triangle ADE$ to $\triangle ABC$.
14. Two similar triangles have areas 9 and 25. Find the ratio of their corresponding sides.
15. In right $\triangle ABC$ (right angle at C), the altitude from C meets the hypotenuse at H . Prove $\triangle ACH \sim \triangle ABC$, and find CH if $AH = 4$, $HB = 9$.
16. Diagonals of trapezium $ABCD$ ($AB \parallel CD$) meet at P . Prove $\triangle ABP \sim \triangle CDP$, and if $AB = 8$, $CD = 12$, find the ratio $AP : PC$.
17. A 6 ft person walks away from a 15 ft lamppost. When they are 12 ft from the post, how long is their shadow?
18. In $\triangle ABC$, points D, E on AB, AC give $AD/DB = AE/EC = 2/3$. Prove $DE \parallel BC$ and find DE/BC .

19. In $\triangle ABC$ and $\triangle ADE$ the angle at A is common, and $AD/AB = AE/AC = 2/3$. Name the criterion that makes them similar, and if $BC = 12$ find DE .
20. A 1.6 m surveyor sights so that her line of sight to a tower-top just grazes the top of a 4 m pole 5 m in front of her; the tower is 45 m away. Use similar triangles to estimate the tower's height above eye level.
21. In Figure 8.5, with $AH = 4$ and $HB = 9$, find the two legs CA and CB (use $CA^2 = AH \cdot AB$ and $CB^2 = HB \cdot AB$), and verify $CA^2 + CB^2 = AB^2$.

Challenge.

22. (Angle Bisector Theorem.) The bisector of $\angle A$ in $\triangle ABC$ meets BC at D . Prove $BD/DC = AB/AC$. (Hint: draw through C a line parallel to the bisector.)
23. In $\triangle ABC$, a line meets AB at D , AC at E , and is parallel to BC . If DE further has $BD = DA$, find the ratio in which DE divides the area of the triangle.
24. Prove that the diagonals of a trapezium divide each other in the same ratio as the parallel sides.
25. A square is inscribed in a right triangle of legs a, b so that one corner is at the right angle and the opposite corner lies on the hypotenuse. Find the side of the square in terms of a and b .
26. $\triangle ABC$ has D on BC . Prove that $AD^2 = AB \cdot AC - BD \cdot DC$ exactly when AD bisects $\angle A$ (a taste of Stewart's theorem, Chapter 20).
27. (Crossing poles.) Two vertical poles of heights a and b stand on level ground. The line from the top of each to the foot of the other cross at height h . Prove $\frac{1}{h} = \frac{1}{a} + \frac{1}{b}$, and note that h does not depend on the distance between the poles.

★ OLYMPIAD SPARK

(Olympiad.) In $\triangle ABC$, points D and E trisect side BC (so $BD = DE = EC$). Let M be the midpoint of AC , and let BM meet AD at P . Find the ratio $AP : PD$. (Mass points or similar triangles both work; try similarity first.)

(Second spark.) Through a point P inside $\triangle ABC$, three lines are drawn parallel to the three sides, cutting the triangle into three small triangles and three parallelograms. If the three small triangles have areas 4, 9, and 16, find the area of $\triangle ABC$. (A gorgeous use of $\text{area} \propto k^2$.)

9

Pythagoras, Four Ways

★ WONDER

Of all the theorems in mathematics, one is so important that over *four hundred* distinct proofs of it have been collected — by schoolchildren, by presidents, by Leonardo da Vinci. Why would anyone need four hundred proofs of a single fact? And what is it about a humble right triangle that has fascinated every civilisation that ever measured land?

We meet at last the most famous relationship in geometry. We will not be content to state it; we will prove it *four* different ways, because each proof reveals a different face of the same gem — and learning to see one truth from many directions is exactly the habit that makes a problem-solver.

9.1 The theorem

Theorem 9.1 (Pythagoras). In a right triangle, the square on the hypotenuse equals the sum of the squares on the other two sides. If the legs are a, b and the hypotenuse is c , then

$$a^2 + b^2 = c^2.$$

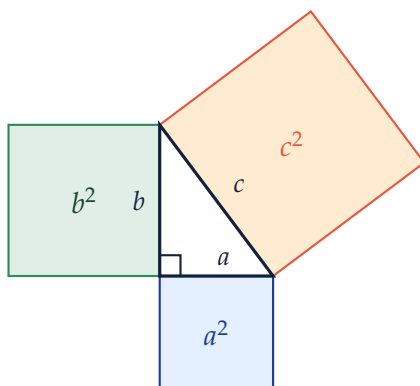


Figure 9.1 — The classical statement: the two smaller squares ($a^2 + b^2$) together hold exactly as much area as the large one (c^2).

9.2 Proof 1 — by similar triangles

Drop the altitude from the right angle C to the hypotenuse, meeting it at H . This single line splits the triangle into two smaller ones, each *similar* to the original.

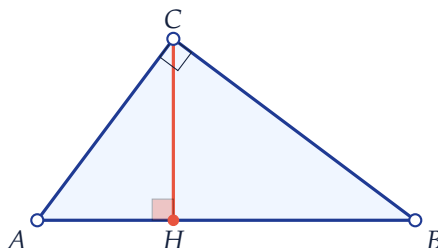


Figure 9.2 — The altitude CH creates $\triangle ACH \sim \triangle ABC \sim \triangle CBH$. Three similar triangles from one stroke.

Proof. In $\triangle ACH$ and $\triangle ABC$: both have a right angle (at H and at C) and share $\angle A$. By **AA**, $\triangle ACH \sim \triangle ABC$, giving $\frac{AC}{AB} = \frac{AH}{AC}$, hence $AC^2 = AB \cdot AH$. Similarly $\triangle CBH \sim \triangle ABC$ gives $BC^2 = AB \cdot BH$. Add:

$$AC^2 + BC^2 = AB \cdot AH + AB \cdot BH = AB (AH + BH) = AB \cdot AB = AB^2.$$

With $AC = b$, $BC = a$, $AB = c$, this is $a^2 + b^2 = c^2$. ■

9.3 Proof 2 — by rearrangement

Take a square of side $a + b$. Place four copies of the right triangle inside it, one along each edge. What is left in the middle is a tilted square of side c .

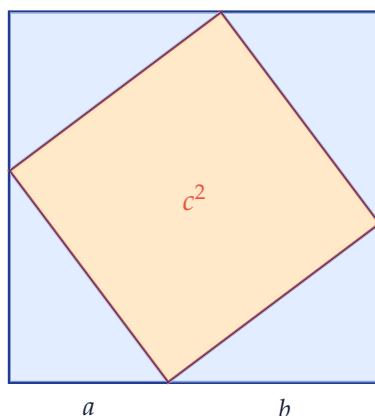


Figure 9.3 — A square of side $a + b$ holds four right triangles plus a tilted square of side c .

Proof. The big square's area is $(a + b)^2$. It is also the four triangles plus the inner square:

$$(a + b)^2 = 4 \cdot \frac{1}{2}ab + c^2.$$

Expanding the left side, $a^2 + 2ab + b^2 = 2ab + c^2$, and cancelling $2ab$ from both sides leaves $a^2 + b^2 = c^2$. ■

9.4 Proof 3 — President Garfield's trapezoid

In 1876, five years before he became U.S. President, James Garfield found a proof using *half* of that picture: a trapezoid built from two copies of the triangle and one right isosceles triangle.

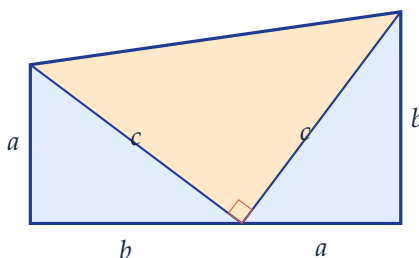


Figure 9.4 — Garfield's trapezoid. The middle triangle (peach) is right-angled and isosceles with legs c .

Proof. The trapezoid has parallel sides a and b and height $a + b$, so its area is $\frac{1}{2}(a + b)(a + b)$. It is also the sum of three triangles: two copies of our right triangle, each $\frac{1}{2}ab$, and the middle right-isosceles triangle, $\frac{1}{2}c^2$. Thus

$$\frac{1}{2}(a + b)^2 = 2 \cdot \frac{1}{2}ab + \frac{1}{2}c^2.$$

Multiply by 2: $(a + b)^2 = 2ab + c^2$, and again $a^2 + b^2 = c^2$. ■

9.5 Proof 4 — Bhāskara's dissection

The twelfth-century Indian mathematician Bhāskara gave a proof so vivid that, legend has it, he drew the figure, wrote a single word beneath it — “*Behold!*” — and considered the matter settled. Pack four copies of the right triangle into a square built on the *hypotenuse* c ; they leave a small square hole of side $b - a$.

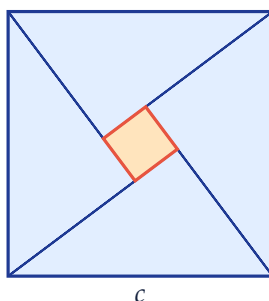


Figure 9.5 — Bhāskara’s “Behold!”: four triangles fill the square on the hypotenuse c , leaving a hole of side $b - a$.

Proof. The large square has area c^2 . It is equally the four triangles plus the small central square:

$$c^2 = 4 \cdot \frac{1}{2}ab + (b - a)^2 = 2ab + (a^2 - 2ab + b^2) = a^2 + b^2.$$

AHA!

Four proofs, one truth. Proof 1 leaned on *similarity*; Proofs 2–4 used *area*, three different ways — two dissections and a trapezoid. And these are a mere sample: over four hundred proofs are known, among them Euclid’s majestic “windmill” of Book I, which slices the hypotenuse-square directly into the two leg-squares. When you can prove a fact several ways, you do not just *know* it — you *own* it, and you carry several tools instead of one.

9.6 The converse, and a warning

Theorem 9.2 (Converse). If a triangle's sides satisfy $a^2 + b^2 = c^2$, then the angle opposite c is a right angle.

This is what made the Egyptian rope-stretchers' 3–4–5 trick work: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$, so the corner *must* be square. Whole-number solutions (a, b, c) are called **Pythagorean triples**; $(3, 4, 5)$, $(5, 12, 13)$, and $(8, 15, 17)$ are the famous small ones.

X COMMON TRAP

Pythagoras holds *only* for right triangles. Before writing $a^2 + b^2 = c^2$, be sure there is a right angle, and that c is the *hypotenuse* (the side opposite it — always the longest). For a general triangle you need the Law of Cosines (Chapter 18), of which Pythagoras is the special right-angled case.

 **Worked Example.** A right triangle has legs 9 and 12. Find the hypotenuse — and then confirm, by the converse, that $(9, 12, 15)$ really is a right triangle.

By Pythagoras, $c^2 = 9^2 + 12^2 = 81 + 144 = 225$, so $c = \sqrt{225} = 15$. Running it the other way (the converse): since $9^2 + 12^2 = 15^2$, the angle opposite the side 15 *must* be a right angle. (Note $(9, 12, 15) = 3 \times (3, 4, 5)$ — every multiple of a Pythagorean triple is again one.) ■

 **Worked Example.** Find the distance between the points $(1, 2)$ and $(4, 6)$ in the plane.

Build the right triangle whose legs are the horizontal and vertical gaps: $\Delta x = 4 - 1 = 3$ and $\Delta y = 6 - 2 = 4$. The straight-line distance is the hypotenuse, so

$$d = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

The *distance formula* you will meet in Chapter 21 is nothing more than Pythagoras wearing coordinates. ■

Four hundred proofs and counting, for a single line of algebra about a corner. Some truths run so deep that every generation wants to discover them again in its own hand.

→ YOUR TURN

Is the triangle with sides 8, 15, 17 right-angled? If so, which side is the hypotenuse?

Set-up (you finish it): by the converse, the triangle is right exactly when the two smaller squares add to the largest. Check whether $8^2 + 15^2$ equals 17^2 .

Answer: $8^2 + 15^2 = 64 + 225 = 289 = 17^2$, so yes — it is right-angled, with the hypotenuse the longest side, 17.

KEY IDEAS & FORMULAS

- **Pythagoras:** in a right triangle $a^2 + b^2 = c^2$, with c the *hypotenuse* (opposite the right angle, the longest side). Holds *only* for right triangles.
- **Converse:** if $a^2 + b^2 = c^2$, the angle opposite c is right — the standard test for a right angle from side lengths.
- **Triples:** $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$ and all their multiples; the family $(2mn, m^2 - n^2, m^2 + n^2)$ generates them.
- **Distance formula** (Chapter 21) is Pythagoras in coordinates: $d = \sqrt{(\Delta x)^2 + (\Delta y)^2}$.
- **Classify** a triangle with $a \leq b \leq c$: acute / right / obtuse as $a^2 + b^2$ is $>$, $=$, or $<$ c^2 .

Exercises

Warm-up.

1. A right triangle has legs 6 and 8. Find the hypotenuse.
2. A right triangle has hypotenuse 13 and one leg 5. Find the other leg.
3. Is a triangle with sides 9, 12, 15 right-angled? Which is the hypotenuse?
4. Find the length of the diagonal of a rectangle 5 by 12.
5. Is $(7, 24, 25)$ a Pythagorean triple? Check.
6. A square has side 1. How long is its diagonal? Leave the answer exact.

Practice.

7. Find the distance between the points $(1, 2)$ and $(7, 10)$ in the plane.
8. An equilateral triangle has side 6. Find its height exactly, then its area.
9. A ladder 10 m long leans against a wall with its foot 6 m out. How high up the wall does it reach?
10. The diagonal of a square is 8. Find the side and the area.
11. In right $\triangle ABC$ (right angle C), the altitude to the hypotenuse is CH . If $AH = 9$ and $HB = 16$, find CH , AC , and BC .
12. Two legs of a right triangle are x and $x + 7$, hypotenuse 13. Find x .
13. A rectangular box is $3 \times 4 \times 12$. Find the length of its space diagonal. (*Hint: Pythagoras twice.*)
14. Show that $(2mn, m^2 - n^2, m^2 + n^2)$ is a Pythagorean triple for any integers $m > n > 0$, and use $m = 2, n = 1$ and $m = 3, n = 2$ to generate two triples.

Challenge.

15. Point P is inside rectangle $ABCD$. Prove the “British Flag” theorem: $PA^2 + PC^2 = PB^2 + PD^2$.
16. In $\triangle ABC$ with right angle at C , prove that the radius of the inscribed circle is $r = \frac{1}{2}(a + b - c)$.
17. A right triangle has integer sides and its perimeter equals its area (numerically). Find all such triangles.
18. Squares are drawn outward on the two legs of a right triangle. Prove that the line joining their centres passes through the right-angle vertex. (Set up coordinates with the right angle at the origin.)
19. Prove that for any triangle with sides $a \leq b \leq c$: the triangle is acute, right, or obtuse according as $a^2 + b^2$ is greater than, equal to, or less than c^2 .

★ OLYMPIAD SPARK

(Olympiad.) Let ABC be a right triangle with the right angle at C . Squares $BCDE$ and $ACFG$ are erected outward on the two legs. Prove that the segments AD and BF are equal in length and meet at right angles. (*Coordinates make it routine; a rotation makes it beautiful.*)

(Second spark.) A point P lies inside equilateral $\triangle ABC$ with $PA = 3$, $PB = 4$, $PC = 5$. Find the side length of the triangle. (*Rotate the triangle 60° about one vertex; a 3–4–5 right angle appears as if by magic.*)

10

The Four Centres

★ WONDER

Cut any triangle out of stiff card and try to balance it on the tip of a pencil. There is exactly *one* point where it balances perfectly. Now ask a different question: where is the one point equally far from all three *corners*? And the one equally far from all three *edges*? Astonishingly, each of these questions has a clean answer — and the points are, in general, all different. A single triangle secretly carries a whole constellation of special centres.

Every triangle hides four classical centres, each born from a different family of three lines that — against all odds — always meet at a single point. The fact that three lines pass through one point is called **concurrency**, and it is never an accident; each is a small miracle we can prove.

10.1 The centroid: where medians meet

A **median** joins a vertex to the midpoint of the opposite side. A triangle has three.

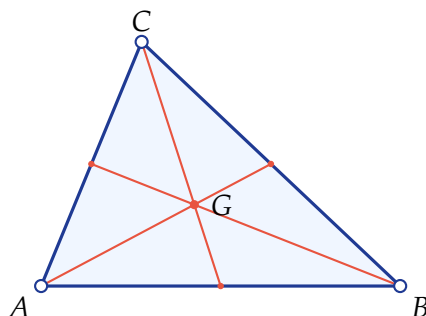


Figure 10.1 — The three medians meet at the centroid G , the triangle's balance point.

 **Worked Example.** Prove that two medians of a triangle cut each other in the ratio $2 : 1$ — and deduce that all three medians pass through one point.

Let E, F be the midpoints of AC, AB , and let medians BE and CF meet at G . Join E and F . By the Midsegment Theorem (Corollary 8.4), $EF \parallel BC$ and $EF = \frac{1}{2}BC$.

Now look at triangles GEF and GBC . Because $EF \parallel BC$, the alternate angles give $\angle GEF = \angle GBC$ and $\angle GFE = \angle GCB$; so $\triangle GEF \sim \triangle GBC$ by **AA**. Their ratio is $EF : BC = 1 : 2$. Therefore

$$\frac{BG}{GE} = \frac{CG}{GF} = \frac{BC}{EF} = 2.$$

So G divides each of BE and CF in the ratio $2 : 1$ from the vertex.

The wow: we never assumed the third median. Take *any* two medians: they meet at the point $2/3$ of the way down each. So the median from A must also pass through that same $2/3$ point — there is nowhere else for it to go. The three medians *cannot help* but be concurrent. One clean similar-triangle argument, and a “coincidence” becomes inevitable.

Definition 10.1 (Centroid). The three medians of a triangle meet at the **centroid** G , which divides each median $2 : 1$ from the vertex. It is the triangle's centre of mass — the balance point of the Wonder.

10.2 The circumcentre and incentre

Two more centres come from the idea of *equidistance*, and each carries its own circle.

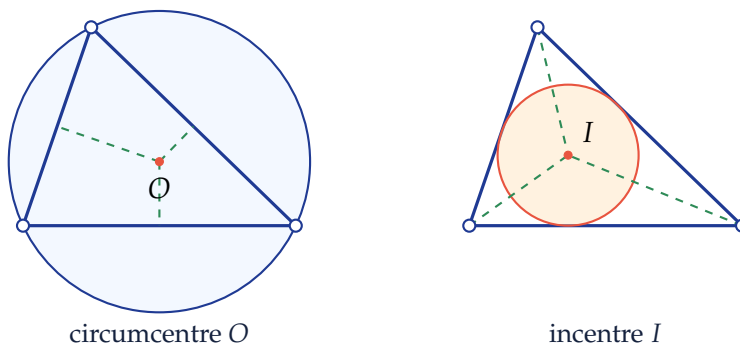


Figure 10.2 — Left: perpendicular bisectors meet at the circumcentre O , equidistant from the vertices (circumscribed circle). Right: angle bisectors meet at the incentre I , equidistant from the sides (inscribed circle).

Circumcentre. A point on the perpendicular bisector of AB is equidistant from A and B (Exercise 5.12). Where the perpendicular bisectors of AB and BC cross, the point is equidistant from A, B and from B, C — hence from all three. So it also lies on the third bisector: the three are concurrent, at the **circumcentre** O . Being equidistant from all vertices, O is the centre of the unique circle through A, B, C , the *circumscribed circle*.

Incentre. The same argument with *angle* bisectors: a point on the bisector of $\angle A$ is equidistant from sides AB and AC . The intersection of two angle bisectors is equidistant from all three sides, so it lies on the third bisector too. This is the **incentre** I , centre of the *inscribed circle* that touches all three sides.

Worked Example. Prove the strikingly simple formula $\angle BIC = 90^\circ + \frac{1}{2}\angle A$, where I is the incentre.

Since I lies on the bisectors of $\angle B$ and $\angle C$, we have $\angle IBC = \frac{1}{2}\angle B$ and $\angle ICB = \frac{1}{2}\angle C$. In triangle BIC the angles sum to 180° :

$$\angle BIC = 180^\circ - \frac{1}{2}\angle B - \frac{1}{2}\angle C = 180^\circ - \frac{1}{2}(\angle B + \angle C).$$

But $\angle B + \angle C = 180^\circ - \angle A$, so

$$\angle BIC = 180^\circ - \frac{1}{2}(180^\circ - \angle A) = 90^\circ + \frac{1}{2}\angle A.$$

The angle at the incentre depends on $\angle A$ *alone* — the other two angles cancel out completely. Slide B and C around however you like; as long

as $\angle A$ holds steady, $\angle BIC$ never moves. A messy-looking quantity turns out to be governed by a single number.

10.3 The orthocentre, and a vector miracle

The three **altitudes** (perpendiculars from each vertex to the opposite side) also concur, at the **orthocentre** H .

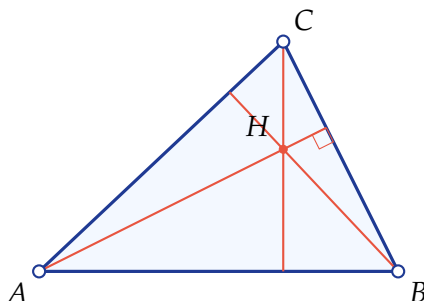


Figure 10.3 — The three altitudes meet at the orthocentre H .

Why do the altitudes concur? The cleanest proof is a piece of pure magic with vectors — and it hands us a second, far deeper, theorem for free.

Worked Example. Place the circumcentre O at the origin, so the position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ of A, B, C all have the same length R (the circumradius). Show that the point H with position vector $\mathbf{a} + \mathbf{b} + \mathbf{c}$ is the orthocentre. Consider the segment from A to H : its direction is $\mathbf{h} - \mathbf{a} = \mathbf{b} + \mathbf{c}$. The opposite side BC has direction $\mathbf{c} - \mathbf{b}$. Take their dot product:

$$(\mathbf{b} + \mathbf{c}) \cdot (\mathbf{c} - \mathbf{b}) = \mathbf{c} \cdot \mathbf{c} - \mathbf{b} \cdot \mathbf{b} = |\mathbf{c}|^2 - |\mathbf{b}|^2 = R^2 - R^2 = 0.$$

So $AH \perp BC$: the point H lies on the altitude from A . By identical reasoning it lies on the altitudes from B and C . Hence all three altitudes pass through $H = \mathbf{a} + \mathbf{b} + \mathbf{c}$ — they are concurrent.

That little computation does more than prove concurrency. Recall the centroid is $\mathbf{g} = \frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c})$. Comparing,

$$\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c} = 3\mathbf{g}, \quad \text{while } O \text{ is the origin.}$$

So O, G, H have position vectors $\mathbf{0}, \frac{1}{3}\mathbf{h}, \mathbf{h}$ — three points on a single straight line through the origin, with $OG : GH = 1 : 2$.

AHA!

This line is the celebrated **Euler line**. In *every* triangle, the circumcentre, centroid, and orthocentre are collinear, with the centroid sitting one-third of the way from O to H . Leonhard Euler discovered this in 1765, and it still feels like a conjuring trick: three centres defined in three completely different ways, yet they always line up. (The only exception is the equilateral triangle, where all three coincide.)

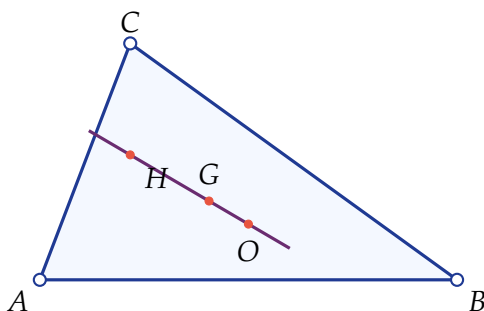


Figure 10.4 — The Euler line (plum) carries O, G , and H , with $OG : GH = 1 : 2$.

Worked Example. For the triangle $A(0,0), B(8,0), C(0,6)$, find the centroid and the circumcentre.

The centroid is the average of the vertices: $G = \left(\frac{0+8+0}{3}, \frac{0+0+6}{3}\right) = \left(\frac{8}{3}, 2\right)$. For the circumcentre, notice the right angle sits at A (the legs lie along the axes), so by the right-triangle fact the circumcentre is the *midpoint of the hypotenuse* BC : $O = \left(\frac{8+0}{2}, \frac{0+6}{2}\right) = (4, 3)$. The circumradius is then $OB = \frac{1}{2}BC = \frac{1}{2}\sqrt{8^2 + 6^2} = \frac{1}{2} \cdot 10 = 5$. Two centres, read straight off the coordinates — the kind of computation Chapter 21 turns into a reflex. ■

Four centres, each the “middle” of a triangle by its own rule, and three of them threaded forever onto a single line. A figure this familiar has no right to hide so much.

→ YOUR TURN

In a triangle, the incentre I satisfies $\angle BIC = 120^\circ$. Find $\angle A$.

Set-up (you finish it): use the incentre formula $\angle BIC = 90^\circ + \frac{1}{2}\angle A$. Set it equal to 120° and solve for $\angle A$.

Answer: $120^\circ = 90^\circ + \frac{1}{2}\angle A \Rightarrow \frac{1}{2}\angle A = 30^\circ \Rightarrow \angle A = 60^\circ$.

KEY IDEAS & FORMULAS

- **Centroid** G (medians): divides each median $2 : 1$ from the vertex; $G = \frac{1}{3}(A + B + C)$ — the average of the vertices.
- **Circumcentre** O (perpendicular bisectors): equidistant from the *vertices*; centre of the circumscribed circle. In a right triangle it is the midpoint of the hypotenuse.
- **Incentre** I (angle bisectors): equidistant from the *sides*; centre of the inscribed circle. $\angle BIC = 90^\circ + \frac{1}{2}\angle A$.
- **Orthocentre** H (altitudes): with O at the origin, $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$.
- **Euler line:** O, G, H collinear with $OG : GH = 1 : 2$ (and the nine-point centre N gives $OG : GN : NH = 2 : 1 : 3$); all coincide only for the equilateral triangle.

Exercises

Warm-up.

1. Define median, altitude, perpendicular bisector, and angle bisector — one sentence each.
2. Which centre is the balance point? Which is equidistant from the vertices? From the sides?
3. The centroid divides a median in what ratio, and from which end?
4. A median has length 12. How far is the centroid from the vertex? From the midpoint?
5. In a triangle with $\angle A = 50^\circ$, find $\angle BIC$.
6. For which triangle do all four centres coincide?

Practice.

7. Triangle with vertices $A(0,0)$, $B(6,0)$, $C(0,9)$. Find the centroid.
8. For the same triangle, find the circumcentre. (*Hint: it is the midpoint of the*

hypotenuse — why?)

9. If $\angle BIC = 130^\circ$, find $\angle A$.
10. In a right triangle, where is the orthocentre? Where is the circumcentre?
11. The medians of a triangle have a common point dividing them 2 : 1. If one median is 9 cm and a vertex is at distance d from G along it, find d .
12. Prove that the circumcentre of a right triangle is the midpoint of the hypotenuse.
13. Vertices $A(0,0)$, $B(4,0)$, $C(0,3)$. Find the orthocentre directly, then verify O , G , H are collinear.

Challenge.

14. Prove that the incentre, the vertex A , and the midpoint of arc BC (of the circumcircle, not containing A) are collinear. (You will appreciate this more after Chapter 13.)
15. Using $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ with O at the origin, find the position vector of the nine-point centre N (the midpoint of OH), and verify that O, G, N, H are collinear with $OG : GN : NH = 2 : 1 : 3$.
16. Prove $\angle BHC = 180^\circ - \angle A$ for the orthocentre H . (Compare with the incentre formula — the contrast is lovely.)
17. Show that reflecting the orthocentre H over a side of the triangle lands on the circumcircle.
18. The three medians cut the triangle into six small triangles. Prove they all have equal area.

★ OLYMPIAD SPARK

(Olympiad.) Let H be the orthocentre of acute triangle ABC . Prove that the reflections of H across the three *midpoints* of the sides all lie on the circumcircle, and identify the diametrically opposite points. (*Vectors with O at the origin make this a two-line miracle.*)

(Second spark.) In triangle ABC , the incentre is I and the incircle touches BC at X . Prove that $BX = s - b$, where $s = \frac{1}{2}(a + b + c)$ is the semiperimeter. (*The two tangent lengths from each vertex are equal — name them and solve the little system.*)

11

Area as a Tool

★ WONDER

A farmer marks out a field by hammering pegs at points of a square grid and running rope between them. No edge is straight-on-the-axis, no corner is square. Yet you can tell her the field's area *exactly* — not by measuring lengths, but simply by *counting pegs*: the ones on the fence, and the ones inside. No ruler ever touches the grass, yet the scattered dots alone already know the answer.

Most students meet area as a set of formulas to memorise. But area is far more than a number at the end of a problem — it is a *tool for proof*. In this chapter area becomes a way of *reasoning*: comparing, slicing, and counting our way to results that would be painful to reach any other way.

11.1 The one formula, and its hidden flexibility

The area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$. The subtle part is that *any* side may serve as the base, paired with the perpendicular height to it. This freedom is the source of all the power.

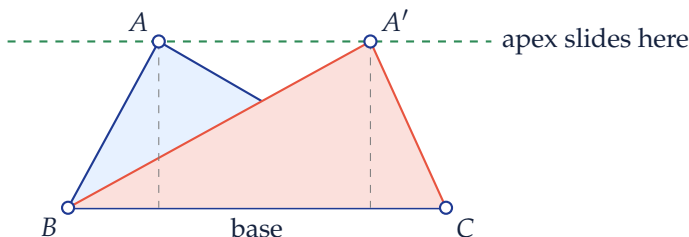


Figure 11.1 — Same base BC , same height: as the apex slides along the dashed parallel, the area never changes, though the triangle is reshaped.

 **Worked Example.** Prove that a cevian from A meeting BC at D splits the triangle into two pieces whose areas are in the ratio $BD : DC$.

Triangles ABD and ACD share the *same height* — the perpendicular distance from A to line BC . With equal heights, area is proportional to base:

$$\frac{[ABD]}{[ACD]} = \frac{\frac{1}{2} \cdot BD \cdot h}{\frac{1}{2} \cdot DC \cdot h} = \frac{BD}{DC}.$$

A ratio of *areas* (two-dimensional) has collapsed into a ratio of *lengths* (one-dimensional), just because the height cancels. Setting D to the midpoint gives equal areas at once: **a median always halves the area**. (Here $[XYZ]$ denotes the area of triangle XYZ .)

 **Worked Example.** Prove that the three medians cut a triangle into six smaller triangles of *equal* area.

Let the medians AD , BE , CF meet at the centroid G , with D, E, F the midpoints of BC , CA , AB . *Step 1:* since F is the midpoint of AB , triangles AGF and BGF have equal bases ($AF = FB$) and share apex G , so $[AGF] = [BGF]$; the same holds at D and at E . *Step 2:* the median CF gives $[ACF] = [BCF]$ (equal bases $AF = FB$, apex C); subtracting the equal pieces $[AGF] = [BGF]$ leaves $[AGC] = [BGC]$, and the median AD likewise gives $[AGB] = [AGC]$. So the three “corner” triangles are equal: $[AGB] = [BGC] = [CGA] = \frac{1}{3}[ABC]$. *Step 3:* each corner triangle is halved by a median into two of the little triangles (e.g. GF halves $[AGB]$). Hence all six little triangles have area $\frac{1}{6}[ABC]$.

The surprise: the balance point doesn’t just balance *mass* — it balances *area* into six exactly equal slivers. Symmetry you could never have guessed from the lopsided picture.

11.2 Area by coordinates: the shoelace formula

When points are given by coordinates, area needs no heights at all.

Theorem 11.1 (Shoelace formula). A triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ has area

$$[T] = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

 **Worked Example.** Find the area of the triangle with vertices $(1, 1), (5, 2), (3, 6)$. Then test whether $(0, 0), (2, 3), (4, 6)$ form a triangle at all.

First: $\frac{1}{2} |1(2 - 6) + 5(6 - 1) + 3(1 - 2)| = \frac{1}{2} |-4 + 25 - 3| = \frac{1}{2}(18) = 9$.

Second: $\frac{1}{2} |0(3 - 6) + 2(6 - 0) + 4(0 - 3)| = \frac{1}{2} |0 + 12 - 12| = 0$.

The wow: an area of zero means the “triangle” is squashed flat — the three points are **collinear**. So the shoelace formula is secretly a *collinearity detector*: three points lie on a line exactly when this expression vanishes. One formula quietly answers two different questions.

11.3 Counting your way to area: Pick’s Theorem

Now the Wonder’s promise. Draw any polygon whose corners sit on grid points (a *lattice polygon*). Let I be the number of grid points strictly *inside* it and B the number *on its boundary*. Then its area is given by a formula of breathtaking simplicity.

Theorem 11.2 (Pick, 1899). For a lattice polygon, $\text{Area} = I + \frac{B}{2} - 1$.

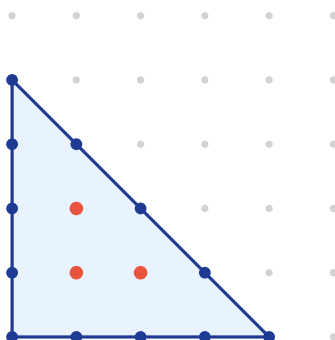


Figure 11.2 — A lattice triangle. Boundary points $B = 12$ (blue), interior points $I = 3$ (coral).

 **Worked Example.** Find the area of the triangle in Figure 11.2 by counting, and check it.

Count the dots: interior $I = 3$ (coral), boundary $B = 12$ (blue). Pick's theorem gives

$$\text{Area} = I + \frac{B}{2} - 1 = 3 + \frac{12}{2} - 1 = 3 + 6 - 1 = 8.$$

Check with the ordinary formula: the triangle has legs 4 and 4, so area $= \frac{1}{2} \cdot 4 \cdot 4 = 8$. ✓

We computed a continuous quantity — area — with nothing but whole-number *counting*. No lengths, no heights, no multiplication of measured sides. Pick's theorem builds a bridge between the discrete world of dots and the continuous world of area, and it works for *any* lattice polygon, however jagged.

MATHEMATICIAN'S MARGIN

Pick's theorem feels like a magic trick, but it rests on a sturdy idea: area is *additive*. Glue two lattice polygons along an edge and both the area and the quantity $I + \frac{B}{2} - 1$ simply add. Since the formula holds for the basic building block (a single grid square, and a single lattice triangle), additivity carries it to everything built from them.

Area began as a number to memorise at the end of a problem. You leave this chapter holding it as an instrument: for slicing, comparing, and even counting your way to a proof.

→ YOUR TURN

A lattice triangle has 6 grid points strictly inside it and 10 on its boundary. Find its area.

Set-up (you finish it): apply Pick's theorem, $\text{Area} = I + \frac{B}{2} - 1$, with $I = 6$ and $B = 10$.

Answer: $\text{Area} = 6 + \frac{10}{2} - 1 = 6 + 5 - 1 = 10$.

KEY IDEAS & FORMULAS

- **Triangle area** $= \frac{1}{2} \times \text{base} \times \text{height}$, with *any* side allowed as the base.
- **Same base, same height** (apex on a parallel to the base) $\Rightarrow$ **same area**; a **median** halves the area, and the three medians cut the triangle into *six* equal pieces.
- **Shoelace**: $[T] = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$; the value is 0 exactly when the three points are collinear.
- **Pick**: for a lattice polygon, $\text{Area} = I + \frac{B}{2} - 1$ (I interior, B boundary grid points).
- **Area as a tool**: ratios of areas with a shared height become ratios of bases — the engine behind cevian and concurrency problems.

Exercises**Warm-up.**

1. A triangle has base 10 and height 6. Find its area.
2. Two triangles share the same base and lie between the same parallels. How do their areas compare?
3. Use the shoelace formula on $(0, 0), (6, 0), (0, 5)$.
4. A lattice polygon has $I = 10$ interior and $B = 8$ boundary points. Find its area.
5. A median divides a triangle of area 30 into two pieces. Find each piece's area.
6. Are $(1, 1), (3, 5), (2, 3)$ collinear? Use the shoelace test.

Practice.

7. D lies on BC with $BD : DC = 3 : 2$ and $[ABC] = 40$. Find $[ABD]$ and $[ACD]$.
8. Find the area of the quadrilateral $(0, 0), (4, 0), (5, 3), (1, 4)$ by the shoelace formula.
9. In $\triangle ABC$, D is the midpoint of BC and E the midpoint of AD . Find $[BEC] : [ABC]$.
10. A triangle has vertices on lattice points with $B = 14$ and area 20. How many interior lattice points does it have?
11. The medial triangle is formed by joining the midpoints of the sides. Prove its area is $\frac{1}{4}$ of the original.

12. Points P, Q divide side BC into three equal parts. Compare $[ABP]$, $[APQ]$, $[AQC]$.
13. A cevian AD and a cevian BE meet at P . If $[APE] = [BPD]$, what can you say about DE and AB ?

Challenge.

14. (Area ratios as a machine.) In $\triangle ABC$, D on BC with $BD : DC = 1 : 2$ and E on CA with $CE : EA = 1 : 2$. Lines AD and BE meet at P . Find $[\triangle PAB] : [\triangle ABC]$ using only area ratios.
15. Prove that a quadrilateral's two diagonals split it into four triangles whose areas satisfy $[PAB] \cdot [PCD] = [PBC] \cdot [PDA]$, where P is the diagonal intersection.
16. Use Pick's theorem to prove that no equilateral triangle has all three vertices on lattice points. (*Hint: its area would be irrational, but Pick forces a rational area.*)
17. A convex lattice polygon has no interior points ($I = 0$). What are the possible areas, and what shapes achieve them?
18. Prove the shoelace formula for a triangle by enclosing it in an axis-aligned rectangle and subtracting the three corner right triangles.

★ OLYMPIAD SPARK

(Olympiad — the one-seventh triangle.) In $\triangle ABC$, cevians AD , BE , CF are drawn so that D, E, F divide the sides each in ratio $1 : 2$ (with $BD : DC = CE : EA = AF : FB = 1 : 2$). The three cevians bound a small central triangle. Prove that its area is exactly $\frac{1}{7}$ of $[ABC]$. (*Pure area-ratio reasoning cracks it — a famous and delightful result.*)

(Second spark.) Prove that among all triangles inscribed in a given circle, the equilateral one has the greatest area. (*Fix the base; the “sliding apex” idea of Figure 11.1 shows where the area is maximised, then iterate.*)

PART III

The Circle

12

Chords, Arcs, and the Circle's Anatomy

★ WONDER

Why are manhole covers round? Try a square lid: turn it diagonally and it falls straight through the hole. A round lid *cannot* fall in, no matter how you turn it. The reason is a single property the circle alone possesses so perfectly — every point of its rim is exactly the same distance from one central point. That one fact, pursued carefully, blossoms into the richest chapter of elementary geometry.

We now turn from straight lines to the curve. The **circle** is the set of all points at a fixed distance (the *radius*) from a fixed point (the *centre*). Everything else — chords, arcs, tangents, and the beautiful angle theorems of the next chapters — flows from that definition. First, the vocabulary.

12.1 The parts of a circle

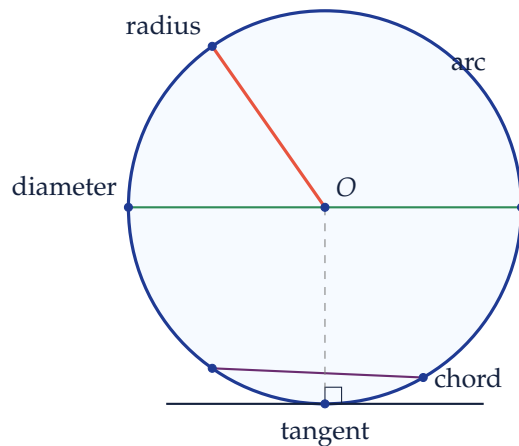


Figure 12.1 — The anatomy of a circle. A *chord* joins two points of the circle; a *diameter* is a chord through the centre; a *tangent* touches at exactly one point, meeting the radius there at a right angle.

The key terms: a **chord** joins two points on the circle; a **diameter** is a chord through the centre (the longest chord, twice the radius); an **arc** is a piece of the circle; a **tangent** line touches the circle at exactly one point; a **secant** cuts it at two. We record one fact about tangents now and prove its companions later: *a tangent is perpendicular to the radius drawn to the point of contact*.

12.2 The centre, the chord, and the right angle

The single most useful fact about chords links them to the centre.

Theorem 12.1 (Perpendicular from the centre). The perpendicular from the centre of a circle to a chord bisects the chord. Conversely, the line from the centre to the midpoint of a chord is perpendicular to it.

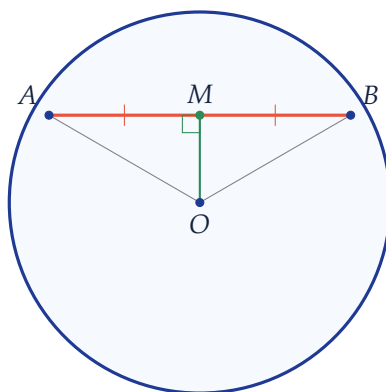


Figure 12.2 — $OM \perp AB$ forces $AM = MB$. The radius, the half-chord, and the centre-distance make a right triangle.

 **Worked Example.** Prove the theorem.

Let M be the foot of the perpendicular from O to chord AB . Compare right triangles OMA and OMB : $OA = OB$ (both radii), $OM = OM$ (shared), and both have a right angle at M . By **RHS**, $\triangle OMA \cong \triangle OMB$, so $AM = MB$. The chord is bisected.

This hands you a right triangle whenever you see a chord — legs (half-chord) and (distance from centre), hypotenuse (radius). Pythagoras then ties the three together as $r^2 = (\text{half-chord})^2 + (\text{distance from centre})^2$. A chord problem is a Pythagoras problem in disguise.

 **Worked Example.** A chord of length 24 lies in a circle of radius 13. How far is it from the centre?

The perpendicular from the centre bisects the chord, giving a half-chord of 12. With radius 13 as hypotenuse,

$$d = \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5.$$

The chord lies 5 units from the centre. (Notice the 5–12–13 triple quietly at work.)

 **Worked Example. Reconstructing a broken plate.** An archaeologist finds a curved shard. Across the broken rim she measures a straight chord

of length $2a$, and at its midpoint the rim bulges out a height s (the *sagitta*) beyond the chord. Show that the whole plate's radius was

$$R = \frac{a^2 + s^2}{2s}.$$

Drop the perpendicular from the centre O to the chord; it passes through the chord's midpoint M and continues to the nearest rim point N , so O, M, N are collinear along this line, with $MN = s$ and $ON = R$. Hence the centre is at distance $OM = R - s$ from the chord. The half-chord a , the distance $R - s$, and the radius R form a right triangle:

$$a^2 + (R - s)^2 = R^2 \Rightarrow a^2 + R^2 - 2Rs + s^2 = R^2 \Rightarrow a^2 + s^2 = 2Rs,$$

$$\text{so } R = \frac{a^2 + s^2}{2s}.$$

The wow: from a chip of pottery no bigger than your palm, you recover the size of the *entire* original plate — the circle “remembers” its radius in any tiny arc. This same formula lets opticians, engineers, and astronomers measure the curvature of lenses, rails, and mirrors from a short span.

12.3 Equal chords, equal tangents

Two quick companions, each proved by the same right-triangle idea.

Proposition 12.2. In a circle, equal chords are equidistant from the centre, and conversely. Also, the two tangent segments drawn to a circle from an external point are equal in length.

The tangent fact is worth a picture, because it appears constantly.

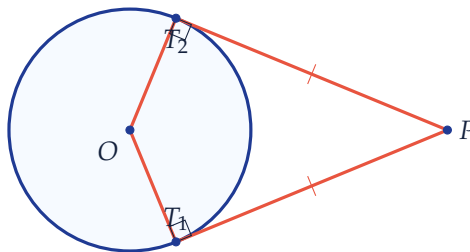


Figure 12.3 — Two tangents from an external point P are equal: $PT_1 = PT_2$ (matching ticks). Each touches at a right angle to the radius.

 **Worked Example.** Prove the two tangents from P are equal.

Join OT_1 , OT_2 (radii to the contact points), and OP . Each radius meets its tangent at a right angle, so $\triangle OT_1P$ and $\triangle OT_2P$ are right-angled at T_1, T_2 . They share hypotenuse OP , and $OT_1 = OT_2$ (radii). By **RHS**, the triangles are congruent, so $PT_1 = PT_2$.

The contact points may sit at wildly different spots, yet the two “reach” distances are always identical — which is exactly why a ball bearing seats symmetrically in a V-groove, and why this equality is the secret engine behind the inscribed-circle tangent lengths you met in Chapter 10.

The circle rests on a single idea, one point and one fixed distance, yet already it is paying out right triangles, equal reaches, and a radius remembered in a shard. We have barely scratched its rim.

→ YOUR TURN

A chord of length 24 lies in a circle of radius 15. How far is the chord from the centre?

Set-up (you finish it): the perpendicular from the centre bisects the chord, giving a half-chord of 12. With the radius 15 as hypotenuse, use $r^2 = (\text{half-chord})^2 + d^2$ to solve for the distance d .

Answer: $d = \sqrt{15^2 - 12^2} = \sqrt{225 - 144} = \sqrt{81} = 9$.

KEY IDEAS & FORMULAS

- **Circle:** all points at a fixed distance r (the radius) from a centre. **Chord** joins two points; **diameter** is a chord through the centre (longest, $2r$); **tangent** touches once; **secant** cuts twice.
- **Perpendicular from the centre bisects a chord** (and conversely): this makes a right triangle with $r^2 = (\text{half-chord})^2 + (\text{distance from centre})^2$ — every chord problem is a Pythagoras problem.
- **Equal chords are equidistant from the centre** (and conversely).
- **Tangent $\perp$ radius** at the point of contact; the **two tangents from an external point are equal** ($PT^2 = OP^2 - r^2$).

- **Three non-collinear points determine exactly one circle** (the perpendicular bisectors of two chords meet at its centre).

Exercises

Warm-up.

1. Name: a chord through the centre; a line touching at one point; a line cutting at two points.
2. A circle has radius 10. How long is its longest chord?
3. A chord of length 16 sits 6 from the centre. Find the radius.
4. Two tangents are drawn from a point 8 cm from the centre of a circle of radius 5 cm. Find each tangent's length.
5. True/false: the perpendicular bisector of any chord passes through the centre.
6. A chord equals the radius in length. Is it longer or shorter than half the diameter?

Practice.

7. A chord of length 30 lies in a circle of radius 17. Find its distance from the centre.
8. Two parallel chords of a circle of radius 13 have lengths 24 and 10, on the same side of the centre. Find the distance between them.
9. A broken arc has a chord of 40 cm and a sagitta of 8 cm. Find the original radius.
10. From an external point P , a tangent of length 12 touches a circle of radius 5. Find OP .
11. Two chords of equal length are drawn in a circle. Prove they are equidistant from the centre.
12. A point is 25 from the centre of a circle of radius 7. Find the length of the tangent from it.
13. Prove that the line joining the centres of two intersecting circles is the perpendicular bisector of their common chord.

Challenge.

14. Three points, not collinear, are given. Prove exactly one circle passes through all three, and describe how to find its centre.

15. A circle of radius R has two perpendicular chords intersecting at distances p and q from the centre along the respective perpendiculars. Relate p, q, R to the four segment lengths.
16. A quadrilateral has an inscribed circle (one tangent to all four sides). Prove that the sums of opposite sides are equal: $AB + CD = BC + DA$. (*Use equal tangents.*)
17. Two circles touch externally. Prove that the length of their common external tangent is $2\sqrt{r_1 r_2}$.
18. A chord is drawn at random distance from the centre. Show that the locus of chord midpoints, as the chord direction varies but its length stays fixed, is a circle concentric with the original.

★ OLYMPIAD SPARK

(Olympiad.) Two circles intersect at A and B . A line through A meets the circles again at P and Q . Prove that the ratio $BP : BQ$ does *not* depend on the choice of line through A . (*Equal arcs subtend equal angles — a perfect lead-in to the next chapter.*)

(Second spark.) A belt runs tautly around two circular pulleys of radii R and r whose centres are a distance d apart (no crossing). Find the total length of the belt. (*Tangent segments plus two arcs; the equal-tangent idea sets it up.*)

13

The Inscribed Angle Theorem

★ WONDER

A striker runs straight down the wing toward the goal line. The goal is a fixed width, off to the side. As she runs, the goalmouth seems to “open up” and then “close” again — there is some best spot where the goal looks widest. Meanwhile a teammate notices something stranger: from every point along a certain curved path, the goal appears at *exactly the same width*. What is that magic curve, and why does the angle refuse to change along it?

The answer is one of the most beautiful and useful theorems in all of geometry — the **Inscribed Angle Theorem**. It connects an angle at the *centre* of a circle to an angle at its *rim*, and from that single link cascades a whole toolbox: right angles in semicircles, equal angles on equal arcs, and the secret life of cyclic quadrilaterals. This theorem, more than any other, is the engine of olympiad angle-chasing.

13.1 Central versus inscribed

Fix a chord AB . The angle it makes at the centre, $\angle AOB$, is the **central angle**. The angle it makes at a point P on the circle, $\angle APB$, is an **inscribed angle**. The theorem says one is always exactly twice the other.

Theorem 13.1 (Inscribed Angle Theorem). The inscribed angle is half the central angle subtending the same arc:

$$\angle APB = \frac{1}{2} \angle AOB.$$

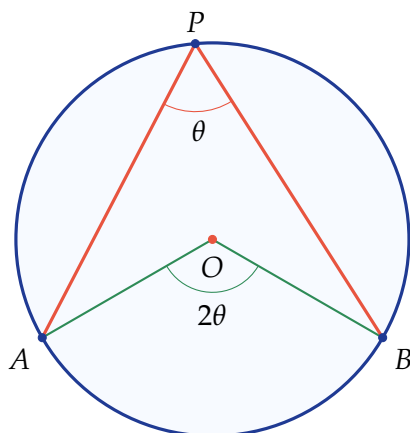


Figure 13.1 — The inscribed angle θ at P is half the central angle 2θ at O , both standing on chord AB .

 **Worked Example.** Prove the theorem (the case where O lies inside $\angle APB$).

Draw the diameter from P through O , reaching the circle again at D . This splits $\angle APB$ into $\angle APD$ and $\angle DPB$. Now $OP = OA$ (radii), so $\triangle OAP$ is isosceles and $\angle OAP = \angle OPA$. The exterior angle of this triangle at O equals the sum of the two equal base angles:

$$\angle AOD = \angle OAP + \angle OPA = 2\angle OPA = 2\angle APD.$$

The same argument on $\triangle OBP$ gives $\angle DOB = 2\angle DPB$. Adding,

$$\angle AOB = \angle AOD + \angle DOB = 2(\angle APD + \angle DPB) = 2\angle APB.$$

So $\angle APB = \frac{1}{2}\angle AOB$. (The cases where O lies on or outside the angle are handled the same way, adding or subtracting instead of just adding.)

The wow: the centre O doesn't appear in the conclusion at all. Move P anywhere on the major arc and the central angle $\angle AOB$ never changes — so the inscribed angle θ is *frozen*, no matter where P stands. That is the striker's magic curve: every point of the arc sees the chord AB at the identical angle.

13.2 Two corollaries that do half the work in geometry

Corollary 13.2 (Angles in the same segment). All inscribed angles standing on the same chord, from the same side, are equal — each being half the one fixed central angle.

Corollary 13.3 (Thales' Theorem). An angle inscribed in a semicircle is a right angle.

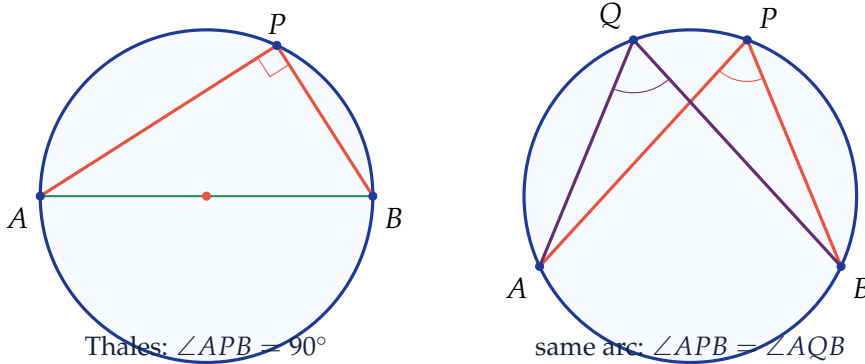


Figure 13.2 — Left: any angle in a semicircle is 90° . Right: P and Q on the same arc see chord AB at equal angles.

 **Worked Example.** Why is the semicircle angle a right angle?

When AB is a diameter, the central angle $\angle AOB$ is a straight angle, 180° . By the Inscribed Angle Theorem, the angle at P is half of that: $\angle APB = \frac{1}{2} \cdot 180^\circ = 90^\circ$.

This gives a stunning test for right angles. *A right angle is exactly an angle that fits inside a semicircle.* So the set of all points from which a fixed segment AB looks like a right angle is precisely the circle with diameter AB . Carpenters use this in reverse: to check a corner is square, see whether it lies on the circle through the two ends.

 **Worked Example.** Chord AB subtends an inscribed angle of 40° at a point P on the major arc. A point Q sits on the same arc; a point R on the *minor* arc. Find $\angle AQB$ and $\angle ARB$.

Q is on the same arc as P , so by “angles in the same segment,” $\angle AQB = 40^\circ$. For R on the opposite arc, $AQBR$ is a cyclic quadrilateral, so (next theorem) $\angle ARB = 180^\circ - 40^\circ = 140^\circ$.

You were told an angle at *one* point and instantly knew it at *every* point of the circle — a single 40° measurement propagates around the entire rim. Information in circle geometry travels astonishingly far.

13.3 Cyclic quadrilaterals: a first look

Four points all lying on one circle are called **concyclic**, and the quadrilateral they form is **cyclic**. The inscribed angle theorem gives them a signature property.

Theorem 13.4 (Opposite angles of a cyclic quadrilateral). If $ABCD$ is cyclic, then $\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$.

Proof. $\angle A = \angle DAB$ and $\angle C = \angle BCD$ are inscribed angles standing on the two arcs BD — one on each side. Together those two arcs make the whole circle, a central total of 360° , so the inscribed angles total half: $\angle A + \angle C = \frac{1}{2} \cdot 360^\circ = 180^\circ$. ■

AHA!

The converse is true too and enormously useful: if a quadrilateral's opposite angles sum to 180° , its four vertices *must* lie on a circle. This is a favourite olympiad weapon — to prove four points concyclic, hunt for two opposite angles that add to a straight angle. We devote all of Chapter 14 to these cyclic quadrilaterals.

→ YOUR TURN

AB is a diameter of a circle, and C is a point on the circle with $\angle CAB = 28^\circ$. Find $\angle ABC$.

Set-up (you finish it): since AB is a diameter, Thales gives $\angle ACB = 90^\circ$; the three angles of $\triangle ABC$ sum to 180° .

Answer: $\angle ABC = 180^\circ - 90^\circ - 28^\circ = 62^\circ$.

KEY IDEAS & FORMULAS

- **Inscribed Angle Theorem:** an inscribed angle is *half* the central angle on the same arc, $\angle APB = \frac{1}{2}\angle AOB$.
- **Same segment:** all inscribed angles on the same chord, from the same side, are equal — the chord looks the same width from every point of its arc.
- **Thales:** an angle in a semicircle is 90° ; conversely the right-angle vertices on a fixed segment AB trace the circle with diameter AB .
- **Cyclic quadrilateral:** opposite angles sum to 180° — and the *converse* (opposite angles supplementary $\Rightarrow$ concyclic) is the standard test for four points on a circle.
- **Equal arcs** subtend equal inscribed angles, and an angle bisector through the rim lands at an arc midpoint.

Exercises**Warm-up.**

1. A central angle is 84° . Find the inscribed angle on the same arc.
2. An inscribed angle is 33° . Find its central angle.
3. PQ is a diameter and R is on the circle. What is $\angle PRQ$?
4. In cyclic quadrilateral $ABCD$, $\angle A = 95^\circ$. Find $\angle C$.
5. Two points on the same arc subtend a chord. One angle is 52° . What is the other?
6. An inscribed angle stands on an arc of 100° (central). Find the angle.

Practice.

7. In a circle, $\angle AOB = 130^\circ$ (central). Find the inscribed angle on the major arc and on the minor arc.
8. AB is a diameter; C on the circle has $\angle ABC = 35^\circ$. Find $\angle BAC$ and $\angle ACB$.
9. In cyclic quadrilateral $ABCD$, $\angle B = 78^\circ$ and $\angle C = 95^\circ$. Find $\angle A$ and $\angle D$.
10. Two chords AB and CD meet inside a circle at X . Prove $\triangle AXC \sim \triangle DXB$. (Inscribed angles!)
11. AB is a diameter and C, D are on the same arc. If $\angle ACD = 64^\circ$, find $\angle ABD$.
12. A quadrilateral has $\angle A = 70^\circ$, $\angle B = 110^\circ$, $\angle C = 110^\circ$. Could it be cyclic? Find $\angle D$ if so.
13. Points A, B, C, D lie on a circle in order. If arc $AB = 80^\circ$ and arc $CD = 120^\circ$

(central measures), find the angle between chords AC and BD at their intersection.

Challenge.

14. Prove the converse of Thales: if $\angle APB = 90^\circ$, then P lies on the circle with diameter AB .
15. Prove that the locus of points P with $\angle APB = \alpha$ (a fixed angle), on one side of AB , is an arc of a circle through A and B .
16. In acute $\triangle ABC$, the feet of the altitudes from B and C are E and F . Prove that B, C, E, F are concyclic, and name the diameter of their circle.
17. $ABCD$ is cyclic. Prove that the exterior angle at one vertex equals the interior angle at the opposite vertex.
18. Two circles meet at A, B . A line through A meets them again at P, P' and a line through B meets them again at Q, Q' (with P, Q on one circle). Prove $PQ \parallel P'Q'$.

★ OLYMPIAD SPARK

(Olympiad.) Let AB be a chord of a circle and P a point on the major arc. The internal bisector of $\angle APB$ meets the circle again at M . Prove that M is the midpoint of arc AB , and hence that M is independent of the position of P . (*The bisected inscribed angles cut off equal arcs.*)

(Second spark — the best seat.) A statue of height h stands with its base a height p above your eye level. As you walk along the ground, your eye sees the statue (base to top) subtend an angle. Prove that this angle is largest when you stand where a circle through the statue's top and bottom is *tangent* to your eye-level line, and find that optimal horizontal distance. (*The "constant angle on an arc" idea pins down the maximum — the famous "museum problem."*)

14

Cyclic Quadrilaterals & Ptolemy

★ WONDER

Drive four nails into a board at random and stretch a loop of string around them: you get a quadrilateral. Now ask — could a single circular hoop be slipped over all four nails at once, touching every one? For three nails the answer is always yes. For four, only sometimes. Hidden among those four positions is a single condition that quietly decides it.

Four points that lie on a common circle are **concyclic**, and they are the secret heroes of olympiad geometry. Spotting a hidden circle — four points you can prove concyclic — unlocks a flood of equal angles via the Inscribed Angle Theorem. This chapter gives you the tests to find those circles, and then Ptolemy's theorem, a relation so powerful it contains the Pythagorean theorem as a single special case.

14.1 Tests for concyclicity

From the Inscribed Angle Theorem we already know two facts about a cyclic quadrilateral $ABCD$: its opposite angles are supplementary, and two vertices on the same side of a chord see it at equal angles. The *converses* are what make these into tools, and they are true:

Theorem 14.1 (Concyclicity tests). Four points A, B, C, D lie on one circle if *either*

- (a) a pair of opposite angles of quadrilateral $ABCD$ sum to 180° ; *or*
- (b) C and D lie on the same side of line AB and $\angle ACB = \angle ADB$ (equal

angles on the same segment).

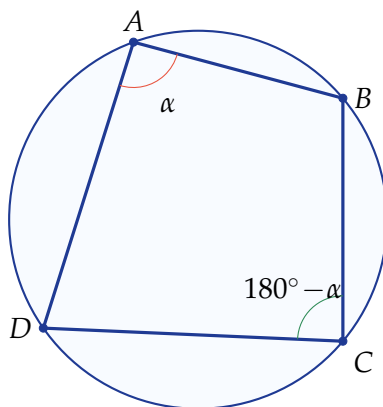


Figure 14.1 — A cyclic quadrilateral: opposite angles are supplementary, $\angle A + \angle C = 180^\circ$.

Worked Example. In triangle ABC , the altitudes from B and C meet the opposite sides at E and F . Prove that B, C, E, F lie on one circle.

The altitude from B is perpendicular to AC , so $\angle BEC = 90^\circ$ (the angle at E in the right triangle BEC). The altitude from C gives $\angle BFC = 90^\circ$ likewise. So E and F both see the segment BC at a right angle — and by the converse of Thales' Theorem, every such point lies on the circle with diameter BC . Hence B, C, E, F are concyclic.

Two right angles standing on the same segment instantly summon a circle — the one with that segment as diameter. “Right angle” and “semicircle” are two views of the same fact, and this single move (“these points see BC at 90° , so they are concyclic”) appears in a startling fraction of olympiad solutions.

14.2 Ptolemy's Theorem

Theorem 14.2 (Ptolemy). For a cyclic quadrilateral $ABCD$, the product of the diagonals equals the sum of the products of opposite sides:

$$AC \cdot BD = AB \cdot CD + AD \cdot BC.$$

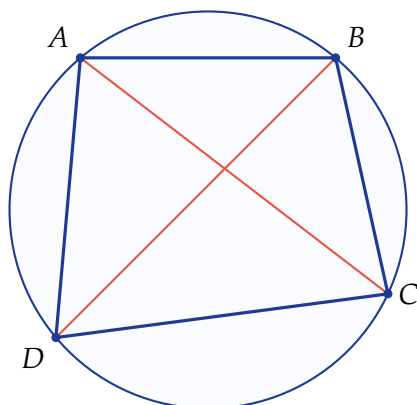


Figure 14.2 — Ptolemy's theorem: $AC \cdot BD = AB \cdot CD + AD \cdot BC$. The two diagonals (coral) "know about" the four sides.

 **Worked Example.** Prove Ptolemy's theorem.

The trick is one clever auxiliary point. On the diagonal AC choose K so that $\angle ABK = \angle DBC$.

First pair. Triangles ABK and DBC share $\angle ABK = \angle DBC$ (construction), and $\angle BAK = \angle BDC$ (inscribed angles on arc BC). So $\triangle ABK \sim \triangle DBC$, giving $\frac{AK}{DC} = \frac{AB}{DB}$, i.e. $AK \cdot DB = AB \cdot DC$.

Second pair. Adding $\angle KBD$ to the equal angles $\angle ABK$ and $\angle DBC$ shows $\angle ABD = \angle KBC$; and $\angle ADB = \angle ACB$ (inscribed angles on arc AB). So $\triangle ABD \sim \triangle KBC$, giving $\frac{KC}{AD} = \frac{BC}{BD}$, i.e. $KC \cdot DB = AD \cdot BC$.

Add them. $AK \cdot DB + KC \cdot DB = AB \cdot DC + AD \cdot BC$, and since $AK + KC = AC$,

$$AC \cdot BD = AB \cdot CD + AD \cdot BC. \quad \blacksquare$$

14.3 Two dazzling consequences

 **Worked Example.** **Ptolemy contains Pythagoras.** Apply Ptolemy to a rectangle.

A rectangle $ABCD$ has all four vertices on one circle (each angle is 90° , so opposite angles sum to 180°). Its sides are $AB = CD = a$ and $BC = DA = b$, and its diagonals are equal, $AC = BD = d$. Ptolemy says

$$AC \cdot BD = AB \cdot CD + AD \cdot BC \Rightarrow d \cdot d = a \cdot a + b \cdot b,$$

that is, $d^2 = a^2 + b^2$.

The wow: that is the Pythagorean theorem, falling out of Ptolemy as the special case of a rectangle. A theorem we proved four ways in Chapter 9 is just one corner of this larger truth.

 **Worked Example. Van Schooten's theorem.** Let ABC be equilateral, inscribed in a circle, and let P lie on the arc BC not containing A . Prove $PA = PB + PC$.

The four points, in order around the circle, are B, P, C, A , so $BPCA$ is a cyclic quadrilateral with diagonals BC and PA . Ptolemy gives

$$BC \cdot PA = BP \cdot CA + PC \cdot BA.$$

Since the triangle is equilateral, $AB = BC = CA = s$. Dividing through by s ,

$$PA = PB + PC. \quad \blacksquare$$

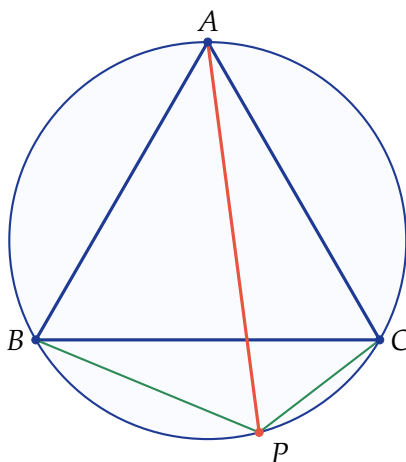


Figure 14.3 — Van Schooten: for P on arc BC of an equilateral triangle, the long chord equals the sum of the two short ones, $PA = PB + PC$.

AHA!

Notice the pattern of this chapter. A *circle* converts *position* facts (conconcyclic points) into *angle* facts (equal or supplementary angles), and Ptolemy converts them further into *length* facts (products of sides). Each theorem

is a translator between the three languages of geometry — position, angle, and length — and fluency in all three is what problem-solving really is.

Four nails hammered into a board at random, and out of them falls a relation that quietly swallows Pythagoras whole. The circle has been generous today.

→ YOUR TURN

$ABCD$ is a cyclic quadrilateral with $AB \cdot CD = 30$ and $AD \cdot BC = 12$. One diagonal is $AC = 7$. Find the other diagonal BD .

Set-up (you finish it): Ptolemy says $AC \cdot BD = AB \cdot CD + AD \cdot BC$, so $7 \cdot BD = 30 + 12$.

Answer: $BD = 42/7 = 6$.

KEY IDEAS & FORMULAS

- **Concyclicity tests:** four points are on one circle if a pair of opposite angles sums to 180° , or two points see a segment at equal angles from the same side. Two right angles on a segment $\Rightarrow$ all on the circle with that segment as diameter.
- **Ptolemy:** for cyclic $ABCD$, $AC \cdot BD = AB \cdot CD + AD \cdot BC$ — diagonals' product = sum of products of opposite sides.
- **Ptolemy's inequality:** for *any* four points $AC \cdot BD \leq AB \cdot CD + AD \cdot BC$, with equality exactly when $ABCD$ is cyclic (in order).
- **Ptolemy $\supset$ Pythagoras:** applied to a rectangle it gives $d^2 = a^2 + b^2$; applied to an equilateral triangle's circumcircle it gives van Schooten's $PA = PB + PC$.
- **Big picture:** a circle translates *position* (concyclic) into *angle* (equal/-supplementary) into *length* (Ptolemy).

Exercises

Warm-up.

1. In cyclic quadrilateral $ABCD$, $\angle A = 85^\circ$ and $\angle B = 95^\circ$. Find $\angle C$ and $\angle D$.
2. Is a parallelogram (that is not a rectangle) ever cyclic? Explain using opposite angles.
3. State Ptolemy's relation for cyclic quadrilateral $PQRS$.

4. A cyclic quadrilateral has sides $AB = 4$, $BC = 5$, $CD = 6$, $DA = 3$ and one diagonal $AC = 7$. Use Ptolemy to find the other diagonal BD .
5. Why is every rectangle cyclic? What is the centre of its circle?
6. P lies on arc BC of an equilateral triangle with $PB = 2$ and $PC = 5$. Find PA .

Practice.

7. Prove that an *isosceles trapezium* (a trapezium with equal legs) is always cyclic.
8. In quadrilateral $ABCD$, $\angle BAC = \angle BDC$. Prove A, B, C, D are concyclic.
9. Two circles meet at P and Q . Through P , a line meets the circles again at A and B ; through Q , a line meets them again at C and D (with A, C on the same circle). Prove $AC \parallel BD$.
10. A cyclic quadrilateral has perpendicular diagonals. If the sides in order are a, b, c, d , prove $a^2 + c^2 = b^2 + d^2$.
11. In cyclic quadrilateral $ABCD$, prove $\frac{AC}{BD} = \frac{AB \cdot AD + CB \cdot CD}{BA \cdot BC + DA \cdot DC}$. (Apply Ptolemy twice, to two different "re-labellings".)
12. Points A, B, C, D, E lie on a circle in order, with $ABCDE$ a regular pentagon of side 1. Use Ptolemy on $ABCD$ to find the length of a diagonal (and hence the golden ratio).
13. Prove that in any cyclic quadrilateral the product of the diagonals is at least the larger of $AB \cdot CD$ and $AD \cdot BC$.

Challenge.

14. (Ptolemy's inequality.) For *any* four points in the plane (not necessarily concyclic), prove $AC \cdot BD \leq AB \cdot CD + AD \cdot BC$, with equality exactly when $ABCD$ is a cyclic quadrilateral in that order.
15. In triangle ABC , let P be any point on the circumcircle's arc BC . Generalise van Schooten: express PA in terms of PB, PC and the side lengths.
16. Prove that the four *angle bisectors* of a (non-cyclic) quadrilateral's interior angles meet to form a cyclic quadrilateral.
17. Let $ABCD$ be cyclic. Prove that the incentres of triangles ABC, BCD, CDA, DAB form a rectangle.
18. A point P inside square $ABCD$ satisfies $PA : PB : PC = 1 : 2 : 3$. Find $\angle APB$. (Rotate, then chase a circle into the picture.)

★ OLYMPIAD SPARK

(Ptolemy builds trigonometry.) Place a chord AC as a diameter of a circle of diameter 1, and let B, D be two more points on the circle. Using the fact that a chord subtending an inscribed angle θ in a unit-diameter circle has length $\sin \theta$, apply Ptolemy to $ABCD$ to *derive the sine addition formula*

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

This is how Ptolemy himself built the first trigonometric tables, eighteen centuries ago.

(Second spark.) Let P be a point and ABC a triangle. Prove that $PA \cdot BC$, $PB \cdot CA$, $PC \cdot AB$ are the three side lengths of some triangle — and that this “pedal” triangle degenerates to a segment exactly when P lies on the circumcircle of ABC (recovering Ptolemy’s equality).

15

Power of a Point & the Radical Axis

★ WONDER

Stand anywhere near a circle and draw *any* straight line through your spot; it cuts the circle at two points. Multiply the two distances from you to those points. Now swing the line to a new direction and do it again — a completely different pair of distances. Astonishingly, the product comes out *exactly the same*. Every line through your point carries the same secret product. What is this number that the point “remembers” about the circle?

That number is the **power** of the point, and it turns a whole family of length problems into one clean idea. It also leads to the *radical axis*, the hidden line of balance between two circles, and to one more surprising concurrency.

15.1 The power of a point

Theorem 15.1 (Power of a Point). Let P be a point and consider a circle of centre O , radius r . For *any* line through P meeting the circle at X and Y , the product $PX \cdot PY$ is the same; its value is $|OP^2 - r^2|$. In particular:

- if P is inside, two chords through it give $PA \cdot PB = PC \cdot PD$;
- if P is outside, two secants give $PA \cdot PB = PC \cdot PD$, and a tangent PT gives $PT^2 = PA \cdot PB$.

Why $OP^2 - r^2$? Take the one special line through P and the centre O . It meets the circle at distances $OP - r$ and $OP + r$ from P (one on each side, for P outside), so the product is $(OP - r)(OP + r) = OP^2 - r^2$. The theorem says

every other line gives the same value — so this single quantity, the **power** $\text{pow}(P) = OP^2 - r^2$, captures it all. It is negative inside, zero on the circle, positive outside.

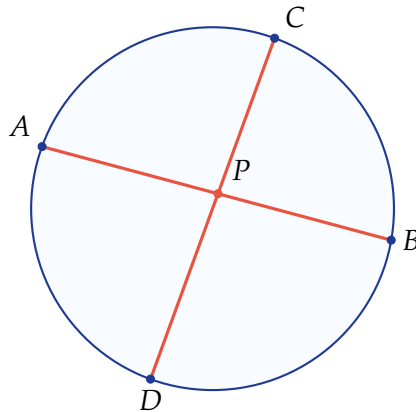


Figure 15.1 — Two chords through an interior point P : $PA \cdot PB = PC \cdot PD$.

Worked Example. Prove the interior case: two chords AB and CD meet at P inside the circle, then $PA \cdot PB = PC \cdot PD$.

Join AC and DB . In triangles PAC and PDB : the angles at P are vertical, so $\angle APC = \angle DPB$; and $\angle CAB = \angle CDB$, since both are inscribed angles standing on the same arc CB . By **AA**, $\triangle PAC \sim \triangle PDB$. Hence

$$\frac{PA}{PD} = \frac{PC}{PB} \implies PA \cdot PB = PC \cdot PD.$$

The messy fact “every chord through P has the same product” is really just *one* pair of similar triangles, hiding behind the Inscribed Angle Theorem. The same picture, read for a point *outside* the circle, proves the secant case word-for-word.

Worked Example. From an external point P , a secant meets a circle at A (near) and B (far), and a tangent touches at T . Show $PT^2 = PA \cdot PB$, and use it: if $PA = 4$ and $AB = 5$, find the tangent length PT .

A tangent is the limiting case of a secant whose two intersection points have merged into the single point T ; setting $A = B = T$ in $PA \cdot PB$ gives $PT \cdot PT = PT^2$. So $PT^2 = PA \cdot PB$.

Here $PB = PA + AB = 4 + 5 = 9$, so $PT^2 = 4 \cdot 9 = 36$ and $PT = 6$.

The wow: the tangent is the “geometric mean” of the two secant distances. This is exactly how a surveyor finds the distance to an unreachable circular wall, or how you compute the distance to the horizon: the line of sight is a tangent to the Earth, so (your height) $\times$ (your height + diameter) is the square of how far you can see.

15.2 The radical axis

Two circles sit in the plane. Is there a point with the *same* power with respect to both? There are infinitely many — and they form a line.

Theorem 15.2 (Radical axis). The set of points having equal power with respect to two non-concentric circles is a straight line, the **radical axis**, perpendicular to the line joining the centres. If the circles intersect, it is the line through their two intersection points.

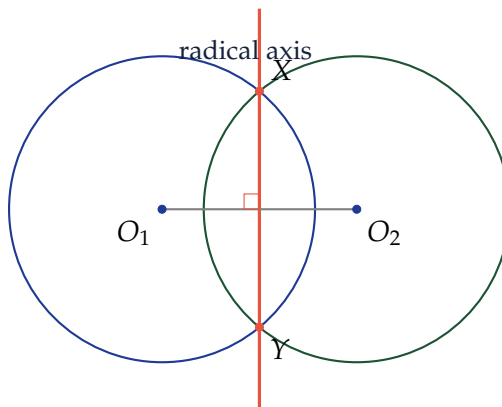


Figure 15.2 — The radical axis (coral) of two intersecting circles is their common chord line, perpendicular to the line of centres O_1O_2 .

 **Worked Example.** Prove that the three radical axes of three circles (taken in pairs) pass through one point — the **radical centre**.

Let the circles be $\omega_1, \omega_2, \omega_3$. The radical axis of ω_1, ω_2 is where $\text{pow}_1 = \text{pow}_2$; of ω_2, ω_3 where $\text{pow}_2 = \text{pow}_3$. Let R be the point where these two lines cross. At R we have $\text{pow}_1 = \text{pow}_2$ and $\text{pow}_2 = \text{pow}_3$, so by transitivity $\text{pow}_1 = \text{pow}_3$ as well. But that is exactly the condition for R to lie on the radical axis of ω_1, ω_3 . So all three axes pass through R .

The proof is pure logic — transitivity of equality — with no computation at all. And it is genuinely useful: the radical centre is the one point from which the tangents to all three circles are equal, so it is the centre of the unique circle cutting all three at right angles.

AHA!

The power of a point quietly unifies three separate-looking facts (two chords, two secants, tangent-and-secant) into one number, $OP^2 - r^2$. Whenever a problem multiplies two distances along a line through a circle, think “power,” and the circle’s centre and radius are secretly in charge.

A single number, carried unseen by every point in the plane, and three separate-looking facts melt into one. The more geometry you learn, the fewer things there turn out to be.

→ YOUR TURN

Two chords of a circle cross at an interior point P . The first is divided into segments of length 6 and 4; the second into a segment of length 3 and an unknown x . Find x .

Set-up (you finish it): equal products along the two chords give $6 \cdot 4 = 3 \cdot x$.

Answer: $x = 24/3 = 8$.

KEY IDEAS & FORMULAS

- **Power of a point:** $\text{pow}(P) = OP^2 - r^2$ — negative inside, zero on, positive outside the circle.
- **Constant product:** for any line through P meeting the circle at X, Y , the product $PX \cdot PY = |OP^2 - r^2|$ is the same. Interior: $PA \cdot PB = PC \cdot PD$; exterior secants likewise.
- **Tangent:** $PT^2 = PA \cdot PB = OP^2 - r^2$ — the tangent is the geometric mean of the two secant distances.
- **Converse** (concyclicity test): if lines AB, CD meet at P with $PA \cdot PB = PC \cdot PD$, then A, B, C, D are concyclic.
- **Radical axis:** the equal-power locus of two circles — a line perpendicular to the line of centres (the common chord when they meet). The

three radical axes of three circles concur at the *radical centre*.

Exercises

Warm-up.

1. Two chords meet inside a circle. One is cut into 3 and 8; the other into 4 and x . Find x .
2. A tangent from P has length 6; a secant from P has near distance 4. Find the far distance.
3. Find the power of a point 10 units from the centre of a circle of radius 6.
4. Is the power of a point inside a circle positive or negative?
5. A point lies on a circle of radius 5. What is its power with respect to that circle?
6. Where is the radical axis of two circles that touch externally?

Practice.

7. From external point P , two secants are drawn. The first meets the circle at distances 5 and 12; the second has near distance 4. Find its far distance.
8. A tangent of length t and a secant through the centre are drawn from P , with the circle of radius r and $OP = d$. Verify $t^2 = d^2 - r^2$ two ways.
9. Chords AB and CD meet at P with $PA = PC$. Prove $PB = PD$ and that $ABDC$ is an isosceles trapezium.
10. Point P is 13 from the centre of a circle of radius 5. A line through P meets the circle at A, B . Find $PA \cdot PB$, and the shortest possible chord AB visible from P .
11. Two circles intersect at A, B . From a point P on line AB (beyond the circles), prove the tangents to the two circles are equal.
12. A circle has a chord AB . The tangents at A and B meet at T . Prove $TA = TB$ using power of a point.
13. Given a circle and an external point P , describe a straightedge-and-compass construction of the tangent from P , and justify it with power of a point.

Challenge.

14. Prove the converse of the power-of-a-point theorem: if A, B, C, D are points with lines AB, CD meeting at P and $PA \cdot PB = PC \cdot PD$, then A, B, C, D are concyclic.

15. Three circles pass through a common point P and meet pairwise again at X, Y, Z . Investigate when X, Y, Z are collinear.
16. Given two circles, construct their radical axis using only straightedge and compass (even when the circles do not intersect). (*Hint: a third, auxiliary circle and two radical axes.*)
17. A point P moves so that the ratio of its distances to two fixed points A, B is constant, $PA : PB = k \neq 1$. Prove the locus is a circle (the *Apollonius circle*), and relate it to power of a point.
18. Prove that the radical centre of the three circles having the sides of a triangle as diameters is the orthocentre of the triangle.

★ OLYMPIAD SPARK

(Olympiad.) Two circles meet at A and B . A common tangent touches them at P and Q . Prove that line AB bisects segment PQ . (*The midpoint of PQ has equal power to both circles — so it lies on their radical axis, which is line AB .*)

(Second spark.) Let ABC be a triangle and ω its circumcircle. A circle through B and C meets AB again at F and AC again at E . Prove that B, C, E, F are concyclic (immediate) and that $AE \cdot AC = AF \cdot AB$; deduce that EF is antiparallel to BC . *Power of the point A does all the work.*

16

Tangents & the Tangent–Chord Angle

★ WONDER

A cyclist rounds a perfectly circular track. At the very instant she passes you, her direction of travel points along a line that grazes the circle — touching it at exactly her position and nowhere else. That line is a *tangent*. Now here is the riddle: the angle her path makes with the straight line back to any landmark on the track seems to “know” about the far side of the circle. A line that grazes the rim at a single point should be a stranger to everything beyond it. This one is not.

The tangent is the circle’s most subtle line — meeting it just once — and it hides two gems: a right angle that is always there, and the beautiful *Alternate Segment Theorem*, which links the tangent to inscribed angles clear across the circle.

16.1 The tangent is perpendicular to the radius

Theorem 16.1 (Tangent–radius). A tangent to a circle is perpendicular to the radius drawn to the point of contact.

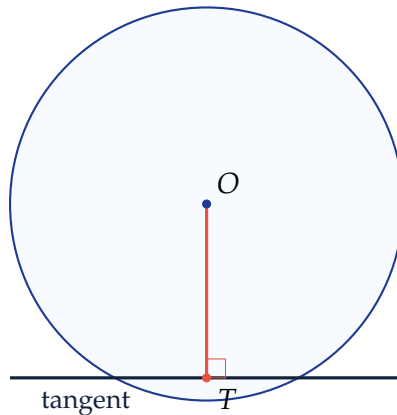


Figure 16.1 — The radius OT meets the tangent at T at a right angle.

 **Worked Example.** Prove it.

Let the tangent touch the circle at T , and let S be any *other* point on the tangent line. Since the tangent meets the circle only at T , the point S lies outside the circle, so $OS > r = OT$. Thus among all points of the tangent line, T is the one *closest* to O . But the closest point of a line to an external point is precisely the foot of the perpendicular from that point. Therefore $OT \perp$ the tangent.

We proved a perpendicularity without ever measuring an angle — purely from “the tangent stays outside.” This single right angle is the seed of everything else about tangents, including the equal-tangents property and the power-of-a-point tangent formula.

16.2 The Alternate Segment Theorem

Here is the tangent’s most surprising gift.

Theorem 16.2 (Alternate Segment / Tangent–Chord Angle). The angle between a tangent to a circle and a chord drawn from the point of contact equals the inscribed angle subtended by that chord in the *alternate* segment.

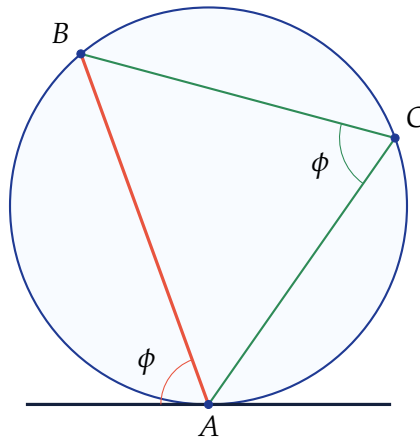


Figure 16.2 — The tangent–chord angle ϕ at A equals the inscribed angle $\angle ACB$ in the alternate segment.

 **Worked Example.** Prove the Alternate Segment Theorem.

Let ϕ be the angle between the tangent at A and the chord AB . Draw the diameter AE from A (so E is the antipode of A). Because the tangent is perpendicular to the radius (hence to the diameter AE), the chord AB makes angle $90^\circ - \phi$ with AE ; that is, $\angle EAB = 90^\circ - \phi$.

Now AE is a diameter, so by Thales' Theorem the inscribed angle $\angle ABE = 90^\circ$. In triangle ABE the angles sum to 180° , so

$$\angle AEB = 180^\circ - 90^\circ - (90^\circ - \phi) = \phi.$$

But $\angle AEB$ is the inscribed angle subtending chord AB from E , and E lies in the same (alternate) segment as C , so $\angle ACB = \angle AEB = \phi$.

The bridge: the tangent direction at A is an inscribed angle in disguise. So a tangent–chord angle can be used *exactly* like any inscribed angle in an angle chase — it is the bridge that lets a tangent line “talk” to points on the far side of the circle, answering the Wonder.

16.3 Tangent lengths and the incircle

Recall that the two tangents to a circle from an external point are equal. Applied to a triangle's incircle, this yields one of the most useful little formulas in all of geometry.

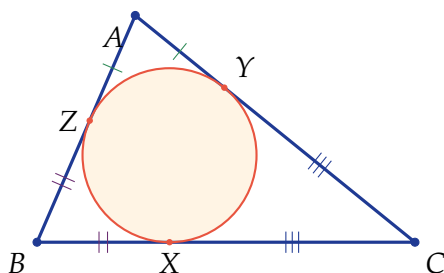


Figure 16.3 — The incircle touches the sides at X, Y, Z . Equal tangents from each vertex (matching ticks) give the lengths $s - a, s - b, s - c$.

Worked Example. Prove that the tangent length from vertex A to the incircle is $s - a$, where $s = \frac{1}{2}(a + b + c)$ is the semiperimeter and a, b, c are the sides opposite A, B, C .

Let the incircle touch BC, CA, AB at X, Y, Z . The two tangents from each vertex are equal, so set

$$AY = AZ = x, \quad BZ = BX = y, \quad CX = CY = z.$$

Reading the side lengths off the figure,

$$a = BC = y + z, \quad b = CA = z + x, \quad c = AB = x + y.$$

Adding all three: $a + b + c = 2(x + y + z)$, so $x + y + z = s$. Then

$$x = (x + y + z) - (y + z) = s - a.$$

Likewise $y = s - b$ and $z = s - c$.

Three messy tangent lengths collapse into the clean trio $s - a, s - b, s - c$. This is the engine behind the incircle radius $r = \text{Area}/s$, behind Heron's formula, and behind a host of olympiad length-chases — all from "tangents from a point are equal."

AHA!

Part III has been one long demonstration that the circle is a *machine for converting between angles and lengths*. Inscribed angles, cyclic quadrilaterals, Ptolemy, power of a point, the tangent–chord angle: each is a different gear in that machine. With Part III complete, you now hold the entire circle toolkit — and you are ready for the techniques of Part IV, where

these tools combine into the art of the angle chase.

→ YOUR TURN

A triangle has sides $a = 9$, $b = 10$, $c = 11$. Find the tangent length from vertex A to the incircle (the distance from A to each nearby point of contact).

Set-up (you finish it): the tangent length from A is $s - a$, where $s = \frac{1}{2}(a + b + c)$.

Answer: $s = \frac{1}{2}(9 + 10 + 11) = 15$, so the tangent length is $s - a = 15 - 9 = 6$.

KEY IDEAS & FORMULAS

- **Tangent $\perp$ radius** at the point of contact — the seed of every other tangent fact.
- **Two tangents from an external point are equal**; the tangent length is $\sqrt{OP^2 - r^2}$.
- **Alternate Segment Theorem:** the tangent–chord angle equals the inscribed angle subtended by that chord in the alternate segment — so a tangent acts exactly like an inscribed angle in an angle chase.
- **Incircle tangent lengths:** from the vertices they are $s - a$, $s - b$, $s - c$ (with s the semiperimeter) — the engine behind $r = \text{Area}/s$ and Heron's formula.
- **Pitot:** a quadrilateral has an inscribed circle iff $AB + CD = BC + DA$.

Exercises

Warm-up.

1. A tangent touches a circle of radius 7 at T . What is the angle between the tangent and the radius OT ?
2. The tangent–chord angle on one side of a chord is 54° . What inscribed angle does the chord subtend in the alternate segment?
3. A triangle has sides $a = 6$, $b = 7$, $c = 9$. Find the semiperimeter and the three tangent lengths $s - a$, $s - b$, $s - c$.
4. From a point 13 from the centre of a circle of radius 5, find the length of the tangent.

5. Two tangents from a point P touch a circle at U, V with $\angle UPV = 50^\circ$. Find $\angle UOV$ (centre O).
6. In Figure 16.3, if $a = 8, b = 5, c = 7$, find AZ, BX, CY .

Practice.

7. A tangent at A and a chord AB make an angle of 63° . The chord AB subtends a central angle $\angle AOB$. Find $\angle AOB$.
8. Two circles touch externally at T . A line through T meets one circle at P and the other at Q . Prove that the tangents to the circles at P and Q are parallel. (*Alternate segment, twice.*)
9. In triangle ABC the incircle touches BC at X . Prove $BX = s - b$ and hence X divides BC into $s - b$ and $s - c$.
10. PA and PB are tangents to a circle from P . A third tangent touches the circle at C and meets PA, PB at D, E . Prove that the perimeter of $\triangle PDE$ equals $2PA$.
11. The tangent to a circle at A meets a chord BC (extended) at P . Prove $PA^2 = PB \cdot PC$ and that $\triangle PAB \sim \triangle PCA$.
12. A circle is inscribed in a right triangle with legs a, b and hypotenuse c . Prove the inradius is $r = \frac{1}{2}(a + b - c)$ using the tangent lengths.
13. Through a point A on a circle, a tangent and a chord AB are drawn. A second chord AC makes the tangent–chord angles ϕ_1 (with AB) and ϕ_2 (with AC). Relate ϕ_1, ϕ_2 to arcs AB and AC .

Challenge.

14. A quadrilateral $ABCD$ has an inscribed circle touching all four sides. Prove $AB + CD = BC + DA$ (Pitot's theorem), and prove the converse for convex quadrilaterals.
15. Two circles are tangent internally at T . A chord of the larger circle is tangent to the smaller. Prove that T lies on the angle bisector from T of the chord's endpoints — i.e. TT bisects the relevant angle. (Use the alternate segment theorem at T .)
16. Prove that the tangents to the circumcircle of $\triangle ABC$ at B and C meet at a point P on the symmedian from A . (Tangent–chord angles reveal the equal angles.)
17. In $\triangle ABC$, the incircle touches BC at X . Let X' be the point where the *excir- cle* opposite A touches BC . Prove $BX = CX'$ (so X and X' are reflections in the midpoint of BC).
18. Circles ω_1, ω_2 meet at A, B . The tangent to ω_1 at A meets ω_2 again at C ; the tangent to ω_2 at A meets ω_1 again at D . Prove $AB^2 = BC \cdot BD$.

★ OLYMPIAD SPARK

(Olympiad.) Let $\triangle ABC$ have incircle touching BC, CA, AB at X, Y, Z . Prove that the lines AX, BY, CZ are concurrent. (Their common point is the *Gergonne point*.) *The tangent lengths $s - a, s - b, s - c$ plus a concurrency criterion from Part IV make this clean; for now, set it up with the equal tangents.*

(Second spark.) A circle ω is tangent to a line ℓ at P . A second circle is tangent to ℓ at Q and tangent to ω externally at R . Prove that P, R , and the “top” of the configuration are arranged so that line PR passes through the midpoint of the arc — more concretely, prove QR bisects the tangent–chord angle at R . (*The alternate segment theorem at the tangency R is the key.*)

PART IV

The Toolbox

17

Angle Chasing, the Workhorse

★ WONDER

You are handed a crowded diagram — circles, chords, a triangle, a tangent — and asked for one particular angle, buried deep in the tangle. There is no single formula for it. Yet a grandmaster glances at the figure and writes the answer in seconds. They are not guessing. They are *chasing*: stepping from one known angle to the next along a chain of tiny, certain deductions, until the unknown is cornered. How do you learn to see that chain?

Angle chasing is the most-used skill in all of olympiad geometry. It is not a theorem but a *craft*: combining the angle facts you already own into chains that march across a figure. This chapter assembles your toolkit, adds two powerful new results about circles, and then hands you the professional's secret weapon — directed angles.

17.1 The angle-chaser's toolkit

Everything you need is already proved. Keep this checklist in your mind's eye; each item turns one known angle into another.

AHA!

The toolkit. (1) Angles on a straight line sum to 180° ; around a point, 360° . (2) Vertical angles are equal. (3) Across parallels: corresponding and alternate angles equal, co-interior supplementary. (4) Triangle angles sum to 180° ; the exterior angle equals the sum of the two remote interior angles. (5) Isosceles triangle: equal sides face equal angles.

(6) Inscribed Angle Theorem; angles in the same segment are equal; angle in a semicircle is 90° . (7) Cyclic quadrilateral: opposite angles supplementary (and its converse, to *find* circles). (8) Tangent–chord (alternate segment) angle.

The art is choosing *which* tool, and *where*. Two habits help enormously: mark every angle you can find with the same symbol when they are equal, and always ask “do four of these points lie on a circle?”

17.2 The angle between two chords

Two results synthesise the inscribed-angle theorem into exactly what a chaser needs.

Theorem 17.1 (Chord angle, interior). If two chords meet *inside* a circle at P , the angle between them is half the *sum* of the two arcs it intercepts.

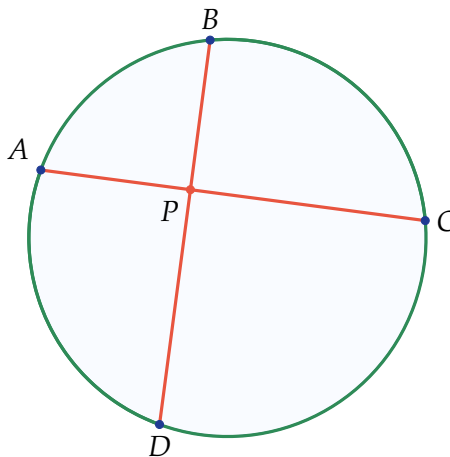


Figure 17.1 — Two chords meeting at P : the angle $\angle APB$ is half the sum of arcs AB and CD (green).

 **Worked Example.** Prove it, and apply it: arcs AB and CD measure 50° and 30° . Find $\angle APB$.

Draw the chord BC . Then $\angle APB$ is the *exterior* angle at P of triangle BPC ,

so it equals the sum of the two remote interior angles:

$$\angle APB = \angle PBC + \angle PCB.$$

Now $\angle PCB = \angle ACB$ (same ray CA), an inscribed angle on arc AB , so it is $\frac{1}{2}$ arc AB . And $\angle PBC = \angle DBC$, inscribed on arc CD , equal to $\frac{1}{2}$ arc CD . Hence

$$\angle APB = \frac{1}{2}(\text{arc } AB + \text{arc } CD) = \frac{1}{2}(50^\circ + 30^\circ) = 40^\circ.$$

The inscribed-angle theorem (an angle is half *one* arc) and the exterior-angle theorem combine into “half the *sum* of two arcs.” A two-step chase becomes a one-line rule — and the intersection point “averages” the arcs facing it.

Theorem 17.2 (Chord angle, exterior). If two secants meet *outside* a circle at P , the angle between them is half the *difference* of the two intercepted arcs.

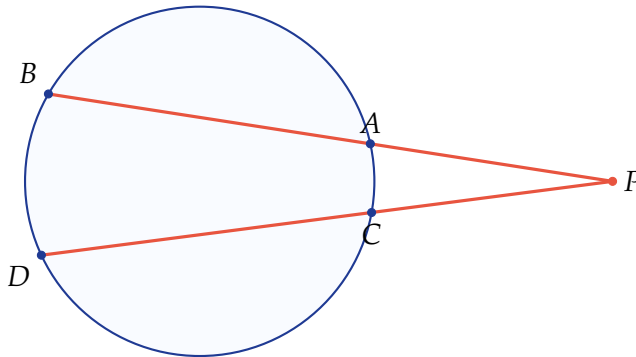


Figure 17.2 — Two secants from an external point P : $\angle P = \frac{1}{2}(\text{arc } BD - \text{arc } AC)$, the far arc minus the near arc.

 **Worked Example.** Prove the exterior case.

Let the secants be $P-A-B$ and $P-C-D$ (with A, C the near points). Draw chord BC . In triangle PBC , the angle at P is $\angle P$, and extending PC to D makes $\angle BCD$ an exterior angle:

$$\angle BCD = \angle P + \angle PBC \Rightarrow \angle P = \angle BCD - \angle PBC.$$

Here $\angle PBC = \angle ABC$ (same ray BA), inscribed on arc AC , so $\frac{1}{2}$ arc AC ; and $\angle BCD$ is inscribed on arc BD , so $\frac{1}{2}$ arc BD . Therefore

$$\angle P = \frac{1}{2}(\text{arc } BD - \text{arc } AC).$$

The wow: inside the circle you *add* the arcs; outside you *subtract*. Slide P from inside to outside and the rule smoothly switches sum $\rightarrow$ difference — and when P sits *on* the circle (one arc shrinks to nothing), it collapses to the plain inscribed-angle theorem. One idea, three faces.

17.3 Directed angles: the professional's shortcut

A frustration of angle chasing is *configuration dependence*: the same theorem needs “ θ ” in one picture and “ $180^\circ - \theta$ ” in another, depending on which side a point falls. **Directed angles** sweep this away. Write $\angle(\ell, m)$ for the angle of rotation *from* line ℓ *to* line m , measured modulo 180° . Then:

- $\angle(\ell, m) = -\angle(m, \ell)$, and $\angle(\ell, m) + \angle(m, n) = \angle(\ell, n)$ (angles just add);
- **Four points** A, B, C, D **are concyclic or collinear** exactly when

$$\angle(CA, CB) = \angle(DA, DB).$$

 **Worked Example.** Why directed angles are worth the trouble.

The ordinary statement “angles in the same segment are equal” needs a caveat: if C and D are on *opposite* arcs, the angles are *supplementary*, not equal — two cases, two diagrams. In directed angles, both cases are the single identity

$$\angle(CA, CB) = \angle(DA, DB) \pmod{180^\circ},$$

because working modulo 180° identifies θ with $\theta - 180^\circ$. So a directed-angle chase, once set up, is valid for *every* configuration at once.

You prove the hardest part of many olympiad problems — “it holds in all cases” — for free, simply by computing modulo 180° . The bookkeeping that sinks careless solutions just disappears.

X COMMON TRAP

Directed angles are measured modulo 180° , never 360° : a line has no “arrow.” So $\angle(\ell, m) = 80^\circ$ and $\angle(\ell, m) = -100^\circ$ mean the same thing. Never feed a directed-angle result into a formula (like the Law of Sines) that needs the *actual* angle — convert back to an ordinary angle first, reading the final figure.

Angle chasing is less a theorem than a way of walking a figure: one certain step after another, until the unknown has nowhere left to hide.

→ YOUR TURN

Two chords of a circle cross at an interior point P , intercepting arcs of 84° and 36° . Find the angle between the chords.

Set-up (you finish it): for chords meeting *inside*, the angle is half the *sum* of the intercepted arcs, $\frac{1}{2}(84^\circ + 36^\circ)$.

Answer: $\frac{1}{2}(84^\circ + 36^\circ) = \frac{1}{2}(120^\circ) = 60^\circ$.

KEY IDEAS & FORMULAS

- **The toolkit:** straight-line/point sums, vertical angles, parallels, triangle sum and exterior angle, isosceles, the inscribed-angle family, cyclic quadrilaterals (and the converse, to *find* circles), and the tangent–chord angle.
- **Interior chord angle** $= \frac{1}{2}(\text{sum of the two intercepted arcs})$; **exterior** (two secants) $= \frac{1}{2}(\text{difference of arcs})$. On the circle they collapse to the inscribed-angle theorem.
- **Directed angles** $\angle(\ell, m)$, taken mod 180° , simply add: $\angle(\ell, m) + \angle(m, n) = \angle(\ell, n)$.
- **One concyclicity test for all configurations:** A, B, C, D are concyclic (or collinear) iff $\angle(CA, CB) = \angle(DA, DB)$.
- **Two habits:** mark equal angles with one symbol, and keep asking “do four of these points lie on a circle?”

Exercises**Warm-up.**

1. Two chords meet inside a circle, intercepting arcs of 70° and 40° . Find the

angle between them.

2. Two secants from an external point intercept arcs of 100° (far) and 30° (near). Find the angle at the point.
3. A chord and a tangent meet on a circle; the chord cuts off an arc of 86° . Find the tangent–chord angle.
4. In a cyclic quadrilateral $ABCD$, $\angle A = 78^\circ$. Find $\angle C$.
5. State the directed-angle test for four points to be concyclic.
6. An inscribed angle and the central angle on the same arc differ by what factor?

Practice.

7. Two chords meet inside a circle at 80° . One intercepted arc is 50° . Find the other.
8. Two tangents from an external point meet at 40° . Find the two arcs they cut off.
9. In triangle ABC , I is the incentre. Re-derive $\angle BIC = 90^\circ + \frac{1}{2}\angle A$ by angle chasing.
10. AB is a diameter; C, D are on the circle with $\angle CAB = 25^\circ$ and $\angle DBA = 40^\circ$. Find the angle between chords AC and BD .
11. In triangle ABC , the altitude from A and the diameter of the circumcircle through A are drawn. Prove they make equal angles with the two sides AB, AC (they are *isogonal*).
12. Two circles meet at P, Q . A variable line through P meets the circles again at A and B . Prove that $\angle AQB$ is the same for every such line (it does not depend on the line's direction).
13. Quadrilateral $ABCD$ is cyclic; diagonals meet at P . Prove $\angle APB = \frac{1}{2}(\text{arc } AB + \text{arc } CD)$ and use it to locate P 's angle when the four arcs are $70^\circ, 80^\circ, 110^\circ, 100^\circ$.

Challenge.

14. (Miquel point.) On the sides BC, CA, AB of a triangle choose points X, Y, Z . Prove that the three circles AYZ, BZX, CXY pass through a common point. (Directed angles make it a three-line chase.)
15. Prove that the reflection of the orthocentre over a side lies on the circumcircle, using only angle chasing.
16. In a circle, AB and CD are chords meeting at P ; M and N are the midpoints of arcs AC and BD . Prove MN is perpendicular to the bisector of $\angle APC$.
17. Two circles are internally tangent at T . A chord AB of the larger is tangent

to the smaller at S . Prove TS bisects $\angle ATB$.

18. Points A, B, C, D, E, F lie on a circle. Prove that the three “long diagonals” AD, BE, CF relate to the alternating arc-sums (a hexagon angle identity); state and prove the cleanest version you can.

★ OLYMPIAD SPARK

(Olympiad-style.) A circle passes through vertices B and C of triangle ABC and meets AB, AC again at C', B' . Prove that $B'C'$ is *antiparallel* to BC (that is, $\angle AB'C' = \angle ABC$), and deduce that the line through A perpendicular to $B'C'$ is exactly the circumdiameter from A — it passes through the circumcentre O and meets the circumcircle again at the antipode of A . (*A short directed-angle chase reveals the hidden cyclic quadrilateral; then recall that the tangent at A is antiparallel to BC .*)

(Second spark.) Let H be the orthocentre of acute triangle ABC . Prove that the four points B, C , and the reflections of H over AB and over AC are concyclic, and that the circle is the circumcircle of ABC .

18

Trigonometry in Geometry

★ WONDER

You want the height of a distant mountain you cannot climb, or the width of a river you cannot cross. You are allowed to walk along your own bank and to measure *angles* with a sighting instrument — nothing more. With two angle readings and one pacing-off of distance along your side, you can name the far dimension exactly. How do angles alone reach across the uncrossable?

Synthetic geometry chases angles; *trigonometry* turns angles into *lengths* and back, by way of the ratios $\sin$, $\cos$, $\tan$. Three relations — the Law of Sines, the Law of Cosines, and the area formula — let you compute any triangle from partial data, and they connect a triangle to its circumscribed and inscribed circles in a startlingly tidy way.

18.1 Ratios in a right triangle

In a right triangle, fix one acute angle θ . The three side ratios depend on θ alone:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

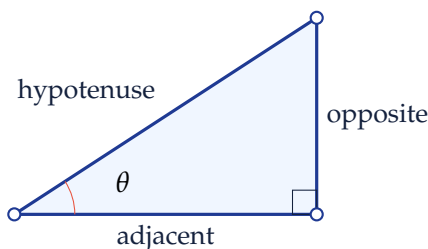


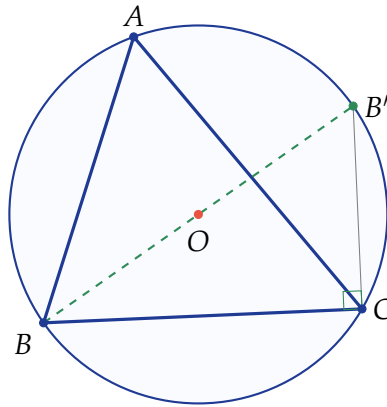
Figure 18.1 — The three ratios at angle θ : SOH-CAH-TOA.

These ratios are the dictionary between an angle and the shape of its right triangle. The whole chapter extends them to *any* triangle.

18.2 The Law of Sines — and a hidden circle

Theorem 18.1 (Law of Sines, extended). In any triangle with sides a, b, c opposite angles A, B, C , and circumradius R ,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Figure 18.2 — A triangle in its circumcircle. The diameter BB' exposes why $a = BC = 2R \sin A$.

 **Worked Example.** Prove the extended law, and see why the ratio is the circumdiameter.

Draw the diameter BB' from B through the circumcentre O . Since BB' is a diameter, $\angle BCB' = 90^\circ$ (angle in a semicircle). And $\angle BB'C = \angle BAC = A$, because both are inscribed angles standing on the same chord BC . In the right triangle BCB' ,

$$\sin A = \sin \angle BB'C = \frac{BC}{BB'} = \frac{a}{2R}, \quad \text{so} \quad a = 2R \sin A.$$

The same argument at the other vertices gives $b = 2R \sin B$ and $c = 2R \sin C$, hence $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$.

The wow: the common ratio in the Law of Sines is not just “some constant” — it is the *diameter of the circumscribed circle*. A single number, read off any one side and its opposite angle, tells you the size of the circle through all three vertices. Angle and length and circle, bound together in one equation.

18.3 The Law of Cosines

Theorem 18.2 (Law of Cosines). In any triangle, $c^2 = a^2 + b^2 - 2ab \cos C$, where C is the angle opposite side c .

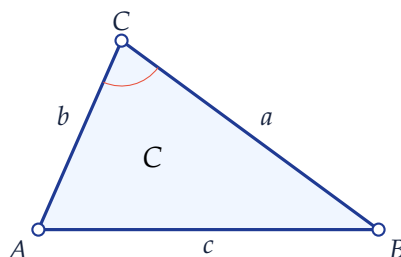


Figure 18.3 — The Law of Cosines relates the three sides through the angle C opposite c .

 **Worked Example.** A triangle has sides 5, 8, and 7. Find the angle opposite the side of length 7.

With $c = 7$, $a = 5$, $b = 8$, the Law of Cosines gives

$$7^2 = 5^2 + 8^2 - 2 \cdot 5 \cdot 8 \cos C \Rightarrow 49 = 89 - 80 \cos C \Rightarrow \cos C = \frac{40}{80} = \frac{1}{2},$$

so $C = 60^\circ$.

Look at the formula when $C = 90^\circ$: $\cos 90^\circ = 0$ and it becomes $c^2 = a^2 + b^2$ — Pythagoras exactly. The Law of Cosines is Pythagoras with a correction term $-2ab \cos C$ that measures how far the angle is from a right angle. Acute C makes c shorter than Pythagoras predicts; obtuse C makes it longer.

18.4 Area, and the bridge to both circles

Theorem 18.3 (Area). A triangle with two sides a, b enclosing angle C has area $[ABC] = \frac{1}{2}ab \sin C$. Equivalently, $[ABC] = \frac{abc}{4R} = rs$, where R is the circumradius, r the inradius, and s the semiperimeter.

 **Worked Example.** Derive $[ABC] = \frac{abc}{4R}$ from the two facts you now have.

The area is $\frac{1}{2}ab \sin C$. By the extended Law of Sines, $\sin C = \frac{c}{2R}$. Substituting,

$$[ABC] = \frac{1}{2}ab \cdot \frac{c}{2R} = \frac{abc}{4R}.$$

The reward: two ideas from this chapter snap together to reveal that a triangle's area, its three sides, and its circumradius obey one clean law.

Rearranged, $R = \frac{abc}{4[ABC]}$ — the circumradius computed from sides and area alone. And pairing with $[ABC] = rs$ links the *incircle* into the same web.

AHA!

Synthetic geometry and trigonometry are two languages for one subject. A hard angle chase sometimes melts in one line of the Law of Sines; a stubborn length yields to the Law of Cosines. The expert keeps *both* channels open and switches the instant one looks easier — that fluency is much of what Part IV is teaching you.

A sighting instrument and a single pace along the bank, and the far mountain can no longer hide its height. Angles, it turns out, are a currency you can spend for length.

→ YOUR TURN

Two sides of a triangle, of length 5 and 3, enclose an angle of 120° . Find the third side.

Set-up (you finish it): the Law of Cosines gives $c^2 = 5^2 + 3^2 - 2 \cdot 5 \cdot 3 \cos 120^\circ$, and $\cos 120^\circ = -\frac{1}{2}$.

Answer: $c^2 = 25 + 9 - 30(-\frac{1}{2}) = 34 + 15 = 49$, so $c = 7$.

KEY IDEAS & FORMULAS

- **Right-triangle ratios:** $\sin, \cos, \tan$ — SOH-CAH-TOA.
- **Law of Sines (extended):** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$ — the common ratio is the *circumdiameter*.
- **Law of Cosines:** $c^2 = a^2 + b^2 - 2ab \cos C$ — Pythagoras plus a correction that vanishes at $C = 90^\circ$.
- **Area:** $[ABC] = \frac{1}{2}ab \sin C = \frac{abc}{4R} = rs$ — one quantity tying together sides, circumradius R , and inradius r .
- **Switch languages freely:** a hard angle chase can melt in one line of the Law of Sines; a stubborn length yields to the Law of Cosines.

Exercises

Warm-up.

1. In a right triangle, the side opposite a 30° angle is 5. Find the hypotenuse.
2. In triangle ABC , $a = 10$, $A = 30^\circ$. Find the circumradius R .
3. Two sides 6 and 9 enclose an angle of 30° . Find the area.
4. A triangle has sides 3, 5, 7. Find the largest angle. (*It is opposite the longest side.*)
5. State the Law of Cosines for the angle A opposite side a .
6. A triangle has $a = 8$, $b = 6$, $c = 10$. What is angle C , and what does that make the triangle?

Practice.

7. In triangle ABC , $A = 45^\circ$, $B = 60^\circ$, $a = 10$. Find b and the circumradius.
8. A river: from a point on one bank you sight a tree on the far bank at 90° to the bank; walking 50 m along the bank, the tree now sits at 35° to the bank. Find the river's width.
9. Sides 7 and 5 enclose a 120° angle. Find the third side.
10. A triangle has sides 13, 14, 15. Find its area (Law of Cosines, then $\frac{1}{2}ab \sin C$),

and then its circumradius via $abc/4R$.

11. Prove the projection formula $a = b \cos C + c \cos B$.
12. In triangle ABC , prove $\frac{a^2 - b^2}{c^2} = \frac{\sin(A - B)}{\sin(A + B)}$.
13. A regular pentagon is inscribed in a circle of radius 1. Use the Law of Cosines (or Sines) to find its side length.

Challenge.

14. Derive **Heron's formula** $[ABC] = \sqrt{s(s-a)(s-b)(s-c)}$ from $[ABC] = \frac{1}{2}ab \sin C$ and the Law of Cosines.
15. Prove the Law of Cosines by dropping an altitude and using Pythagoras on the two pieces (handle the obtuse case too).
16. In triangle ABC , prove $a \cos A + b \cos B + c \cos C = \frac{2[ABC]}{R}$... or find and prove the cleanest identity of this type you can.
17. Prove that $r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$, linking inradius and circumradius.
18. Three circles of radius 1 are mutually tangent. Find the area of the curved region between them (use a 60° sector and an equilateral triangle).

★ OLYMPIAD SPARK

(Olympiad.) In triangle ABC , prove the identity $\tan A + \tan B + \tan C = \tan A \tan B \tan C$, and explain why it requires $A + B + C = 180^\circ$. Then use it to show that an acute triangle with two of its angle-tangents equal must be isosceles.

(Second spark — triangulation in the sky.) Two observers 20 km apart on flat ground both measure the angle of elevation to a high-altitude balloon, lying in the vertical plane through both of them: 52° and 44° . Find the balloon's altitude. (*One application of the Law of Sines, then a sine of the elevation — exactly how early surveyors mapped continents.*)

19

Ceva, Menelaus, and Concurrency

★ WONDER

The three medians of a triangle meet at one point. So do the three altitudes, and the three angle bisectors. Each time, three lines — with no obvious reason to cooperate — pass through a single point. Is this luck repeated three times, or is there one master rule that decides, for *any* three cevians, whether they will meet? And a mirror question: when do three points, one on each side, line up straight?

There *is* a master rule for each question. **Ceva's theorem** settles concurrency of cevians; **Menelaus' theorem** settles collinearity of points on the sides. They are twins — almost identical equations — and together they reduce a huge class of “do these meet / line up?” problems to a single multiplication.

MATHEMATICIAN'S MARGIN

Giovanni Ceva published this criterion in Milan in 1678. Its mirror-twin is far older: Menelaus of Alexandria proved the collinearity version around the year 100, while setting down the geometry of spheres for astronomers. The two results sat sixteen centuries apart before anyone noticed they were a single idea seen from two sides.

19.1 Ceva's Theorem

A **cevian** is a segment from a vertex to a point on the opposite side. Let cevians AD , BE , CF have feet $D \in BC$, $E \in CA$, $F \in AB$.

Theorem 19.1 (Ceva). The cevians AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1.$$

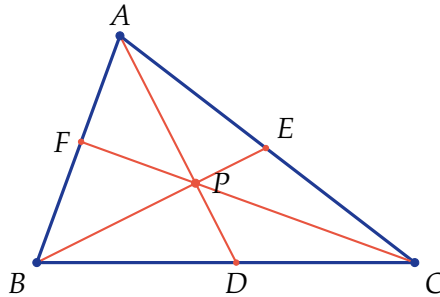


Figure 19.1 — Three cevians meet at P exactly when the product of the three side-ratios equals 1.

 **Worked Example.** Prove the “concurrent $\Rightarrow$ product = 1” direction using areas.

Suppose the cevians meet at P . The cevian AD from A gives, by the area-ratio rule (triangles with the same height are in the ratio of their bases),

$$\frac{BD}{DC} = \frac{[ABD]}{[ACD]} = \frac{[PBD]}{[PCD]}.$$

Two equal fractions subtract cleanly (if $\frac{p}{q} = \frac{r}{s}$ then both equal $\frac{p-r}{q-s}$), so

$$\frac{BD}{DC} = \frac{[ABD] - [PBD]}{[ACD] - [PCD]} = \frac{[ABP]}{[ACP]}.$$

By the same reasoning at the other two vertices, $\frac{CE}{EA} = \frac{[BCP]}{[BAP]}$ and $\frac{AF}{FB} = \frac{[CAP]}{[CBP]}$. Multiply all three:

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = \frac{[ABP]}{[ACP]} \cdot \frac{[BCP]}{[BAP]} \cdot \frac{[CAP]}{[CBP]} = 1,$$

since every area appears once on top and once on the bottom.

The proof is a telescoping cancellation — every triangle’s area cancels its own twin. A statement about three lines *meeting* is settled purely by *areas*, with no angle or length computed. (The converse follows: if the product is 1, the third cevian must pass through the intersection of the first two.)

 **Worked Example.** One theorem, two classic concurrences. Show that the medians are concurrent, and that the angle bisectors are concurrent.

Medians. Each foot is a midpoint, so $\frac{BD}{DC} = \frac{CE}{EA} = \frac{AF}{FB} = 1$, and the product is $1 \cdot 1 \cdot 1 = 1$. By Ceva, concurrent (this point is the centroid).

Angle bisectors. The Angle Bisector Theorem gives $\frac{BD}{DC} = \frac{AB}{AC} = \frac{c}{b}$, and cyclically $\frac{CE}{EA} = \frac{a}{c}$, $\frac{AF}{FB} = \frac{b}{a}$. The product is

$$\frac{c}{b} \cdot \frac{a}{c} \cdot \frac{b}{a} = 1,$$

so by Ceva the bisectors concur (at the incentre).

Watch what happens: two famous facts that each took real work in Part II now fall out of one criterion in a single line apiece — and the same one line would prove the altitudes concurrent too (try it). Ceva turns “prove these three lines meet” into “multiply three fractions.”

19.2 Menelaus' Theorem

Theorem 19.2 (Menelaus). Let a straight line (a *transversal*) meet the lines BC , CA , AB at X , Y , Z respectively. Then

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = 1 \quad (\text{unsigned}),$$

where the transversal meets the three sides in an odd number of *external* points. With signed ratios, the product is -1 — the single sign that distinguishes collinearity from Ceva's concurrency ($+1$).

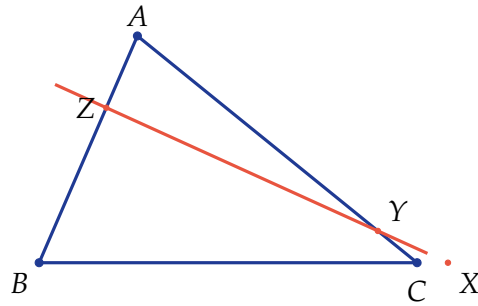


Figure 19.2 — A transversal meets two sides internally (Y, Z) and the third side's extension (X , beyond C).

Worked Example. Prove Menelaus by dropping perpendiculars.

Drop perpendiculars from A, B, C to the transversal line, of lengths p, q, r (these are the distances from the vertices to the line). Where the transversal crosses a side, similar right triangles appear. At X on line BC , triangles formed by the feet from B and C are similar, so

$$\frac{BX}{XC} = \frac{q}{r}.$$

Likewise $\frac{CY}{YA} = \frac{r}{p}$ and $\frac{AZ}{ZB} = \frac{p}{q}$. Multiplying,

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = \frac{q}{r} \cdot \frac{r}{p} \cdot \frac{p}{q} = 1.$$

Again the proof *telescopes* — the three perpendicular distances each appear once up, once down, and cancel. Ceva and Menelaus are the same cancellation wearing two hats; the only real difference is a sign, $+1$ for “meet at a point,” -1 for “lie on a line.”

AHA!

Keep the twins side by side. Same three fractions $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB}$; product $+1$ means the cevians *concur* (Ceva), product -1 (signed) means the points are *collinear* (Menelaus). When a problem asks you to prove a concurrency or a collinearity, reach first for these.

Two theorems, one product of three fractions, divided only by a sign: concur at

a point, or fall along a line. Geometry is fondest of the pairs that are secretly one thing.

→ YOUR TURN

In triangle ABC , cevians AD, BE, CF have $\frac{BD}{DC} = 2$ and $\frac{CE}{EA} = 3$. For the three to be concurrent, what must $\frac{AF}{FB}$ be?

Set-up (you finish it): Ceva requires $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$, so $2 \cdot 3 \cdot \frac{AF}{FB} = 1$.

Answer: $\frac{AF}{FB} = \frac{1}{6}$.

KEY IDEAS & FORMULAS

- **Cevian:** a segment from a vertex to the opposite side.
- **Ceva:** cevians AD, BE, CF concur iff $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$.
- **Menelaus:** points X, Y, Z on the three side-lines are *collinear* iff the same product equals -1 (signed). One sign separates the twins: $+1$ meet, -1 line up.
- **Trig form of Ceva:** concurrency iff $\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} = 1$.
- **Both telescope:** the area proof (Ceva) and the perpendicular-distance proof (Menelaus) each cancel every term once up, once down. Reach for these on any concurrency or collinearity — medians, bisectors, altitudes, Gergonne, Nagel.

Exercises

Warm-up.

1. Write Ceva's product for cevians AD, BE, CF , naming each ratio.
2. Cevians have $\frac{BD}{DC} = 2$ and $\frac{CE}{EA} = \frac{3}{2}$. For concurrency, find $\frac{AF}{FB}$.
3. Why do the three medians satisfy Ceva automatically?
4. State the difference between Ceva ($+1$) and Menelaus (-1).
5. In a triangle, an angle bisector from A meets BC at D with $AB = 6$, $AC = 9$. Find $BD : DC$.

6. A transversal meets AB at its midpoint and AC at its midpoint. Where (in principle) does it meet line BC ?

Practice.

7. Prove that the three altitudes of a triangle are concurrent, using Ceva. (*Express each foot-ratio with cosines or with the right-triangle legs.*)
8. In triangle ABC , D on BC with $BD : DC = 2 : 3$, and E on CA with $CE : EA = 1 : 2$. The cevians AD and BE meet at P ; line CP meets AB at F . Find $AF : FB$.
9. A transversal meets BC at X with $BX : XC = 3 : 1$ (externally), and CA at Y with $CY : YA = 2 : 3$. Find $AZ : ZB$ on AB .
10. Use Menelaus to prove that the midpoints argument: a line parallel to BC cannot meet line BC , and see how the theorem signals this.
11. In triangle ABC , cevians divide the sides with $\frac{BD}{DC} = \frac{CE}{EA} = \frac{AF}{FB} = k$. For which k are they concurrent? For which are D, E, F collinear?
12. Prove **the trigonometric form of Ceva**: cevians AD, BE, CF concur iff

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} = 1.$$

13. Use Ceva to prove the medians of a triangle cut it into six equal-area triangles (combine with Chapter 11).

Challenge.

14. (Gergonne point.) The incircle touches the sides at X, Y, Z . Prove AX, BY, CZ concur. (Use the tangent lengths from Chapter 16 in Ceva.)
15. (Nagel point.) The excircles touch the sides at X', Y', Z' . Prove AX', BY', CZ' concur.
16. Prove **Desargues' theorem** in the plane: if two triangles are perspective from a point, they are perspective from a line. (Three applications of Menelaus.)
17. In triangle ABC , points D, E, F on the sides make AD, BE, CF concurrent at P . Prove $\frac{PD}{AD} + \frac{PE}{BE} + \frac{PF}{CF} = 1$.
18. Prove that the three lines joining the midpoints of opposite sides of a hexagon inscribed in a circle relate via Carnot's or a Menelaus chain — formulate and prove a clean concurrency you can reach.

★ OLYMPIAD SPARK

(Olympiad.) Let the incircle of triangle ABC touch BC at X . Let X' be the antipode of X on the incircle. Prove that A , X' , and the point where the A -excircle touches BC are collinear. (*Menelaus or a careful homothety; the tangent lengths set up the ratios.*)

(Second spark.) Three cevians of a triangle divide it into six small triangles and a central one (when not concurrent, they bound a central triangle). Using Menelaus three times, find the area of the central triangle when each side is cut in ratio $1 : 2$ — recovering the “one-seventh triangle” of Chapter 11 by a completely different route.

20

Stewart, Apollonius, and Lengths

★ WONDER

You know all three sides of a triangle. Now someone draws a line from a corner to a marked spot on the far side — a median, an angle bisector, any cevian at all — and asks: how long is it? You have no protractor and no coordinates, only the three side lengths and where the cevian lands. Is that enough to pin down its length *exactly*? It is, and one elegant formula does it every time.

Part IV so far has chased angles and used trigonometry; this chapter is about *lengths* — specifically the length of a cevian. **Stewart's theorem** is the master formula, and from it the lengths of medians and angle bisectors drop out as special cases. The proof is a small miracle of cancellation.

MATHEMATICIAN'S MARGIN

The relation is named for Matthew Stewart, a young Scottish mathematician who published it in 1746 — though the special case for medians was already known to Apollonius of Perga some nineteen centuries earlier. Old theorems rarely have a single owner; they have a lineage.

20.1 Stewart's Theorem

Theorem 20.1 (Stewart). Let D lie on side BC of triangle ABC , with $BD = m$, $DC = n$, and cevian $AD = d$. Writing $a = BC = m + n$, $b = CA$,

$$c = AB,$$

$$b^2m + c^2n = a(d^2 + mn).$$

A memorable way to say it: *a man and his dad put a bomb in the sink* — $man + dad = bmb + cnc$, that is, $amn + ad^2 = b^2m + c^2n$.

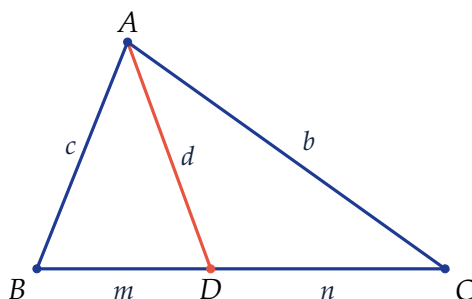


Figure 20.1 — Stewart's setup: cevian $d = AD$ cuts BC into $m = BD$ and $n = DC$, with $a = m + n$.

 **Worked Example.** Prove Stewart's theorem.

Let $\angle ADB = \theta$; then $\angle ADC = 180^\circ - \theta$, and $\cos \angle ADC = -\cos \theta$. Apply the Law of Cosines in each sub-triangle:

$$c^2 = d^2 + m^2 - 2dm \cos \theta, \quad b^2 = d^2 + n^2 + 2dn \cos \theta.$$

Multiply the first by n and the second by m :

$$c^2n = d^2n + m^2n - 2dmn \cos \theta, \quad b^2m = d^2m + n^2m + 2dmn \cos \theta.$$

Add them — the $\cos \theta$ terms are equal and opposite, so they vanish:

$$b^2m + c^2n = d^2(m + n) + mn(m + n) = (m + n)(d^2 + mn) = a(d^2 + mn).$$

The one unknown we could not control — the angle θ at D — *cancels itself out*, because the cevian makes supplementary angles on its two sides and their cosines are negatives. What is left is a pure relation among lengths. That cancellation is the whole reason a cevian's length is determined by the sides alone.

20.2 Medians and angle bisectors for free

 **Worked Example. Apollonius / the median length.** Show that the median from A to the midpoint of BC has length m_a with

$$m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}.$$

For a median, D is the midpoint, so $m = n = \frac{a}{2}$. Stewart gives

$$b^2 \cdot \frac{a}{2} + c^2 \cdot \frac{a}{2} = a \left(m_a^2 + \frac{a^2}{4} \right).$$

Divide by a : $\frac{1}{2}(b^2 + c^2) = m_a^2 + \frac{a^2}{4}$, so $m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}$.

Summing this over all three medians, the cross-terms combine to give $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$ — the three medians “carry” exactly three-quarters of the triangle’s total squared side-length, no matter its shape.

 **Worked Example. The angle-bisector length.** Show that the bisector from A has length t_a with $t_a^2 = bc - BD \cdot DC$, hence $t_a^2 =$

$$bc \left[1 - \left(\frac{a}{b+c} \right)^2 \right].$$

By the Angle Bisector Theorem, $\frac{BD}{DC} = \frac{c}{b}$, so $BD = \frac{ac}{b+c}$ and $DC = \frac{ab}{b+c}$. Then

$$b^2 BD + c^2 DC = \frac{a}{b+c} (b^2 c + c^2 b) = \frac{a bc(b+c)}{b+c} = abc.$$

Stewart says $b^2 BD + c^2 DC = a(t_a^2 + BD \cdot DC)$, so $abc = a(t_a^2 + BD \cdot DC)$, giving

$$t_a^2 = bc - BD \cdot DC = bc - \frac{a^2 bc}{(b+c)^2} = bc \left[1 - \left(\frac{a}{b+c} \right)^2 \right].$$

Notice: the bisector length is the geometric-mean product bc of the two enclosing sides, docked by a single correction $BD \cdot DC$. Notice it is always

less than $\sqrt{bc}$ — a bisector can never reach the length its two flanking sides would suggest.

 **Worked Example.** Compute the median to the longest side of the 13–14–15 triangle.

Take $a = 14$ (the side receiving the median), $b = 13$, $c = 15$. Then

$$m_a^2 = \frac{2(13^2) + 2(15^2) - 14^2}{4} = \frac{338 + 450 - 196}{4} = \frac{592}{4} = 148,$$

so $m_a = \sqrt{148} = 2\sqrt{37} \approx 12.17$.

The wow: three side lengths in, one exact cevian length out — $2\sqrt{37}$, with no diagram measured and no angle ever found. This is the computational power Part IV has been building toward.

AHA!

Stewart's theorem is the length-counterpart of the area and angle tools: feed it the sides and the split point, and it returns the cevian. Medians, bisectors, and the segments cut by a cevian are all one substitution away. Whenever a problem hands you "a cevian of length...", Stewart is your first move.

Three side lengths and a landing point go in; an exact cevian comes out, measured without a protractor or a picture. The sides, it seems, were keeping the answer all along.

→ YOUR TURN

Find the length of the median to the side of length 8 in a triangle whose other two sides are both 5.

Set-up (you finish it): the median to side a is $m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}$, here with $a = 8$, $b = c = 5$.

Answer: $m_a^2 = \frac{2(25) + 2(25) - 64}{4} = \frac{36}{4} = 9$, so $m_a = 3$.

KEY IDEAS & FORMULAS

- **Stewart:** $b^2m + c^2n = a(d^2 + mn)$ for a cevian d splitting a into m, n (“ $man + dad = bmb + cnc$ ”).
- **Median (Apollonius):** $m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}$; the three obey $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$.
- **Angle bisector:** $t_a^2 = bc - BD \cdot DC = bc \left[1 - \left(\frac{a}{b+c} \right)^2 \right]$ — always less than $\sqrt{bc}$.
- **The mechanism:** the angle at the foot makes supplementary angles on the two sides, so its cosine cancels — a cevian’s length is fixed by the sides and the split point alone.
- Whenever a problem hands you “a cevian of length...,” Stewart is the first move.

Exercises**Warm-up.**

1. State Stewart’s theorem and label each letter on a sketch.
2. A triangle has $b = 7, c = 8$, and a cevian to BC with $m = 2, n = 3$. Find d .
3. Find the median to the side of length 10 in a triangle with the other two sides 6 and 8.
4. In a triangle with $b = 5, c = 9$, the bisector from A meets BC . Find $BD : DC$.
5. For an equilateral triangle of side s , find the length of any median two ways (Stewart and Pythagoras).
6. Use $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$ to find the sum of squared medians of a 3–4–5 triangle.

Practice.

7. In the 13–14–15 triangle, find all three medians.
8. Find the length of the bisector from A in a triangle with $a = 6, b = 5, c = 7$.
9. A cevian of length 4 goes from A to D on BC , with $BD = 3, DC = 5$, and $AB = 6$. Find AC .
10. Prove that in any triangle the median to a side is shorter than the average of the other two sides, $m_a < \frac{1}{2}(b + c)$.

11. Show that if two medians of a triangle are equal, the triangle is isosceles (use Apollonius).
12. A triangle has sides a, b, c . Express the distance from a vertex to the centroid in terms of the medians, and then in terms of the sides.
13. Prove that the length of the cevian from A to the point dividing BC in ratio $k : 1$ is $\sqrt{\frac{b^2k + c^2}{k + 1} - \frac{ka^2}{(k + 1)^2}}$.

Challenge.

14. Prove the parallelogram law: the sum of the squares of the four sides of a parallelogram equals the sum of the squares of its two diagonals. (Apollonius on a diagonal.)
15. A triangle has integer sides and an integer-length median to one side. Find a family of such triangles.
16. Prove that the three medians of a triangle can themselves be assembled into a triangle, and that its area is $\frac{3}{4}$ of the original.
17. Use Stewart to derive the distance from a vertex to the point where the incircle touches the opposite side, then to the incentre.
18. Given the three medians of a triangle, reconstruct the side lengths. (Invert the Apollonius relations.)

★ OLYMPIAD SPARK

(Olympiad.) In triangle ABC , the cevian AD is both a median and makes $AD = \frac{1}{2}BC$. Prove that $\angle BAC = 90^\circ$. (*Apollonius plus a clean algebraic step — and notice the converse of Thales lurking.*)

(Second spark.) Point P is inside triangle ABC . Prove the **Leibniz relation** $PA^2 + PB^2 + PC^2 = GA^2 + GB^2 + GC^2 + 3PG^2$, where G is the centroid, and deduce that the sum $PA^2 + PB^2 + PC^2$ is least exactly when $P = G$. (*Vectors with G at the origin, or Stewart/Apollonius applied cleverly.*)

21

Coordinates: Geometry by Algebra

★ WONDER

The story goes that René Descartes, lying ill in bed, watched a fly wander across the ceiling and realised he could pin down its position at every instant by just *two numbers*: its distance from each of two walls. In that moment geometry and algebra became one subject. If every point is a pair of numbers, then every line is an equation and every theorem is a calculation. What does geometry feel like when you can simply *compute* the answer?

Coordinates turn pictures into algebra. A clever choice of axes can reduce a theorem that resisted every synthetic trick to a few lines of routine computation — the technique fondly known as *bashing*. It is not always elegant, but it is reliable, and a strong solver keeps it ready as a fallback that never fails.

21.1 The coordinate toolkit

Place the plane on axes. For points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \quad \text{midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

The point dividing PQ in ratio $m : n$ (from P) is $\left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$.

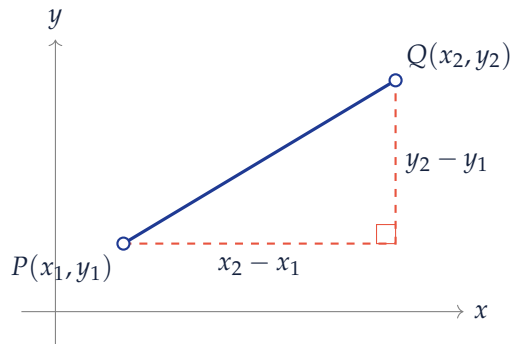


Figure 21.1 — The distance formula is just Pythagoras on the dashed legs $x_2 - x_1$ and $y_2 - y_1$.

For **lines**, the slope between P and Q is $m = \frac{y_2 - y_1}{x_2 - x_1}$; parallel lines share a slope, and **perpendicular** lines have slopes multiplying to -1 . A **circle** of centre (h, k) , radius r has equation $(x - h)^2 + (y - k)^2 = r^2$.

AHA!

The art of bashing is the choice of axes. You are free to place the origin and axes anywhere. Spend that freedom wisely: put a right angle at the origin, a key side along the x -axis, a centre of symmetry at the origin. A good placement fills your figure with zeros and turns pages of algebra into a few clean lines.

21.2 Bashing in action

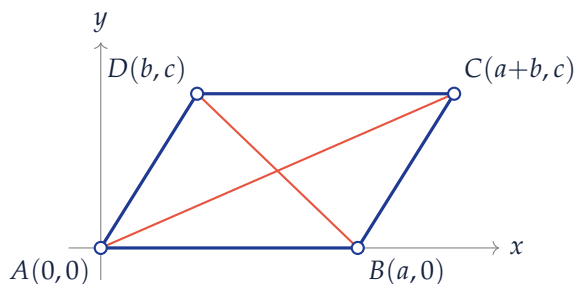


Figure 21.2 — A parallelogram placed to make the algebra vanish: A at the origin, AB along the x -axis.

 **Worked Example.** Prove that the diagonals of a parallelogram bisect each other.

Place it as in Figure 21.2: $A = (0,0)$, $B = (a,0)$, and since $ABCD$ is a parallelogram, $D = (b,c)$ and $C = B + D - A = (a+b,c)$. The midpoint of diagonal AC is

$$\left(\frac{a+b}{2}, \frac{c}{2}\right),$$

and the midpoint of diagonal BD is $\left(\frac{a+b}{2}, \frac{c}{2}\right)$ — *the same point*. So the diagonals share a midpoint: each bisects the other.

A synthetic proof needs congruent triangles and several steps; in coordinates the two midpoints are simply computed and seen to be identical. The clever placement (origin at A , side on the axis) did all the work — the proof is two midpoint formulas.

 **Worked Example.** Show the three medians of a triangle are concurrent, and find the meeting point.

Let $A = (x_1, y_1)$, $B = (x_2, y_2)$, $C = (x_3, y_3)$. The midpoint of BC is $M_a = \left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right)$. The point two-thirds of the way from A to M_a is

$$A + \frac{2}{3}(M_a - A) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

This expression is *symmetric* in the three vertices — it does not care which vertex we started from. So the same point lies two-thirds along *every* median. The three medians therefore meet, at the centroid $G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$.

The elegance: the centroid is simply the *average* of the three vertices. The hard-won concurrency of Chapter 10 becomes obvious the instant you notice the formula is symmetric — the algebra cannot tell the vertices apart, so neither can the meeting point.

 **Worked Example. The Apollonius circle.** A point P moves so that its distances to two fixed points are in a constant ratio $PA : PB = k$ (with $k \neq 1$). Prove the locus is a circle.

Place $A = (0,0)$ and $B = (d,0)$, and let $P = (x,y)$. The condition

$PA^2 = k^2 PB^2$ reads

$$x^2 + y^2 = k^2((x - d)^2 + y^2).$$

Expand and gather: $(1 - k^2)x^2 + (1 - k^2)y^2 + 2k^2dx - k^2d^2 = 0$. Since $k \neq 1$ we may divide by $1 - k^2$, leaving

$$x^2 + y^2 + (\text{linear in } x) + \text{const} = 0,$$

which is exactly the equation of a circle. (When $k = 1$ the x^2, y^2 terms cancel and the locus degenerates to the perpendicular bisector — a “circle of infinite radius.”)

The wow: a purely *metric* condition — “twice as far from here as from there” — secretly traces a perfect circle, the *Apollonius circle*. You would never guess it from the wording, but the algebra makes it inevitable: a single $x^2 + y^2$ with matching coefficients *is* a circle.

✗ COMMON TRAP

Bashing can bury you in algebra if you place the axes carelessly. Before computing, invest in the setup: exploit every right angle, midpoint, and symmetry to seed zeros. And once you have an answer, sanity-check it against the picture — a sign slip in coordinates produces a confidently wrong number with no warning bell.

Descartes’ fly gave us a world where every point is a pair of numbers and every theorem a computation you can simply grind out. Sometimes the bravest move in geometry is to stop being clever and start counting.

→ YOUR TURN

Find the distance between $P = (1, 2)$ and $Q = (4, 6)$, and the midpoint of PQ .

Set-up (you finish it): $PQ = \sqrt{(4 - 1)^2 + (6 - 2)^2}$, and the midpoint is $(\frac{1+4}{2}, \frac{2+6}{2})$.

Answer: $PQ = \sqrt{9 + 16} = \sqrt{25} = 5$; midpoint = $(\frac{5}{2}, 4)$.

KEY IDEAS & FORMULAS

- **Distance** $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$; **midpoint** $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$; **section** (ratio $m : n$) $\left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n}\right)$.
- **Slopes**: parallel lines share a slope; perpendicular slopes multiply to -1 .
- **Circle**: $(x - h)^2 + (y - k)^2 = r^2$; complete the square to read off centre and radius.
- **Centroid** = the average of the three vertices, $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$.
- **Bashing strategy**: choose axes to seed zeros — right angle at the origin, a side on the x -axis, symmetry at the origin — then compute and sanity-check against the figure.

Exercises**Warm-up.**

1. Find the distance between $(2, 3)$ and $(7, 15)$.
2. Find the midpoint of the segment from $(-1, 4)$ to $(5, -2)$.
3. Find the slope of the line through $(1, 2)$ and $(4, 8)$, and the slope of any line perpendicular to it.
4. Write the equation of the circle with centre $(3, -1)$ and radius 5.
5. Find the centroid of the triangle with vertices $(0, 0)$, $(6, 0)$, $(3, 9)$.
6. Where does the point dividing $(0, 0)$ to $(10, 5)$ in ratio $2 : 3$ land?

Practice.

7. Prove that the triangle with vertices $(0, 0)$, $(4, 0)$, $(0, 3)$ is right-angled, two ways (slopes; Pythagoras).
8. Find the circumcentre of the triangle $(0, 0)$, $(6, 0)$, $(0, 8)$ and its circumradius.
9. Prove that the midpoints of the sides of any quadrilateral form a parallelogram (Varignon), using coordinates.
10. Show the points $(1, 2)$, $(3, 6)$, $(5, 10)$ are collinear (slopes or shoelace).
11. A circle has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. Find its centre and radius.
12. Find the foot of the perpendicular from $(4, 5)$ to the line $y = 2x$.
13. Prove the **British Flag** theorem with coordinates: for P and rectangle $ABCD$, $PA^2 + PC^2 = PB^2 + PD^2$.

Challenge.

14. Prove that the perpendicular bisectors of the three sides of a triangle are concurrent, by coordinates (locate the circumcentre).
15. Place a triangle with $B = (-a, 0)$, $C = (a, 0)$, $A = (p, q)$. Find the orthocentre and verify O, G, H are collinear (rediscover the Euler line by algebra).
16. Find the locus of points the sum of whose squared distances to two fixed points is constant.
17. Three vertices of a square are $(0, 0)$, $(4, 1)$, and one more; find all possible fourth vertices.
18. Prove that the medians of a triangle, taken as vectors from the centroid, sum to zero.

★ OLYMPIAD SPARK

(Olympiad.) Let $A = (0, 0)$ and $B = (1, 0)$. A point P satisfies $PA^2 + PB^2 = 3 AB^2$. Describe the locus of P exactly, and find the area it encloses. Then generalise: for what constant λ is the locus $PA^2 + PB^2 = \lambda AB^2$ a single point, and where?

(Second spark — a coordinate proof worth savouring.) Prove that in any triangle, the foot of the altitude from A , the midpoint of BC , and the two points where the circle on BC as diameter meets AB, AC are all concyclic. Set $B = (-1, 0)$, $C = (1, 0)$, $A = (p, q)$ and let the algebra reveal the circle.

22

Vectors & Complex Numbers

★ WONDER

Coordinates pin a point with two numbers, but they break the symmetry of a problem — you must choose which way is “x.” Vectors keep the freedom of direction without the bookkeeping. And the plane has a second, almost magical, arithmetic: treat each point as a single *complex number*, and suddenly *multiplication* performs a rotation. A whole category of problems, the ones built on turning, is about to fall open in your hands.

This chapter offers two more algebraic engines. **Vectors** handle midpoints, centroids, and perpendicularity with effortless symmetry. **Complex numbers** turn the plane into a number system in which translation is addition and rotation is multiplication — the cleanest possible language for any problem about turning.

22.1 Vectors

Write $\vec{a}, \vec{b}, \vec{c}$ for the position vectors of points A, B, C from a chosen origin. Then the midpoint of AB is $\frac{1}{2}(\vec{a} + \vec{b})$, the point dividing AB in ratio $m : n$ is $\frac{n\vec{a} + m\vec{b}}{m+n}$, and — with delightful symmetry — the centroid of $\triangle ABC$ is $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$.

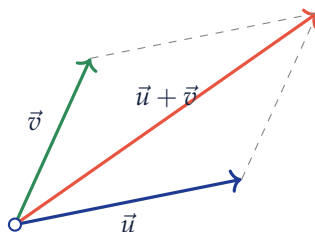


Figure 22.1 — Vector addition: $\vec{u} + \vec{v}$ is the diagonal of the parallelogram on $\vec{u}$ and $\vec{v}$.

The **dot product** $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$ measures both length ($\vec{u} \cdot \vec{u} = |\vec{u}|^2$) and angle: crucially, $\vec{u} \perp \vec{v}$ exactly when $\vec{u} \cdot \vec{v} = 0$.

 **Worked Example.** Prove that the diagonals of a rhombus are perpendicular.

Let two adjacent sides from one vertex be the vectors $\vec{u}$ and $\vec{v}$. A rhombus has all sides equal, so $|\vec{u}| = |\vec{v}|$. Its diagonals are $\vec{u} + \vec{v}$ and $\vec{v} - \vec{u}$. Their dot product is

$$(\vec{u} + \vec{v}) \cdot (\vec{v} - \vec{u}) = \vec{v} \cdot \vec{v} - \vec{u} \cdot \vec{u} = |\vec{v}|^2 - |\vec{u}|^2 = 0.$$

A zero dot product means the diagonals are perpendicular.

The entire proof is one line of algebra, and it lays bare *why* the result is true — the diagonals are $\vec{u} \pm \vec{v}$, and a sum-times-difference is a difference of squares, which vanishes precisely because the sides are equal. The dot product converts “perpendicular” into “equals zero.”

22.2 Complex numbers: the plane that multiplies

Identify the point (x, y) with the complex number $z = x + iy$. Then:

- **Addition** $z \mapsto z + w$ is *translation* by w .
- **Multiplication** by $e^{i\theta}$ is *rotation* by θ about the origin (and multiplying by a real r scales by r). In particular, multiplying by i rotates 90° counter-clockwise.
- **Rotation about a point** p by angle θ is $z \mapsto p + e^{i\theta}(z - p)$.

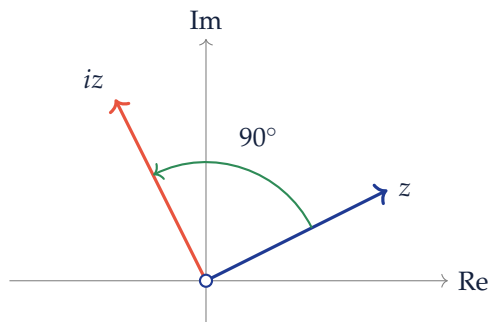


Figure 22.2 — Multiplying by i rotates z by 90° about the origin. Multiplication *is* rotation.

 **Worked Example.** Square $ABCD$ is labelled counter-clockwise, and you know A and B as complex numbers. Find C .

The side BC is the side AB turned 90° counter-clockwise at B . As a vector, $AB = b - a$, and turning it left is multiplying by i , so $C - B = i(b - a)$. Hence

$$C = b + i(b - a).$$

(Then $D = a + i(b - a)$ similarly, or $D = A + (C - B)$.)

The delight: constructing a perpendicular of equal length — the heart of “erect a square” — is just *multiplying by i* . Every rotation problem that would need careful trigonometry in coordinates becomes a single complex multiplication.

 **Worked Example.** Prove that a triangle with vertices a, b, c (as complex numbers) is equilateral if and only if

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

Let $\zeta = e^{2\pi i/3}$, a primitive cube root of unity, so $\zeta^3 = 1$ and $1 + \zeta + \zeta^2 = 0$. A triangle is equilateral exactly when one vertex is carried to the next by a 120° rotation about the centroid — algebraically, when $a + \zeta b + \zeta^2 c = 0$ (one orientation) or $a + \zeta^2 b + \zeta c = 0$ (the other). Now multiply those two expressions:

$$(a + \zeta b + \zeta^2 c)(a + \zeta^2 b + \zeta c) = a^2 + b^2 + c^2 - ab - bc - ca,$$

using $1 + \zeta + \zeta^2 = 0$ to collapse the cross-terms. The product is zero iff one factor is zero iff the triangle is equilateral. Hence equilateral $\iff a^2 + b^2 + c^2 = ab + bc + ca$.

The wow: “equilateral” — a statement about equal lengths and 60° angles — turns out to be a single *symmetric polynomial* equation in the vertices. The cube roots of unity, the natural “thirds of a full turn,” are exactly the right tool, and they make the condition fall out in two lines.

AHA!

Three algebraic languages now stand beside synthetic geometry: *coordinates* (best for distances and circles), *vectors* (best for midpoints, centroids, and perpendicularity), and *complex numbers* (best for anything involving rotation or equilateral/square configurations). Part of mastery is sensing, at a glance, which language will make a given problem dissolve.

Three algebraic tongues now answer to you, and a rotation that once meant a page of trigonometry has shrunk to a single multiplication. Choosing the right language is already half of solving the problem.

→ YOUR TURN

Find the image of the complex number $z = 4 + i$ under a 90° counter-clockwise rotation about the point $p = 1 + i$.

Set-up (you finish it): rotation about p by 90° is $z \mapsto p + i(z - p)$, so the image is $(1 + i) + i((4 + i) - (1 + i))$.

Answer: $(1 + i) + i(3) = (1 + i) + 3i = 1 + 4i$.

KEY IDEAS & FORMULAS

- **Vectors:** midpoint $\frac{1}{2}(\vec{a} + \vec{b})$, centroid $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$; dot product $\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos \theta$, with $\vec{u} \perp \vec{v} \iff \vec{u} \cdot \vec{v} = 0$.
- **Complex numbers:** a point is $z = x + iy$; *addition* is translation, *multiplication by $e^{i\theta}$* is rotation by θ (by i , a 90° turn).
- **Rotation about p :** $z \mapsto p + e^{i\theta}(z - p)$.
- **Equilateral test:** a, b, c form an equilateral triangle iff $a^2 + b^2 + c^2 = ab + bc + ca$ (via the cube roots of unity).
- **Pick the language:** coordinates for distances and circles, vectors for midpoints and perpendicularity, complex numbers for anything that turns.

Exercises

Warm-up.

1. Given $\vec{a} = (1, 2)$ and $\vec{b} = (3, -1)$, find $\vec{a} + \vec{b}$, $\vec{a} \cdot \vec{b}$, and $|\vec{a}|$.
2. Are $(2, 1)$ and $(-1, 2)$ perpendicular as vectors? Check with the dot product.

3. Write the centroid of A, B, C as a vector expression.
4. What does multiplying a complex number by i do geometrically? By -1 ? By $2i$?
5. If $A = 1 + i$ and $B = 4 + 2i$ are adjacent vertices of a counter-clockwise square, find C .
6. State the complex-number rule for rotating z about a point p by angle θ .

Practice.

7. Use vectors to prove the midpoint of the hypotenuse of a right triangle is equidistant from all three vertices.
8. Prove $\vec{u} \cdot \vec{v} = \frac{1}{4}(|\vec{u} + \vec{v}|^2 - |\vec{u} - \vec{v}|^2)$ (the polarisation identity).
9. With the circumcentre at the origin, show the orthocentre is $\vec{a} + \vec{b} + \vec{c}$ (recall Chapter 10) and re-derive the Euler line.
10. Using complex numbers, find the image of $z = 3 + i$ under a 60° rotation about $1 + i$.
11. Prove that the diagonals of a square are equal and perpendicular, using complex numbers.
12. Points a, b, c form an equilateral triangle. Verify the identity $a^2 + b^2 + c^2 = ab + bc + ca$ for $a = 0, b = 1, c = \frac{1}{2} + \frac{\sqrt{3}}{2}i$.
13. Prove with vectors that the segment joining the midpoints of two sides of a triangle is parallel to, and half, the third side.

Challenge.

14. (Napoleon's theorem.) Equilateral triangles are erected outward on the three sides of an arbitrary triangle. Prove that their three centres form an equilateral triangle. (Complex numbers with $\zeta = e^{2\pi i/3}$.)
15. Prove that the centroid of the outer Napoleon triangle coincides with the centroid of the original triangle.
16. Let a, b, c, d be complex numbers on the unit circle ($|z| = 1$). Prove that $a + b + c + d = 0$ if and only if the four points form a rectangle.
17. Using vectors, prove that $\vec{GA} + \vec{GB} + \vec{GC} = \vec{0}$ for the centroid G of $\triangle ABC$, and use it to show that the point P minimising $PA^2 + PB^2 + PC^2$ is exactly G .
18. Prove that $z \mapsto \frac{az + b}{cz + d}$ (a Möbius map) sends circles-and-lines to circles-and-lines (investigate the special cases first).

★ OLYMPIAD SPARK

(Olympiad — the rotation that solves it.) Equilateral triangles ABF and ACG are erected outward on sides AB, AC of triangle ABC . Prove that $BG = CF$, and that the angle between lines BG and CF is 60° . (*A single 60° rotation about A carries $F \rightarrow B$ and $C \rightarrow G$; everything follows.*)

(Second spark.) Let z_1, z_2, z_3 be the cube roots of a complex number w . Prove they form an equilateral triangle centred at the origin, and conversely that every equilateral triangle centred at the origin arises this way. (*Roots of $z^3 = w$ and the identity above.*)

23

Transformations

★ WONDER

A cattle-hand at point A must ride to the straight river, fill a bucket anywhere along the bank, then carry it to the barn at point B on the same side. Where should she meet the river so the total ride is as short as possible? Trying points one by one is hopeless. But a single *reflection* turns the bent path into a straight one — and a straight line is the shortest thing there is. How can moving the whole picture solve a problem the picture refused to give up?

A **transformation** moves the plane — sliding, turning, flipping, or scaling it. The deep idea of this chapter is that the right motion can make a hard problem easy, because it converts the configuration into one you already understand. Reflections straighten paths; rotations create equilateral triangles out of thin air; scalings (homotheties) explain why certain figures are automatically similar.

23.1 Isometries: motions that preserve distance

A motion that preserves all distances is an **isometry**. There are exactly four kinds in the plane: *translation* (slide), *rotation* (turn about a centre), *reflection* (flip across a line), and *glide reflection*. Each preserves lengths and angles, so it carries any figure to a congruent copy — which is why an isometry never changes what we are trying to prove, only how easy it is to see.

Reflection straightens a path

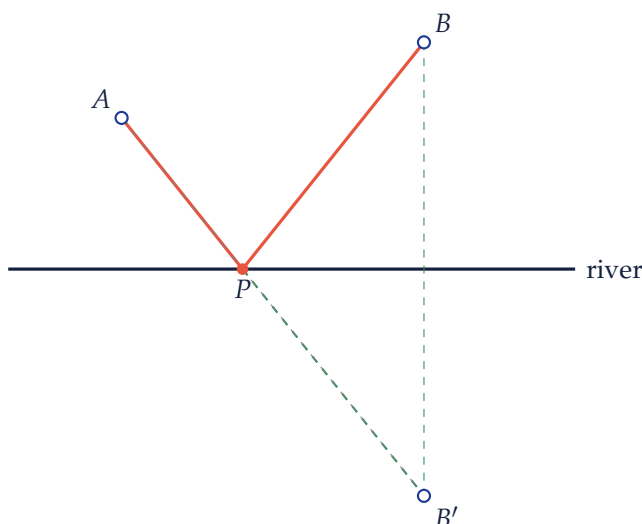


Figure 23.1 — Reflect B across the river to B' . The shortest bent path $A-P-B$ corresponds to the straight segment $A-P-B'$, so P is where line AB' crosses the river.

Worked Example. Solve the Wonder: find the point P on the line minimising $AP + PB$, with A, B on the same side.

Reflect B across the line to its mirror image B' . For *any* point P on the line, $PB = PB'$ (the line is the perpendicular bisector of BB'), so

$$AP + PB = AP + PB'.$$

The right-hand side is the length of a path from A to the fixed point B' via P ; it is smallest when A, P, B' are collinear — a straight line. So the optimal P is exactly where segment AB' crosses the line.

The reflection converts “minimise a bent path” into “a straight line is shortest,” which needs no calculus at all. The same trick proves the law of reflection in optics (angle in = angle out) and solves billiard-ball bank shots — light and bouncing balls both take the reflected straight line.

23.2 Rotation conjures triangles

A rotation about a point preserves distances and turns the whole plane by a fixed angle. Rotating by 60° is special: it manufactures equilateral triangles, because a segment and its image meet at 60° and have equal length.

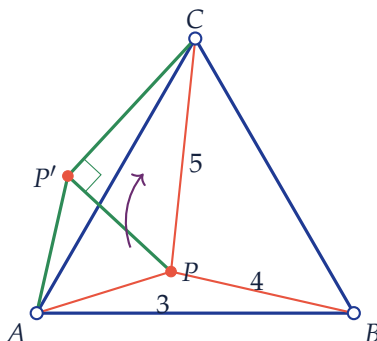


Figure 23.2 — Rotating P by 60° about A (so $B \mapsto C$) creates P' with $PP' = AP = 3$, $P'C = PB = 4$, and $PC = 5$: a 3–4–5 right angle at P' .

Worked Example. A point P inside an equilateral triangle ABC has $PA = 3$, $PB = 4$, $PC = 5$. Find the side length.

Rotate the whole figure by 60° about A , in the direction that sends B to C . Let P map to P' . Then:

- $AP' = AP = 3$ and $\angle PAP' = 60^\circ$, so triangle APP' is equilateral and $PP' = 3$;
- the rotation sends $B \mapsto C$ and $P \mapsto P'$, so $P'C = PB = 4$;
- $PC = 5$ is unchanged.

Now look at triangle $PP'C$: its sides are 3, 4, 5, so by the converse of Pythagoras it has a right angle at P' . Therefore $\angle AP'C = \angle AP'P + \angle PP'C = 60^\circ + 90^\circ = 150^\circ$. Finally, the Law of Cosines in triangle $AP'C$ (with $AP' = 3$, $P'C = 4$) gives the side:

$$AC^2 = 3^2 + 4^2 - 2 \cdot 3 \cdot 4 \cos 150^\circ = 25 + 12\sqrt{3},$$

so $AC = \sqrt{25 + 12\sqrt{3}} \approx 6.77$. *The wow:* the numbers 3, 4, 5 looked unrelated until a 60° rotation *folded them into a single right triangle*. The rotation created exactly the equilateral piece ($PP' = 3$) needed to assemble a 3–4–5. This is the rotation promised back in Chapter 9 — now delivered.

23.3 Homothety: scaling from a centre

A **homothety** with centre O and ratio k sends each point X to the point X' on line OX with $OX' = k \cdot OX$ (on the far side if $k > 0$, the opposite side if $k < 0$). It is not an isometry — it changes size — but it preserves *shape*: every figure

maps to a *similar* figure, and every line to a parallel line.

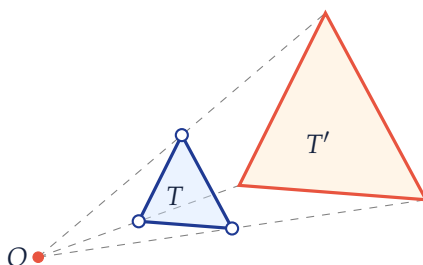


Figure 23.3 — A homothety from centre O with ratio 2: triangle T maps to the twice-as-large parallel triangle T' , with O, X, X' collinear for every point.

Worked Example. Show the medial triangle (joining the midpoints of the sides) is similar to the original, with ratio $\frac{1}{2}$, and explain why it is “upside-down.”

Let G be the centroid. The homothety centred at G with ratio $-\frac{1}{2}$ sends each vertex to the midpoint of the opposite side: indeed G divides each median so that the vertex at distance $\frac{2}{3}$ of the median maps to the point at distance $\frac{1}{3}$ on the *other* side of G — precisely the opposite midpoint. A homothety multiplies all lengths by $|k| = \frac{1}{2}$ and preserves shape, so the medial triangle is similar to $\triangle ABC$ with ratio $\frac{1}{2}$; the negative sign means it is rotated 180° — pointing the opposite way.

The medial triangle isn't just *accidentally* half-size — it *is* the original triangle, shrunk by $\frac{1}{2}$ and flipped through the centroid. One homothety explains the size, the parallel sides, and the inversion all at once. (This same map, iterated and combined, is the key to the nine-point circle in Part V.)

AHA!

The strategic question a transformation answers is: “*what motion makes this configuration familiar?*” Reflect to straighten a path or exploit a mirror symmetry; rotate by the angle hidden in the problem (60° for equilateral, 90° for squares); scale by homothety to turn one circle or triangle into another. The transformation does not change the truth — it changes your *view* until the truth is obvious.

Slide it, turn it, flip it, shrink it: the figure never changes what it is, only how willing it becomes to confess.

→ YOUR TURN

Towns A and B stand 3 and 5 units above a straight road, with a horizontal separation of 8. Find the length of the shortest path from A to the road and on to B .

Set-up (you finish it): reflect B across the road to B' (now 5 units below); the shortest path equals the straight distance AB' , whose horizontal leg is 8 and vertical leg is $3 + 5$.

Answer: $AB' = \sqrt{8^2 + 8^2} = \sqrt{128} = 8\sqrt{2} \approx 11.31$.

KEY IDEAS & FORMULAS

- **Four isometries:** translation, rotation, reflection, glide reflection — each preserves distances, so it never changes what you are proving, only how clearly you see it.
- **Reflection straightens paths** (Heron's shortest-path, optics, billiards): a bent path becomes a straight segment to the mirror image.
- **Rotation conjures figures:** a 60° turn manufactures equilateral triangles; a 90° turn, squares.
- **Homothety** (centre O , ratio k): scales lengths by $|k|$, sends every line to a parallel line and every figure to a similar one.
- **Composing reflections:** two parallel mirrors give a translation (twice their gap); two intersecting mirrors give a rotation (twice their angle).

Exercises

Warm-up.

1. Name the four kinds of isometry of the plane.
2. Reflect the point $(3, 2)$ across the x -axis; across the line $y = x$.
3. A rotation by 90° about the origin sends $(2, 1)$ to which point?
4. A homothety with centre O and ratio 3 sends a segment of length 5 to a segment of what length?
5. Why does a homothety send every line to a parallel line?
6. In the reflection (Heron) problem, what is reflected, and why does $PB = PB'$?

Practice.

7. Towns A and B lie 3 and 5 units above a straight road, with horizontal

- separation 10. Find the shortest path $A \rightarrow \text{road} \rightarrow B$.
8. A billiard ball at A must bounce off one cushion and reach B . Construct the aim point using a reflection.
 9. The composition of two reflections in parallel lines is a translation. By how much, in terms of the distance between the lines?
 10. The composition of two reflections in lines meeting at angle θ is a rotation. By what angle?
 11. Use a 60° rotation to prove that if equilateral triangles are erected outward on AB and AC , then $BG = CF$ (the segments of the previous chapter's spark).
 12. A homothety centre O ratio 2 maps circle ω to circle ω' . Relate their radii and the distance between centres.
 13. Two circles are tangent externally at T . Prove that T is the centre of a homothety (negative ratio) taking one circle to the other.

Challenge.

14. (Fermat point.) For a triangle with all angles below 120° , prove that the point P minimising $PA + PB + PC$ sees all three sides at 120° . (Rotate by 60° to straighten $PA + PB + PC$ into one path.)
15. Prove that the composition of three reflections is either a single reflection or a glide reflection.
16. A point P inside a square $ABCD$ has $PA = 1$, $PB = 2$, $PC = 3$. Find $\angle APB$ (rotate 90° about B).
17. Prove that two circles have (in general) two centres of homothety, one with positive and one with negative ratio, lying on the line of centres; locate them.
18. Use homothety to prove that the centroid, the circumcentre, and the centre of the nine-point circle are collinear, and find the ratios along the Euler line.

★ OLYMPIAD SPARK

(Olympiad classic.) In triangle ABC , equilateral triangles BCX , CAY , ABZ are erected outward. Prove that AX , BY , CZ are concurrent and equal in length (the lines meet at the *Fermat point*). (A 60° rotation about a vertex carries one of these segments exactly onto another.)

(Second spark.) Prove **Napoleon's theorem** by transformations: erect outward equilateral triangles on the sides of any triangle; the composition

of the three 120° rotations about their centres is the identity, which forces the three centres to form an equilateral triangle. (*Compose rotations and track the total angle.*)

PART V

Olympiad Geometry

24

The Configurations You Must Know

★ WONDER

Take any triangle and mark nine special points: the three midpoints of the sides, the three feet of the altitudes, and the three midpoints joining each vertex to the orthocentre. Nine points, defined in three completely different ways, scattered around the figure. Now draw the circle through any three of them — and watch, astonished, as it passes through *all nine*. Why should a triangle be so generous?

From here on, the elementary tools rarely work alone; they combine into *configurations* — recurring constellations of points and circles that appear in problem after problem. Knowing them on sight is half the battle. We begin with the most famous of all: the nine-point circle, and the Euler line it completes.

24.1 The nine-point circle

Theorem 24.1 (Nine-Point Circle). In any triangle, these nine points lie on a single circle: the midpoints of the three sides, the feet of the three altitudes, and the midpoints of the segments from each vertex to the orthocentre H . Its centre N is the midpoint of OH (where O is the circumcentre), and its radius is $\frac{1}{2}R$.

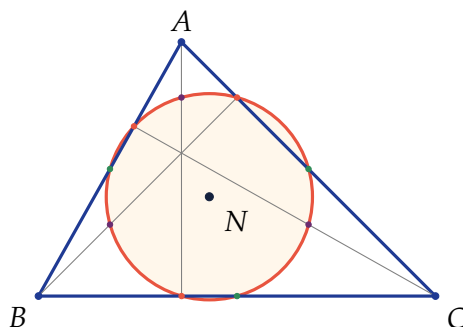


Figure 24.1 — The nine-point circle. Side-midpoints (green), altitude-feet (coral), Euler points (plum) — all nine on one circle, centre N .

 **Worked Example.** Prove the nine points are concyclic, with centre N and radius $\frac{1}{2}R$.

Consider the homothety h centred at the orthocentre H with ratio $\frac{1}{2}$. It sends each vertex X to the midpoint of HX — exactly the three *Euler points*. So h maps the circumcircle (centre O , radius R , through A, B, C) to a circle through the three Euler points, with centre $h(O) =$ the midpoint of HO , call it N , and radius $\frac{1}{2}R$.

It remains to see the side-midpoints and altitude-feet lie on *this same* circle. A known fact (Chapter 10 exercises) is that the reflection of H over a side, and over a side's midpoint, lands on the circumcircle. The foot of an altitude is the midpoint of H and its reflection over that side, and the side-midpoint is the midpoint of H and the reflection over the midpoint — both are therefore images under $h(H, \frac{1}{2})$ of points *on* the circumcircle. Hence they too lie on the circle of centre N , radius $\frac{1}{2}R$. All nine points lie on it.

The wow: the nine-point circle is simply the circumcircle *shrunk by half* toward the orthocentre. One homothety — “halve everything toward H ” — explains all nine coincidences at once. Nine miracles collapse into a single scaling.

24.2 The Euler line, completed

We met the Euler line in Chapter 10: the circumcentre O , centroid G , and orthocentre H are collinear with $OG : GH = 1 : 2$. The nine-point centre now joins them.

MATHEMATICIAN'S MARGIN

Leonhard Euler found this line in 1765. Within a few years he had gone nearly blind — and kept producing mathematics at a furious pace for two more decades, dictating it aloud. The eye that truly *sees* a figure, it seems, was never the one that goes dark.

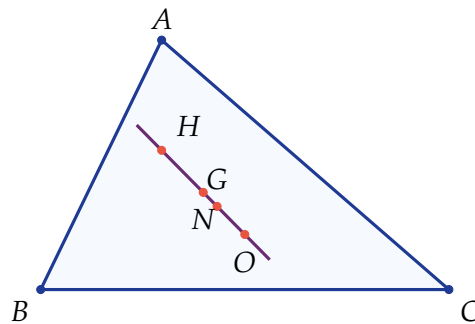


Figure 24.2 — The Euler line carries four centres in fixed proportion: O , then G , then the nine-point centre N , then H .

Worked Example. Locate N on the Euler line and give the ratios $OG : GN : NH$.

Put O at the origin as a position-vector origin. From Chapter 10, $\vec{H} = \vec{a} + \vec{b} + \vec{c}$ and $\vec{G} = \frac{1}{3}(\vec{a} + \vec{b} + \vec{c}) = \frac{1}{3}\vec{H}$. The nine-point centre is the midpoint of OH , so $\vec{N} = \frac{1}{2}\vec{H}$. The four points have position vectors $0, \frac{1}{3}\vec{H}, \frac{1}{2}\vec{H}, \vec{H}$ — all multiples of $\vec{H}$, hence collinear. Reading the coefficients,

$$OG : GN : NH = \frac{1}{3} : \left(\frac{1}{2} - \frac{1}{3}\right) : \left(1 - \frac{1}{2}\right) = \frac{1}{3} : \frac{1}{6} : \frac{1}{2} = 2 : 1 : 3.$$

A *fourth* centre, defined by a circle through nine points, lands exactly on the same line as the other three — and at a clean rational position. The Euler line is one of geometry's great unifiers; the nine-point centre is its midpoint of OH , sitting between G and H .

Worked Example. **A concrete triangle.** For the 13–14–15 triangle, find the circumradius R , the nine-point radius, and the inradius r — and check Euler's inequality.

Here $s = \frac{1}{2}(13 + 14 + 15) = 21$, and Heron gives the area $K =$

$\sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = 84$. Hence

$$r = \frac{K}{s} = \frac{84}{21} = 4, \quad R = \frac{abc}{4K} = \frac{13 \cdot 14 \cdot 15}{4 \cdot 84} = \frac{2730}{336} = \frac{65}{8},$$

and the nine-point radius is $\frac{1}{2}R = \frac{65}{16}$. As a check, $OI^2 = R^2 - 2Rr = \left(\frac{65}{8}\right)^2 - 2 \cdot \frac{65}{8} \cdot 4 = \frac{4225}{64} - 65 = \frac{65}{64} > 0$, so $R = \frac{65}{8} > 8 = 2r$: Euler's $R \geq 2r$ holds, with a little room to spare.

24.3 The incircle–excircle web

The incircle touches the sides at the *contact triangle* XYZ , and (Chapter 16) the tangent lengths are $s - a, s - b, s - c$. Erected outside each vertex are the three *excircles*, each tangent to one side and to the extensions of the other two. These four circles weave a web of equal-tangent relations that power countless problems.

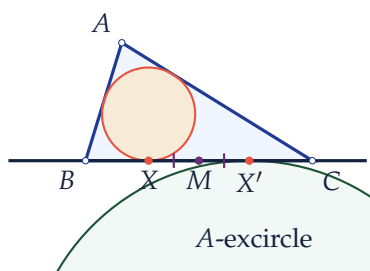


Figure 24.3 — The incircle (coral) meets BC at X ; the A -excircle (green), tangent to BC and to the extensions of the other two sides, meets it at X' . The two touch points are mirror images in the midpoint M of BC .

Worked Example. The incircle touches BC at X ; the excircle opposite A touches BC at X' . Prove X and X' are reflections in the midpoint of BC . From the incircle's equal tangents, $BX = s - b$. For the A -excircle, the analogous tangent-length computation gives $BX' = s - c$. The midpoint M of BC satisfies $BM = \frac{a}{2}$. Now

$$BX + BX' = (s - b) + (s - c) = 2s - b - c = a = BC,$$

so X and X' are equidistant from the two ends of BC in swapped order — that is, $MX = MX'$ with X, X' on opposite sides of M . They are reflections in M .

The incircle and the excircle touch BC at points symmetric about its midpoint — a hidden symmetry between the circle nestled *inside* the triangle and the one bulging *outside*. This single fact is the seed of the Gergonne and Nagel points and of many olympiad length-chases.

AHA!

Olympiad problems rarely announce which configuration is hidden inside them. Training your eye to recognise “that’s a nine-point circle” or “those are incircle tangent lengths” is exactly what separates a solver who flounders from one who, like the grandmaster of Chapter 17, simply *sees* the structure and writes down the key step.

Nine points that had no reason to agree, caught on a single circle; four centres strung along a single line. In this country the coincidences are never coincidences.

→ YOUR TURN

A triangle has circumradius $R = 12$ and inradius $r = 5$. Find its nine-point radius, and check Euler’s inequality $R \geq 2r$.

Set-up (you finish it): the nine-point radius is $\frac{1}{2}R$, and Euler’s inequality compares R with $2r$.

Answer: Nine-point radius = $\frac{1}{2}(12) = 6$; and $R = 12 \geq 2r = 10$, so Euler’s inequality holds.

KEY IDEAS & FORMULAS

- **Nine-point circle:** the 3 side-midpoints, 3 altitude-feet, and 3 Euler points (midpoints of AH, BH, CH) lie on one circle, centre N = midpoint of OH , radius $\frac{1}{2}R$ — the circumcircle halved toward H .
- **Euler line:** O, G, N, H are collinear with $OG : GN : NH = 2 : 1 : 3$.
- **Contact lengths:** the incircle touches the sides at distances $s - a, s - b, s - c$ from the vertices.
- **In/ex symmetry:** the incircle and A -excircle touch BC at points symmetric about its midpoint, a distance $|b - c|$ apart.
- **Feuerbach:** the nine-point circle is tangent to the incircle and all three excircles. *Learn to recognise these configurations on sight.*

Exercises

Warm-up.

1. Name the nine points on the nine-point circle.
2. If a triangle has circumradius $R = 10$, what is its nine-point radius?
3. Where is the nine-point centre relative to O and H ?
4. Give the ratio $OG : GN : NH$ along the Euler line.
5. The incircle touches BC at X with $BX = s - b$. If $a = 13$, $b = 14$, $c = 15$, find BX and CX .
6. For an equilateral triangle, where do O , G , N , H all lie?

Practice.

7. Prove that the reflection of the orthocentre over the midpoint of a side lies on the circumcircle (and is the antipode of the opposite vertex).
8. Show the nine-point circle of a triangle is the circumcircle of its medial triangle.
9. Prove the nine-point circle is also the circumcircle of the orthic triangle (feet of the altitudes).
10. In a right triangle, identify where the nine-point centre falls and why three of the nine points coincide.
11. The incircle has radius r , the A -excircle radius r_a . Prove $rr_a = (s - b)(s - c)$ in a right triangle. (In general $[ABC]^2 = rr_a r_b r_c$.)
12. Prove the distance OH satisfies $OH^2 = R^2 - 8R^2 \cos A \cos B \cos C \dots$ or the cleaner $OH^2 = 9R^2 - (a^2 + b^2 + c^2)$.
13. Show the Gergonne point (lines from vertices to opposite contact points) is concurrent, using Ceva and the tangent lengths.

Challenge.

14. Prove **Feuerbach's theorem**: the nine-point circle is tangent to the incircle and to all three excircles. (One of the loveliest facts in geometry; try inversion after Chapter 27.)
15. Prove that the nine-point centre N is equidistant from the three Euler points and the three side-midpoints directly, via the homothety $h(H, \frac{1}{2})$.
16. Let H be the orthocentre. Prove that triangles ABC , HBC , HCA , HAB all share the same nine-point circle.
17. Prove the A -excircle touch point on BC and the incircle touch point are at distance $|b - c|$ apart.
18. The incentre I and the three excentres I_a, I_b, I_c form an orthocentric system,

with I the orthocentre of the *excentral triangle* $I_a I_b I_c$ and A, B, C the feet of its altitudes. Prove that the circumcircle of ABC is exactly the *nine-point circle* of $I_a I_b I_c$.

★ OLYMPIAD SPARK

(Olympiad.) Let ABC have orthocentre H and circumcircle Ω . The altitude from A meets Ω again at A^* . Prove that A^* is the reflection of H over line BC , and deduce that the nine-point circle bisects every segment from H to a point of Ω .

(Second spark.) Prove that the nine-point centres of the four triangles formed by four points (no three collinear) are themselves concyclic — a “nine-point circle of nine-point circles.”

25

Concurrency & Collinearity, Top Level

★ WONDER

Three circles are sprinkled on the page, overlapping every which way. Each pair shares a “line of balance” — the radical axis along which a point has equal power to both. Draw all three of these lines and they slam through a single point, every time. And there is a dual marvel: every point in the plane secretly names a line, and every line a point, in a way that turns theorems about *meeting* into theorems about *lying on*. Behind both marvels hums a single piece of machinery, and once it is yours these conspiracies stop being surprises.

At the top level, proving that lines *meet* or points *line up* calls for heavier machinery: the radical centre, the duality of poles and polars, and the projective invariance of the cross-ratio. These turn whole families of olympiad problems into near-mechanical checks.

25.1 The radical centre

Theorem 25.1 (Radical Centre). Given three circles with non-collinear centres, the three pairwise radical axes are concurrent. Their common point — the **radical centre** — has equal power with respect to all three circles.

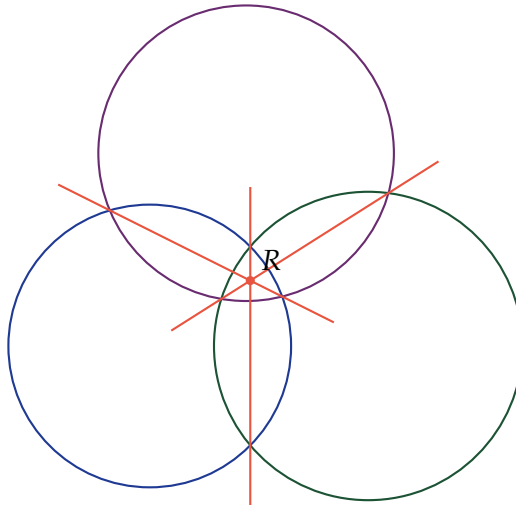


Figure 25.1 — Three circles; their three pairwise radical axes (coral) meet at the radical centre R .

 **Worked Example.** Prove the concurrency, and say what the radical centre is good for.

Let the circles be $\omega_1, \omega_2, \omega_3$. The radical axis of ω_1, ω_2 is the locus where $\text{pow}_1 = \text{pow}_2$; of ω_2, ω_3 , where $\text{pow}_2 = \text{pow}_3$. Let R be where these two lines cross. There $\text{pow}_1 = \text{pow}_2$ and $\text{pow}_2 = \text{pow}_3$, so $\text{pow}_1 = \text{pow}_3$ — meaning R lies on the third radical axis too. All three concur.

Because R has equal power to all three circles, the tangent lengths from R to the three circles are equal; so R is the centre of the unique circle cutting all three orthogonally.

The proof is one line of *transitivity* ($\text{pow}_1 = \text{pow}_2 = \text{pow}_3$) — no computation. To prove three lines concurrent, it often suffices to recognise them as three radical axes; the conspiracy is then automatic.

25.2 Poles and polars

Fix a circle of centre O , radius r . To each point $P \neq O$ we attach a line, its **polar**: the line perpendicular to OP at the inverse point P^* (with $OP \cdot OP^* = r^2$). Conversely each line has a **pole**. For a point outside the circle, the polar is beautifully concrete.

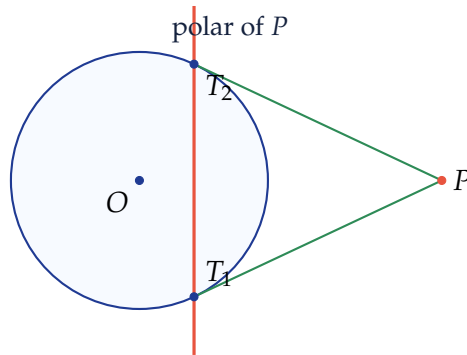


Figure 25.2 — The polar of an external point P is the chord of contact T_1T_2 of the two tangents from P .

Worked Example. Show the polar of an external point is its chord of contact, and prove **La Hire's theorem**: P lies on the polar of Q if and only if Q lies on the polar of P .

From P outside, draw the two tangents, touching at T_1, T_2 . Each contact point satisfies $OT_i \perp PT_i$, and one checks $OP \cdot OP^* = r^2$ where P^* is the midpoint-foot on OP — so the line T_1T_2 is perpendicular to OP at P^* , i.e. it is exactly the polar of P .

For La Hire, write the polar condition algebraically: the polar of P is the set of points X with $\vec{OP} \cdot \vec{OX} = r^2$. This relation is *symmetric* in P and X . So Q on the polar of P means $\vec{OP} \cdot \vec{OQ} = r^2$, which is the very same equation as P on the polar of Q .

The wow: “ P on the polar of Q ” and “ Q on the polar of P ” are *one and the same equation*. This symmetry is a genuine *duality*: it turns any statement about points lying on lines into a statement about lines passing through points. Prove a collinearity, and you have proved a concurrency for free.

25.3 The cross-ratio and harmonic division

Projective geometry's master invariant is the **cross-ratio** of four collinear points,

$$(A, B; C, D) = \frac{AC}{CB} \bigg/ \frac{AD}{DB},$$

which is *unchanged* by projection from one line to another. When it equals -1 (signed), the points are a **harmonic range**, written $(A, B; C, D) = -1$: then C and D divide AB internally and externally in the same ratio.

 **Worked Example.** Why the cross-ratio matters: state its power in one principle.

Projection preserves the cross-ratio. If four collinear points are projected from a centre onto another line, the four images have the *same* cross-ratio. Consequently any property defined purely by a cross-ratio (above all, being harmonic, $= -1$) survives projection — so it can be transported from a convenient figure to the target figure.

A fundamental instance: the two tangent points and the two intersection points cut by any secant through P form a harmonic range with P and the pole. So “ P , its polar, and a secant” always produce a harmonic division — linking this section to the last.

Lengths and angles are mangled by projection, but *one* combination of four lengths is sacred. That single invariant is why a problem can be “projected” into a simpler picture, solved there, and projected back — a technique that dissolves many olympiad configurations.

AHA!

Three master tools for the hardest concurrency/collinearity problems: *radical axes* (when circles are present, look for equal powers), *poles and polars* (to dualise incidences and exploit symmetry), and the *cross-ratio* (to project a hard figure into an easy one). Each converts a fact you cannot see into an equation you can check.

Radical axes, polars, cross-ratios: three machines for seeing the unseeable, each turning a stubborn picture into an equation you can simply check. The hardest problems are not harder than the rest, only better hidden.

→ YOUR TURN

Points $A = 0$, $B = 8$, $C = 2$ lie on a number line. Find the point D for which $(A, B; C, D) = -1$ (the harmonic conjugate of C with respect to A, B).

Set-up (you finish it): C divides AB internally in ratio $AC : CB = 2 : 6 = 1 : 3$; the harmonic conjugate D divides AB *externally* in the same ratio $1 : 3$.

$$\text{Answer: } D = \frac{1 \cdot 8 - 3 \cdot 0}{1 - 3} = \frac{8}{-2} = -4.$$

KEY IDEAS & FORMULAS

- **Radical centre:** the three pairwise radical axes of three circles concur; the centre has equal power (equal tangents) to all three, so it centres the circle orthogonal to all three.
- **Polar of P :** the line $\perp OP$ through the inverse point ($OP \cdot OP^* = r^2$); for an external point it is the chord of contact of the two tangents.
- **La Hire (duality):** P lies on the polar of $Q \iff Q$ lies on the polar of P — prove a collinearity and get a concurrency for free.
- **Cross-ratio** $(A, B; C, D) = \frac{AC/CB}{AD/DB}$ is preserved by projection; $= -1$ is a *harmonic* range.
- **Strategy:** circles present $\rightarrow$ radical axes; incidences $\rightarrow$ poles/polars; awkward figure $\rightarrow$ project via the cross-ratio into an easy one.

Exercises**Warm-up.**

1. Define the radical axis of two circles, and the radical centre of three.
2. What is special about the tangent lengths from the radical centre to the three circles?
3. What is the polar of a point *on* the circle?
4. State La Hire's theorem.
5. Compute the cross-ratio $(A, B; C, D)$ for collinear points at coordinates 0, 6, 2, 9.
6. For which value of the cross-ratio are four points a harmonic range?

Practice.

7. Three circles pairwise intersect. Prove their three common chords are concurrent (a radical-centre instance).
8. Prove that the polar of P is perpendicular to the line OP .
9. Given a circle and external P , construct the polar of P using only a straight-edge. (*Two secants and the diagonal points.*)
10. Show that the cross-ratio $(A, B; C, D) = -1$ exactly when $\frac{2}{AB} = \frac{1}{AC} + \frac{1}{AD}$ (a harmonic mean relation).
11. Prove that if $(A, B; C, D) = -1$ then M , the midpoint of AB , satisfies $MC \cdot MD = MA^2$.

12. Two circles meet at A, B ; a third circle meets the first at C, D and the second at E, F . Prove CD, EF, AB are concurrent.
13. Prove that the pole of a chord is the intersection of the tangents at its endpoints.

Challenge.

14. Prove **Brianchon's theorem**: the three main diagonals of a hexagon circumscribed about a circle are concurrent. (Poles and polars dualise Pascal's theorem.)
15. Prove **Pascal's theorem**: for a hexagon inscribed in a circle, the three intersection points of opposite sides are collinear.
16. In a complete quadrilateral, prove that on each diagonal the two vertices and the two intersection points with the other diagonals form a harmonic range.
17. Prove that the polar of a point P with respect to a circle passes through the harmonic conjugate of P with respect to the two points any line through P cuts on the circle.
18. Use the radical centre to prove that the three altitudes of a triangle concur (build circles on the sides as diameters).

★ OLYMPIAD SPARK

(Olympiad.) Let $ABCD$ be a cyclic quadrilateral whose diagonals meet at P . Let lines AB and CD meet at Q , and AD, BC meet at R . Prove that P is the pole of line QR , and deduce that triangle PQR is *self-polar*: each vertex is the pole of the opposite side.

(Second spark.) Three circles pass pairwise through six points; prove the three radical axes either concur or are all parallel, and characterise exactly when the parallel case occurs (collinear centres).

26

Spiral Similarity

★ WONDER

A single transformation can both *turn* a figure and *resize* it about one fixed point — a “spiral similarity.” Here is the startling part: given *any* two segments in the plane, there is exactly one spiral similarity carrying the first onto the second, and its centre can be *found* as the meeting point of two cleverly chosen circles. Olympiad problems hide this centre everywhere. It waits, every time, at the crossing of two circles you will soon know how to summon.

A **spiral similarity** centred at O is a rotation about O composed with a homothety about the same O : it sends each point z to $O + k e^{i\theta}(z - O)$. It scales by k and turns by θ , all about one centre. Spiral similarities are the secret engine behind a huge fraction of hard olympiad configurations.

26.1 One complex number says it all

Because a spiral similarity is “multiply by a fixed complex number about a fixed point,” complex numbers describe it perfectly.

 **Worked Example.** Show there is a unique spiral similarity sending $A \mapsto C$ and $B \mapsto D$, and find its multiplier and centre (as complex numbers).

A spiral similarity has the form $f(z) = O + w(z - O)$ for a fixed complex $w = k e^{i\theta}$ and centre O . Imposing $f(A) = C$ and $f(B) = D$:

$$C - D = w(A - B) \Rightarrow w = \frac{C - D}{A - B}.$$

Then from $C = O + w(A - O)$ we solve $O = \frac{C - wA}{1 - w}$ (valid unless $w = 1$, i.e. a pure translation). Both w and O are determined, so the spiral similarity is unique.

The entire rotation-and-scaling is captured by the *single complex number* $w = (C - D)/(A - B)$ — its modulus is the scale factor, its argument the rotation angle. Geometry's most flexible motion is, in this language, one division.

26.2 The centre lives at a circle crossing

The complex formula is decisive but hides the geometry. The geometric truth is the lemma every olympiad solver memorises.

Theorem 26.1 (Spiral Similarity Lemma). Let A, B, C, D be four points and let lines AB and CD meet at P . Then the centre of the spiral similarity sending $A \mapsto C$ and $B \mapsto D$ is the *second* intersection point of the circumcircles of $\triangle PAC$ and $\triangle PBD$.

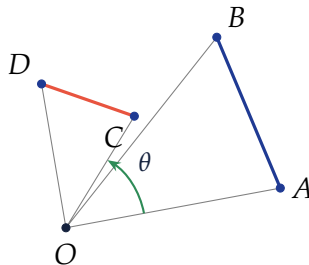


Figure 26.1 — A spiral similarity centred at O carries segment AB to segment CD : a rotation through θ together with a scaling, both about O . (The lemma below locates O as a circle crossing.)

 **Worked Example.** Prove the lemma.

Let O be the centre of the spiral similarity taking $A \mapsto C$, $B \mapsto D$. Because O is fixed and the map scales and rotates equally, triangles OAC and OBD are *similar* (indeed $OA/OC = OB/OD$ and $\angle AOC = \angle BOD$), and so are OAB and OCD .

From $\triangle OAB \sim \triangle OCD$ we get $\angle OAB = \angle OCD$. Now P lies on line

AB and on line CD , so (in directed angles) $\angle OAP = \angle OCP$, which says exactly that O, A, C, P are concyclic — O is on circle (PAC) . The same argument with $\triangle OBA \sim \triangle ODC$ puts O on circle (PBD) . So O is the second common point of those two circles.

The wow: to *locate* a spiral centre you do not compute anything — you draw two circles and take their other intersection. This is the famous “Miquel point” move: in any figure with two segments, the spiral centre is hiding where the two natural circles cross.

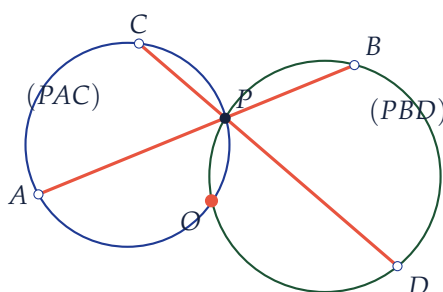


Figure 26.2 — Locating the spiral centre. With $P = AB \cap CD$, the circles (PAC) and (PBD) meet again at O — the centre of the spiral similarity carrying $A \mapsto C$ and $B \mapsto D$.

AHA!

The lemma has a twin reading: the *same* point O is the centre of the spiral similarity sending $A \mapsto B$, $C \mapsto D$ (pairing the other way). So one circle-crossing serves two maps. Whenever you see two segments whose endpoints could correspond, suspect a spiral similarity and hunt for its centre at a circle intersection — it converts ratios and angles between the two segments into one clean transformation.

Two segments that looked unrelated turn out to be one turn-and-scale apart, their pivot hiding at a circle crossing. Once you have seen a spiral similarity, you begin to find it everywhere.

→ YOUR TURN

Find the complex multiplier w of the spiral similarity that sends the segment $A = 0$, $B = 2$ to the segment $C = 0$, $D = 2i$, and read off its scale factor and rotation angle.

Set-up (you finish it): $w = \frac{C - D}{A - B} = \frac{0 - 2i}{0 - 2}$; then $|w|$ is the scale and $\arg w$ the angle.

Answer: $w = \frac{-2i}{-2} = i$, so the scale factor is $|i| = 1$ and the rotation is 90° — a pure quarter-turn.

KEY IDEAS & FORMULAS

- **Spiral similarity:** $f(z) = O + k e^{i\theta}(z - O)$ — a rotation through θ and scaling by k , both about one centre O .
- **One complex number:** the map taking $A \mapsto C, B \mapsto D$ has multiplier $w = \frac{C - D}{A - B}$ ($|w|$ the scale, $\arg w$ the angle) and centre $O = \frac{C - wA}{1 - w}$ — hence it is *unique*.
- **Spiral Similarity Lemma:** with $P = AB \cap CD$, the centre is the *second* intersection of the circumcircles (PAC) and (PBD) — the Miquel point.
- **Twin reading:** the same centre O also realises $A \mapsto B, C \mapsto D$ — one circle-crossing serves two maps.
- **Cue:** two segments whose endpoints might correspond $\Rightarrow$ suspect a spiral similarity and find its centre at a circle intersection.

Exercises

Warm-up.

1. A spiral similarity has multiplier $w = 2e^{i30^\circ}$. What is its scale factor and rotation angle?
2. Find the complex multiplier of the spiral similarity sending $A = 0, B = 2$ to $C = 1 + i, D = 3 + i$.
3. In the Spiral Similarity Lemma, which two circles meet at the centre?
4. What does a spiral similarity with $k = 1$ reduce to? With $\theta = 0$?
5. True/false: a spiral similarity sends every line to a parallel line. Explain.
6. Why is the spiral similarity taking $A \mapsto C, B \mapsto D$ unique?

Practice.

7. Two circles meet at X and Y . A line through X meets them again at A, B ; a line through Y meets them again at C, D . Prove $\triangle XAC \sim \triangle XBD$ via a

- spiral similarity at $X \dots$ (state the map).
8. The spiral similarity sending $A \mapsto C$ and $B \mapsto D$ has centre O . Prove that it sends the midpoint of AB to the midpoint of CD . (*A spiral similarity is affine, so it carries midpoints to midpoints.*)
 9. Use complex numbers to find the centre of the spiral similarity sending $0 \mapsto 4$, $2i \mapsto 4 + 2i$.
 10. Prove that the spiral similarity at O taking $A \rightarrow B$ also takes the circle $(OA \dots)$ to a circle; describe it.
 11. Two equilateral triangles share a vertex. Show one is obtained from the other by a spiral similarity, and find its angle.
 12. Prove the centre of the spiral similarity taking $A \mapsto C$, $B \mapsto D$ equals the centre taking $A \mapsto B$, $C \mapsto D$.
 13. In the lemma's figure, prove $\triangle OAB \sim \triangle OCD$ and find the common ratio in terms of AB, CD .

Challenge.

14. (Miquel point of a complete quadrilateral.) Four lines form four triangles. Prove the circumcircles of all four pass through one common point.
15. Prove that the centre of spiral similarity taking $B \mapsto C$ in two circles through A lies on line $\dots$ (formulate and prove a clean statement about two circles meeting at A).
16. Let $ABCD$ be cyclic with diagonals meeting at P . Prove that the spiral similarity at the Miquel point of $ABCD$ takes $AB \rightarrow DC$, and use it to prove $\frac{AP}{PC} = \frac{AB \cdot AD}{CB \cdot CD}$.
17. Prove that spiral similarities taking $A \rightarrow B$ form a one-parameter family, and describe the locus of their centres.
18. Given triangle ABC and a point P , construct the image of ABC under the spiral similarity centred at P that sends A to B ; describe the image triangle.

★ OLYMPIAD SPARK

(Olympiad.) Two circles ω_1, ω_2 meet at A and B . A line through A meets ω_1 again at P and ω_2 again at Q . Prove that as the line rotates about A , the spiral similarity centred at B taking $\omega_1 \rightarrow \omega_2$ always takes $P \mapsto Q$ — so $\triangle BPQ$ has a fixed shape, and B sees PQ at a constant angle.

(Second spark.) In triangle ABC , points X, Y lie on a line through A with $BX \perp AX$ and $CY \perp AY$. Identify a spiral similarity relating BX and CY ,

and use it to prove a ratio or concyclicity of your choosing.

27

Inversion: The Great Simplifier

★ WONDER

Imagine a transformation that turns the inside of a circle out and the outside in — sending points near the centre flying to infinity and distant points crowding toward the centre. Under this strange map, *circles can become straight lines*. A tangle of circles tangent to one another can be combed into a stack of parallel lines, where the problem is suddenly trivial. There is exactly one motion that does this, and turning it on the right point feels, every time, like a small act of magic.

Inversion is the most powerful single weapon in the olympiad arsenal. It is not an isometry — it badly distorts distances — but it tames circles, tangencies, and angles so beautifully that an impossible-looking circle problem can melt into a line problem. Used well, it feels like cheating.

27.1 What inversion does

Definition 27.1 (Inversion). Fix a centre O and radius r . The inversion with these data sends each point $P \neq O$ to the point P' on ray OP with

$$OP \cdot OP' = r^2.$$

Points inside the circle of inversion go outside and vice versa; points on the circle stay put.

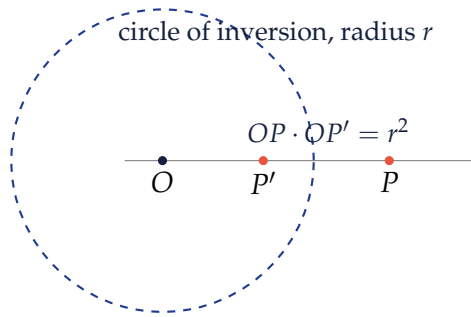


Figure 27.1 — Inversion swaps P and P' along the ray from O , keeping the product of distances equal to r^2 .

The magic is in what happens to whole figures:

Theorem 27.2 (Inversion's behaviour). Under inversion with centre O :

- (a) a line through O maps to itself;
- (b) a line *not* through O maps to a circle *through* O (and vice versa);
- (c) a circle not through O maps to a circle not through O ;
- (d) the angle between two curves is preserved (inversion is *conformal*); tangency is preserved.

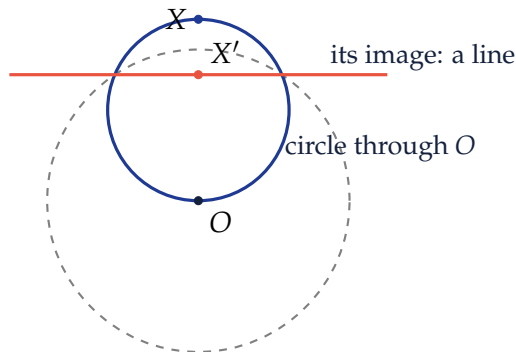


Figure 27.2 — A circle *through* the centre O inverts to a straight line not through O . The point X farthest from O maps to the nearest point X' of the line.

 **Worked Example.** Why does a circle through O become a line?

Let the circle through O have OX as the diameter from O (so X is the antipode of O). For any other point P on the circle, $\angle OPX = 90^\circ$ (angle in a semicircle). Invert: $P \mapsto P'$ and $X \mapsto X'$ with $OP \cdot OP' = OX \cdot OX' = r^2$.

From $OP \cdot OP' = OX \cdot OX'$ we get $\frac{OP}{OX} = \frac{OX'}{OP'}$, so triangles OPX and $OX'P'$ are similar — and the right angle at P transfers to a right angle at X' . Thus every image P' sees the fixed segment OX' ... rather, every P' lies on the fixed line through X' perpendicular to OX . That line is the image of the circle.

Here is the magic: the inscribed right angle in the circle becomes a constant right angle at the *fixed* point X' — pinning every image to a single straight line. Inversion converts “angle in a semicircle” into “perpendicular to a fixed segment.”

27.2 Ptolemy, for free

The reason inversion is prized: it has a clean distance law, and that law makes hard inequalities easy.

Theorem 27.3 (Distance under inversion). If inversion with centre O , radius r sends $P \mapsto P'$ and $Q \mapsto Q'$, then

$$P'Q' = PQ \cdot \frac{r^2}{OP \cdot OQ}.$$

 **Worked Example. Ptolemy’s inequality.** For any four points, $AC \cdot BD \leq AB \cdot CD + AD \cdot BC$, with equality iff $ABCD$ is a cyclic quadrilateral (in order).

Invert about A with any radius r . Let B, C, D map to B', C', D' . For the three image distances, the distance law gives

$$B'C' = BC \cdot \frac{r^2}{AB \cdot AC}, \quad C'D' = CD \cdot \frac{r^2}{AC \cdot AD}, \quad B'D' = BD \cdot \frac{r^2}{AB \cdot AD}.$$

The plain triangle inequality on the images says $B'D' \leq B'C' + C'D'$. Substitute and multiply through by $\frac{AB \cdot AC \cdot AD}{r^2}$:

$$BD \cdot AC \leq BC \cdot AD + CD \cdot AB.$$

That is Ptolemy’s inequality. Equality holds exactly when B', C', D' are *collinear* — which, inverting back, happens precisely when A, B, C, D lie on a circle.

The wow: Ptolemy's theorem — which cost us a clever auxiliary point back in Chapter 14 — here falls out of the *triangle inequality* after one inversion. The cyclic “equality case” is just “three points became collinear.” Inversion turned a subtle circle theorem into the most basic inequality there is.

27.3 Inverting to kill a tangency

AHA!

The strategic move: invert centred at a point where many circles or lines pass, especially a point of tangency. Two circles tangent at O both become *lines* (each is a circle through O), and tangent at O means the lines are *parallel*. A forest of mutually tangent circles can become a simple ladder of parallel lines, where lengths and ratios are obvious — then invert back. Choosing the centre of inversion wisely is the whole art.

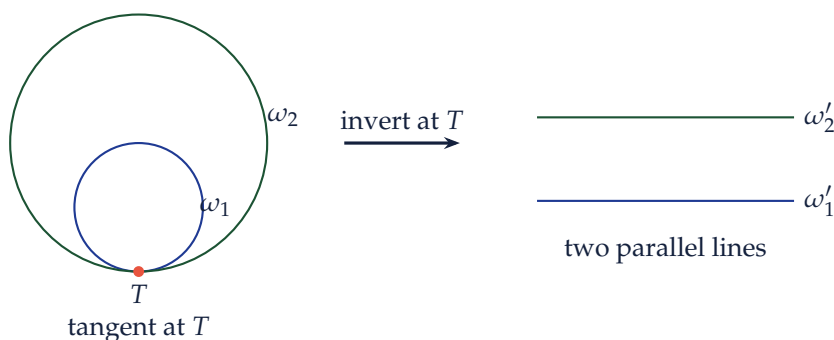


Figure 27.3 — The strategic move. Inverting at the tangency T turns two mutually tangent circles into two *parallel* lines, where lengths and ratios are obvious; solve there, then invert back.

This is how problems about chains of tangent circles (Steiner chains, the arbelos, Descartes' circle theorem) are routinely dismantled: invert at a tangency, solve the easy parallel-line picture, return.

Inversion asks you to turn the plane inside out and trust that the mess will fall into order on the far side. Master it, and the circles that once frightened you become the easiest things in the room.

→ YOUR TURN

An inversion has centre O and radius $r = 6$. Two points satisfy $OP = 3$, $OQ = 4$, and $PQ = 2$. Find the distance $P'Q'$ between their images.

Set-up (you finish it): the distance law gives $P'Q' = PQ \cdot \frac{r^2}{OP \cdot OQ} = 2 \cdot \frac{36}{3 \cdot 4}$.

Answer: $P'Q' = 2 \cdot \frac{36}{12} = 2 \cdot 3 = 6$.

KEY IDEAS & FORMULAS

- **Inversion** (centre O , radius r): $P \mapsto P'$ on ray OP with $OP \cdot OP' = r^2$; inside $\leftrightarrow$ outside, the circle of inversion fixed.
- **Behaviour:** a line through O maps to itself; a line not through $O \leftrightarrow$ a circle through O ; a circle not through $O \rightarrow$ a circle; angles and tangency are preserved (conformal).
- **Distance law:** $P'Q' = PQ \cdot \frac{r^2}{OP \cdot OQ}$.
- **Ptolemy for free:** inverting at A turns Ptolemy's inequality into the plain triangle inequality on the images (equality $\iff$ images collinear $\iff$ concyclic).
- **The strategic move:** invert at a point of tangency — two tangent circles become *parallel lines*, where ratios are obvious; solve there, then invert back.

Exercises

Warm-up.

1. Inversion has centre O and radius $r = 6$. A point is 4 from O ; where is its image?
2. Which points are fixed by an inversion?
3. To what does a line through the centre O invert?
4. To what does a circle through O invert?
5. State the distance-under-inversion formula.
6. Why is a point of tangency a good centre of inversion?

Practice.

7. Under inversion centre O , radius r , a circle of radius ρ centred at distance $d > \rho$ from O maps to a circle; find its radius.
8. Prove that two circles tangent at O invert to two parallel lines.
9. Prove the distance-under-inversion formula using similar triangles $OPQ \sim OQ'P'$.
10. Three circles are mutually tangent, all passing through a point O . Invert at O and describe the image.
11. A circle orthogonal to the circle of inversion maps to itself. Prove it.
12. Use inversion to prove that if four points are concyclic, then a certain Ptolemy *equality* holds.
13. Two circles meet at O at angle θ . What is the angle between their images under inversion at O ?

Challenge.

14. Prove **Feuerbach's theorem** (the nine-point circle is tangent to the incircle) by inverting at a well-chosen point.
15. (Arbelos.) In the "shoemaker's knife" (a big semicircle with two smaller ones removed), prove a chain of inscribed circles climbs in a pattern governed by inversion.
16. Prove **Descartes' Circle Theorem** relating the curvatures (reciprocal radii) of four mutually tangent circles, $(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$, using inversion.
17. Given two non-intersecting circles, prove there is an inversion making them *concentric*, and use it to analyse a Steiner chain.
18. Prove that inversion preserves the cross-ratio of four concyclic points (suitably interpreted).

★ OLYMPIAD SPARK

(Olympiad.) Let ω be a circle and A a point outside it. Two lines through A meet ω at B, C and at D, E . Prove that the circle through A tangent to ω , the circle (ABD) , and the circle (ACE) are related by an inversion centred at A — and use this to prove a tangency in the configuration.

(Second spark — the problem that named a method.) Three circles are mutually externally tangent. Prove there exist exactly two circles tangent to all three, and that inverting at one tangency reduces the whole problem to finding a circle tangent to two parallel lines and one circle — a problem a child could solve.

28

The Computational Arts

★ WONDER

Some olympiad problems are so laden with special points — centroid, incentre, orthocentre, contact points — that synthetic cleverness drowns. For these, the masters reach for a coordinate system custom-built for triangles, in which the centroid is simply $(1 : 1 : 1)$ and the incentre is $(a : b : c)$. In this system the triangle's own structure does the bookkeeping. What would coordinates look like if a triangle, not a pair of axes, were the frame of reference?

Part IV gave you trigonometric, coordinate, vector, and complex “bashing.” This chapter adds the most triangle-friendly engine of all — **barycentric coordinates** — and then steps back to ask the strategic question: with five computational weapons available, which should you draw?

28.1 Barycentric coordinates

Fix a reference triangle ABC . Any point P in the plane can be written as a weighted average of the vertices,

$$P = \alpha A + \beta B + \gamma C, \quad \alpha + \beta + \gamma = 1,$$

and the triple $(\alpha : \beta : \gamma)$ (defined up to scaling) is the **barycentric coordinates** of P . Geometrically α, β, γ are proportional to the signed areas of the sub-triangles PBC, PCA, PAB — which is why these are also called *areal* coordinates.

MATHEMATICIAN'S MARGIN

Barycentric coordinates were the invention of August Möbius, who introduced them in 1827 by imagining weights hung at the triangle's corners and asking where the whole would balance. The same Möbius later described the one-sided strip that still bears his name.

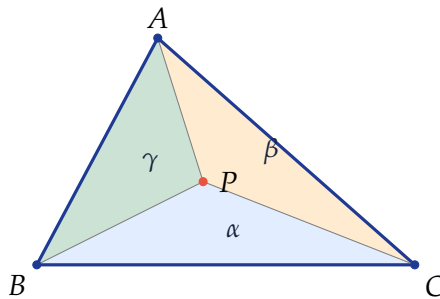


Figure 28.1 — Barycentric (areal) coordinates: $\alpha : \beta : \gamma$ are the areas of $[PBC] : [PCA] : [PAB]$. Here $P = \alpha A + \beta B + \gamma C$.

The reward is a dictionary in which the triangle's centres become memorable triples:

centroid $(1:1:1)$, incentre $(a:b:c)$,

circumcentre $(a^2(b^2+c^2-a^2) : \dots)$, orthocentre $(\tan A : \tan B : \tan C)$.

 **Worked Example.** Prove the incentre has barycentric coordinates $(a : b : c)$.

The incentre I lies on the bisector from A , which meets BC at the point D with $BD : DC = c : b$ (Angle Bisector Theorem). In barycentrics, $D = (0 : b : c)$ (zero weight on A , and weights on B, C inverse to the opposite segments). The bisector from A is the line AD , whose points are $(t : b : c)$. By the symmetric argument the bisector from B is $(a : t : c)$. The common point sets the ratios consistent across both, giving $I = (a : b : c)$.

The incentre — defined by equal distances to the sides — has coordinates that are simply the *side lengths*, $a : b : c$. Every classical centre collapses to a clean triple, and a problem about “the incentre and the centroid and the orthocentre” becomes algebra among $(a : b : c)$, $(1 : 1 : 1)$, $(\tan A : \tan B : \tan C)$.

 **Worked Example.** Show that, in barycentric coordinates, three cevians from the vertices are concurrent precisely when Ceva's product is 1 — recovering Ceva almost for free.

A cevian from A hits BC at $(0 : m : n)$, so the cevian is the line through $A = (1 : 0 : 0)$ and $(0 : m : n)$. Three such cevians (from A, B, C , through $(0 : m_1 : n_1)$, $(p_2 : 0 : q_2)$, $(r_3 : s_3 : 0)$) are concurrent exactly when a certain 3×3 determinant vanishes — and expanding it gives precisely $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$.

The punchline: Ceva's theorem, which we proved by areas in Chapter 19, is *built into* barycentric coordinates — concurrency is a vanishing determinant. The coordinate system was designed around exactly the area-ratios Ceva measures, so the theorem becomes a one-line computation.

28.2 Choosing your weapon

You now hold five computational languages. The skill is matching the tool to the problem.

AHA!

Which bash? *Trigonometry* — when angles and a circumcircle dominate (Law of Sines, $2R$). *Cartesian coordinates* — when there are perpendiculars, midpoints, or one or two circles; place axes to seed zeros. *Complex numbers* — when rotations, equilateral triangles, or a single circle (the unit circle) dominate. *Vectors* — for midpoints, centroids, and dot-product perpendicularity, with full symmetry. *Barycentric* — when the triangle's own centres, cevians, and ratios are the whole story. And always remember the *synthetic* option: the cleanest proof is often no bash at all.

X COMMON TRAP

Bashing trades insight for certainty, and it can be *long*. Before committing, spend thirty seconds asking whether an angle chase, a spiral similarity, or one inversion would finish in three lines. Reach for coordinates when cleverness stalls — not before. And whichever engine you choose, keep the computation organised: define everything up front, and check the answer against the diagram.

Five languages for one geometry, and the quiet wisdom that the best move is sometimes to speak none of them and stay synthetic. A master is fluent in all of them, and loyal to none.

→ YOUR TURN

A triangle has sides $a = 5$, $b = 6$, $c = 7$. Write the barycentric coordinates of its incentre, and find what fraction of the triangle's area is $[IBC]$.

Set-up (you finish it): the incentre is $(a : b : c)$, and the first (normalised) coordinate is the area fraction $\frac{a}{a+b+c}$.

Answer: Incentre = $(5 : 6 : 7)$; $\frac{[IBC]}{[ABC]} = \frac{5}{5+6+7} = \frac{5}{18}$.

KEY IDEAS & FORMULAS

- **Barycentric coordinates:** $P = \alpha A + \beta B + \gamma C$ with $\alpha + \beta + \gamma = 1$; the triple $(\alpha : \beta : \gamma)$ is proportional to the areas $[PBC] : [PCA] : [PAB]$.
- **The dictionary:** centroid $(1:1:1)$, incentre $(a:b:c)$, circumcentre $(a^2(b^2+c^2-a^2) : \dots)$, orthocentre $(\tan A : \tan B : \tan C)$.
- **Cevian concurrency** is a vanishing 3×3 determinant — Ceva's theorem is built in.
- **Choosing the bash:** trig (angles, circumcircle), Cartesian (perpendiculars, a circle or two), complex (rotations, one circle), vectors (midpoints, perpendicularity), barycentric (the triangle's own centres and ratios).
- **But stay synthetic when you can:** the cleanest proof is often no bash at all.

Exercises

Warm-up.

1. Write the barycentric coordinates of the three vertices A , B , C and of the centroid.
2. What are the barycentric coordinates of the midpoint of BC ?
3. Give the incentre's barycentric coordinates for a 3–4–5 triangle.
4. A point has barycentric $(2 : 3 : 5)$. What fraction of the triangle's area is $[PBC]$?
5. Which points have a zero barycentric coordinate?
6. Name the best bashing method for a problem dominated by a single circle

and rotations.

Practice.

7. Verify that $(1 : 1 : 1)$ is the centroid by averaging the vertices.
8. Find the barycentric coordinates of the point dividing BC in ratio $2 : 3$.
9. Using barycentrics, prove the medians concur at $(1 : 1 : 1)$.
10. Show the line BC is exactly the set of points with first barycentric coordinate 0.
11. Find, in barycentric form, the centroid of the medial triangle and compare with the original centroid.
12. Solve by a deliberate coordinate bash: the feet of the perpendiculars from a point on the circumcircle to the three sides are collinear (the *Simson line*).
13. Take one earlier olympiad spark and re-solve it by a different computational method; compare the effort.

Challenge.

14. Derive the barycentric coordinates of the circumcentre, $(a^2(b^2 + c^2 - a^2) : \dots : \dots)$.
15. Prove the *distance formula* in normalised barycentrics and use it to compute $OI^2 = R^2 - 2Rr$ (Euler's formula).
16. Use barycentric coordinates to prove that the Gergonne point is

$$\left(\frac{1}{s-a} : \frac{1}{s-b} : \frac{1}{s-c}\right) = ((s-b)(s-c) : (s-c)(s-a) : (s-a)(s-b)).$$

(The incircle touches BC at $(0 : s-c : s-b)$.)

17. Prove the Nagel point has barycentric $(s-a : s-b : s-c)$.
18. Set up and prove, by any bash you like, that the Euler line has barycentric equation $\sum(b^2 - c^2)(b^2 + c^2 - a^2)x = 0$. (Take the line through the centroid $(1 : 1 : 1)$ and the circumcentre $(a^2(b^2 + c^2 - a^2) : \dots)$; expand the 3×3 determinant. Beware the look-alike $\sum(b^2 - c^2)x = 0$, which is the centroid-symmedian line, not the Euler line.)

★ OLYMPIAD SPARK

(Olympiad, by bash.) In triangle ABC , prove **Euler's formula** $OI^2 = R(R - 2r)$ relating the circumradius R , inradius r , and the distance OI between circumcentre and incentre — and deduce **Euler's inequality** $R \geq 2r$, with equality only for the equilateral triangle. (*Barycentric distance from O to $I = (a : b : c)$, or a clean trigonometric computation.*)

(Second spark.) Choose any single olympiad problem from Chapter 30 and solve it twice — once synthetically, once by your favourite bash — then write a paragraph on which solution you would submit, and why.

29

How to Attack an Olympiad Problem

★ WONDER

You sit down to a contest. Before you is a geometry problem you have never seen, with a statement of one or two lines and a blank page beneath. The clock is running. You know dozens of theorems — but *which* one, and *where*? The difference between the student who freezes and the one who finds the thread is not more theorems. It is *method*. What does a trained solver actually *do* in those first crucial minutes?

Every tool in this book is now in your hands. This chapter is about *wielding* them: the habits, the order of operations, and the inner monologue of a solver who meets a fresh problem and, step by step, takes it apart.

29.1 The five moves

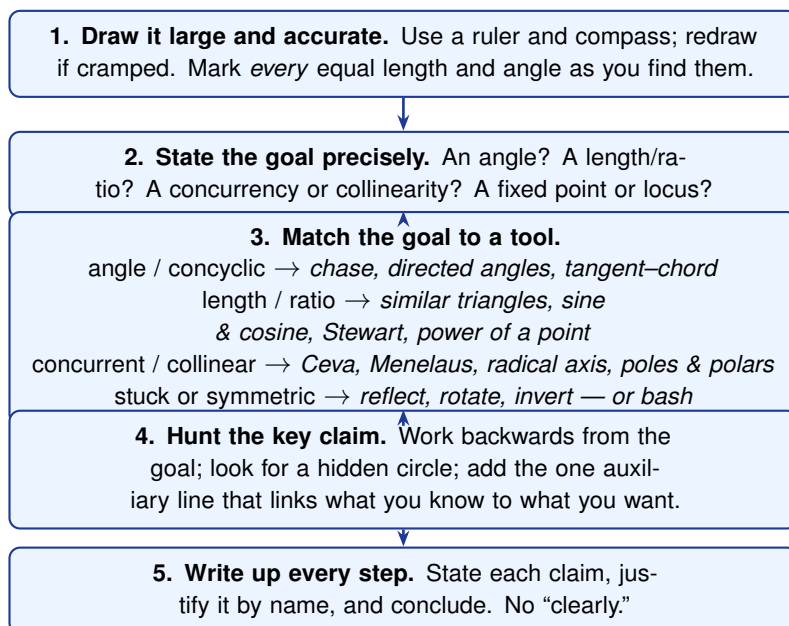


Figure 29.1 — The solver’s loop. Most breakthroughs come at step 1 (a good diagram) or step 4 (the right auxiliary line).

Two of these deserve emphasis. **The diagram is not a sketch — it is your laboratory.** A large, accurate figure reveals coincidences (three points that look collinear, four that look concyclic) which become the conjectures you then prove. And **the auxiliary line** — a parallel, a perpendicular, a circle, a reflection — is where most hard problems turn: the figure as given is a locked door, and the right extra line is the key.

29.2 A live solve

Watch the method in action. The narration is the part that matters.

 **Worked Example. Problem.** In triangle ABC , the incircle touches BC at X , and M is the midpoint of BC . Prove that the reflection of X over M is exactly the point where the A -excircle touches BC .

Move 1 — diagram. Draw a clearly scalene triangle, the incircle, X on BC , mark M .

Move 2 — goal. A statement about two *lengths* on BC being symmetric about M : show $BX = CX'$.

Move 3 — tool. Lengths on the sides of a triangle with incircle/excircle: that is *tangent lengths* (Chapter 16/24).

Move 4 — key claim. Recall $BX = s - b$. For the A -excircle, the tangent-length computation gives $BX' = s - c$. So $BX + BX' = (s - b) + (s - c) = a = BC$.

Move 5 — write-up. Since $BX + BX' = BC$ and both are measured from B along BC , the points X and X' are equidistant from the midpoint M on opposite sides: X' is the reflection of X over M . ■

The takeaway: no brilliance was required — only the discipline to *name the goal* (a length symmetry), *recognise the configuration* (incircle/excircle tangent lengths), and *recall the right fact*. That is what training buys: not magic, but a fast, reliable march from goal to tool to conclusion.

AHA!

When you are stuck, the productive questions are concrete: *What have I not used?* (every hypothesis must earn its place.) *Do four points look concyclic?* *What if I drew the circle / the parallel / the reflection?* *Can I work backwards from the conclusion?* Cycling through these — not staring harder — is how solvers get unstuck.

Method is the quiet engine under every brilliant solution. Drill the loop until it runs without you, and the blank page beneath the problem stops being a threat and becomes an invitation.

→ YOUR TURN

A problem asks you to prove that four given points are concyclic. Name two genuinely different methods you could reach for.

Set-up (you finish it): think about what “concyclic” converts into — an angle condition, or a length-product condition.

Answer: (1) An *angle* method: show two opposite angles of the quadrilateral sum to 180° , or that two points see a segment at equal angles (directed-angle test $\angle(CA, CB) = \angle(DA, DB)$). (2) A *power* method: if lines AB, CD meet at P , show $PA \cdot PB = PC \cdot PD$ (converse power of a point). A third option: exhibit a centre equidistant from all four.

KEY IDEAS & FORMULAS

- **The five-move loop:** (1) draw it *large and accurate*, marking every equal length and angle; (2) state the goal precisely; (3) match goal to tool; (4) hunt the key claim; (5) write up every step — no “clearly.”
- **Goal → tool:** angle/concyclic → chase, directed angles, tangent-chord; length/ratio → similar triangles, sine & cosine, Stewart, power of a point; concurrent/collinear → Ceva, Menelaus, radical axis, poles & polars; stuck/symmetric → reflect, rotate, invert — or bash.
- **The diagram is your laboratory:** an accurate figure reveals the coincidences you then prove.
- **The auxiliary line** (a parallel, perpendicular, circle, or reflection) is where most hard problems turn.
- **When stuck, ask concretely:** What have I not used? Do four points look concyclic? What if I drew the circle/parallel/reflection? Can I work backwards from the conclusion?

Exercises**Warm-up (name the tool, don't solve).**

1. “Prove $\angle XYZ = \angle XWZ$.” Which family of tools?
2. “Prove lines AD , BE , CF concur.” Which two theorems come first to mind?
3. “Prove $PA \cdot PB = PC \cdot PD$.” Which single idea?
4. “Prove four points are concyclic.” Give two different methods.
5. “A configuration full of tangent circles.” Which transformation?
6. “Minimise a sum of two distances to a line.” Which move?

Practice.

7. Take any Olympiad Spark from Part III and write only its *first three moves* (diagram, goal, tool) without finishing.
8. For a problem you have solved, write the one-sentence “key claim” that unlocked it.
9. Given “prove $MI \perp BC$ implies the triangle is isosceles,” identify the hypothesis you must use and the tool.
10. Practise the diagram: draw an accurate triangle with its incircle, circumcircle, and nine-point circle on one figure.
11. Re-solve the live-solve problem above, but target instead the distance XX' in terms of the sides.

12. Write a clean, gap-free write-up of the Vertical Angle Theorem as if for a grader.

Challenge.

13. Take an IMO geometry problem (any year) and, *before* solving, write your full “move 1–3” plan. Then attempt it.
14. Find a problem you can solve two ways (synthetic and bash) and write both; reflect on which is more robust under time pressure.
15. Deliberately solve an easy problem with the “wrong” heavy tool (e.g. invert a problem that needed only an angle chase) to feel why tool choice matters.
16. Keep a “key-claim log”: for ten problems, record only the single insight that cracked each. Look for patterns.

★ OLYMPIAD SPARK

(Meta-spark.) Choose a problem from Chapter 30, cover the solution, and give yourself exactly fifteen minutes with the five-move loop. Whether or not you finish, write down where you were in the loop when the time ran out — diagram, goal, tool, key claim, or write-up. Over many problems, the step where you most often stall is exactly the skill to train next.

30

The Summit: Olympiad Problems

★ WONDER

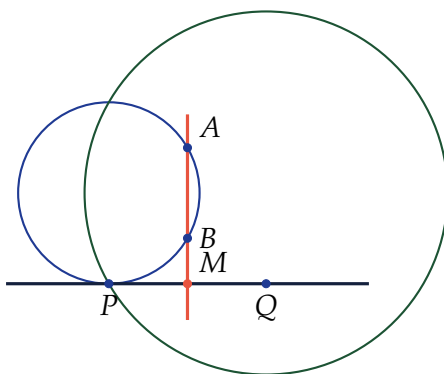
Behind you stand every tool you have forged — angle chasing, the circle theorems, similarity and trigonometry, Ceva and Menelaus, transformations, inversion, the bash. None of them, alone, conquers a real olympiad problem. But *together*, summoned at the right instant, they do. One by one, watch each step onto the field at the exact moment it is needed.

Below are seven problems in the olympiad spirit, each solved in full — and each headlined by the tool that cracks it. Read the *statement*, cover the solution, and try the five-move loop yourself first. Then watch the tool do its work.

Problem 1 — Power of a Point

Two circles meet at A and B . A common tangent touches them at P and Q . Prove that line AB bisects PQ .

Let $M = AB \cap PQ$. Line AB is the *radical axis* of the two circles, so every point of it has equal power to both. At M , the power to the first circle is the squared tangent length MP^2 , and to the second it is MQ^2 . Equal powers give $MP^2 = MQ^2$, so $MP = MQ$: M is the midpoint of PQ . ■



Problem 1 — M on the radical axis AB has equal tangent lengths $MP = MQ$.

Problem 2 — Inscribed Angles

Two circles meet at A and B . A variable line through A meets the first circle again at P and the second at Q . Prove that $\angle PBQ$ is the same for every such line.

P, A, Q are collinear, so $\angle BPQ = \angle BPA$ and $\angle BQP = \angle BQA$. But $\angle BPA$ is an inscribed angle in the first circle on the *fixed* chord BA — so it is constant as P varies (angles in the same segment). Likewise $\angle BQA$ is constant in the second circle. In triangle PBQ ,

$$\angle PBQ = 180^\circ - \angle BPQ - \angle BQP = 180^\circ - \angle BPA - \angle BQA,$$

a sum of two constants. So $\angle PBQ$ never changes. ■ (This constant angle is exactly the spiral similarity at B from one circle to the other.)

Problem 3 — The Tangent–Chord Angle

Two circles are tangent at T . A line through T meets one circle again at P and the other at Q . Prove the tangent to the first circle at P is parallel to the tangent to the second at Q .

Let ℓ be the common tangent at T . In the first circle, the tangent–chord angle between ℓ and chord TP equals the inscribed angle in the alternate segment — equivalently, the angle that the tangent at P makes with PT . In the second circle the same is true at Q . But both tangent–chord angles

at T are the *same* angle, the one between ℓ and the common line PTQ . Hence the tangent at P and the tangent at Q each make this equal angle with line PQ , on the same side — so they are parallel. ■

Problem 4 — A 60° Rotation

Equilateral triangles ABF and ACG are erected outward on sides AB and AC of triangle ABC . Prove that $BG = CF$.

Consider the rotation by 60° about A that carries F to B (it does, since $AF = AB$ and $\angle FAB = 60^\circ$). The very same rotation carries C to G (since $AC = AG$ and $\angle CAG = 60^\circ$). A rotation sends the segment FC to the segment BG , and rotations preserve length. Therefore $FC = BG$. ■ (*The same rotation shows BG and CF meet at 60° — and the construction hides the Fermat point.*)

Problem 5 — Inversion (Ptolemy's Equality)

Prove that if $ABCD$ is a cyclic quadrilateral, then $AC \cdot BD = AB \cdot CD + AD \cdot BC$.

Invert about A with radius r . Since A, B, C, D are concyclic, the circle through them (passing through the centre A) inverts to a *line*; so the images B', C', D' are collinear, with C' between B' and D' . Thus $B'D' = B'C' + C'D'$. Using the distance law $X'Y' = XY \cdot \frac{r^2}{AX \cdot AY'}$,

$$\frac{BD r^2}{AB \cdot AD} = \frac{BC r^2}{AB \cdot AC} + \frac{CD r^2}{AC \cdot AD}.$$

Multiply through by $\frac{AB \cdot AC \cdot AD}{r^2}$ to clear denominators:

$$AC \cdot BD = AD \cdot BC + AB \cdot CD. \quad \blacksquare$$

Problem 6 — Euler's Inequality

Prove that for every triangle, $R \geq 2r$, with equality only for the equilateral triangle.

The power of the incentre I with respect to the circumcircle is $OI^2 - R^2$. Computing this power another way (it equals $-2Rr$) yields **Euler's formula**

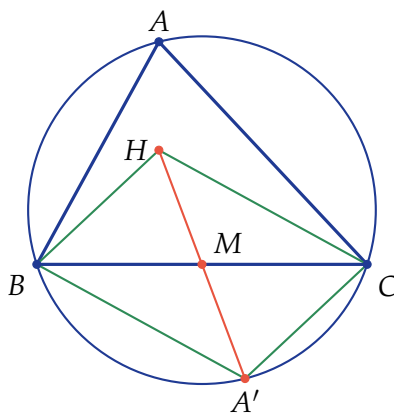
$$OI^2 = R^2 - 2Rr.$$

Since $OI^2 \geq 0$, we get $R^2 - 2Rr \geq 0$, i.e. $R \geq 2r$. Equality holds exactly when $OI = 0$, that is when the circumcentre and incentre coincide — which happens only for the equilateral triangle. ■

Problem 7 — A Configuration Capstone

Let H be the orthocentre of acute triangle ABC , and M the midpoint of BC . Prove that the reflection of H over M is the point A' diametrically opposite A on the circumcircle.

Let A' be the antipode of A . Since AA' is a diameter, $\angle ABA' = \angle ACA' = 90^\circ$ (angle in a semicircle). So $A'B \perp AB$; but the altitude CH is also $\perp AB$, hence $A'B \parallel CH$. Likewise $A'C \perp AC \parallel BH$. Therefore $BHCA'$ has both pairs of opposite sides parallel — it is a parallelogram. Its diagonals HA' and BC bisect each other, so their common midpoint is M . Thus A' is the reflection of H over M . ■



Problem 7 — $BHCA'$ is a parallelogram, so its diagonals HA' and BC share the midpoint M .

AHA!

Look back over the seven. Not one needed a tool you do not own. Each fell to a *single* well-chosen idea — the radical axis, the inscribed angle, the alternate segment, a rotation, an inversion, Euler's formula, a parallelogram — summoned at the right moment. That is the whole secret of olympiad geometry: not knowing more, but *seeing which one*, and trusting it. You can do this now.

→ YOUR TURN

Building straight on Problem 7: in acute $\triangle ABC$ with circumcentre O and orthocentre H , let M be the midpoint of BC . Prove that $AH = 2OM$.

Set-up (you finish it): Problem 7 showed the antipode A' of A is the reflection of H over M , so M is the midpoint of HA' ; and O is the midpoint of AA' (a diameter). Look at triangle AHA' and the segment joining the midpoints of two of its sides.

Answer: In $\triangle AHA'$, O is the midpoint of side AA' and M is the midpoint of side HA' , so OM is a midline: $OM = \frac{1}{2}AH$ (and $OM \parallel AH$). Hence $AH = 2OM$.

Two true ascents

Each of the seven fell to a *single* idea. A real olympiad problem rarely does: it asks you to find *several* of them and chain them in an order nobody hands you. Here are two such climbs — and this time the write-up shows the *search*, not just the finished proof.

Ascent 1 — the lemma you will use most

Let I be the incentre of $\triangle ABC$, and let M be the second point where the bisector of $\angle A$ meets the circumcircle — the midpoint of arc BC not containing A . Prove that $MB = MC = MI$.

The hunt. We must tie I — a creature of *angle bisectors* — to the *circumcircle*. The bisector of $\angle A$ already pierces the circle at M , so M is the natural bridge. One equality is free: M is the midpoint of arc BC , so equal arcs give equal chords, $MB = MC$. Everything now rests on the strange claim $MI = MB$ — strange because I sits inside the triangle and M out on the rim.

A false start. You could drop coordinates, or swing Stewart's theorem at MI . It works, and it teaches you nothing. Pause and ask what *shape* would force $MI = MB$: an isosceles triangle MBI , that is, $\angle MBI = \angle MIB$. So chase those two angles.

The pivot. First notice that A, I, M are collinear — all three lie on the bisector of $\angle A$. Now split the base angle at B :

$$\angle MBI = \angle MBC + \angle CBI.$$

The inscribed angles $\angle MBC$ and $\angle MAC$ stand on the same arc MC , so $\angle MBC = \angle MAC = \frac{1}{2}\angle A$; and $\angle CBI = \frac{1}{2}\angle B$ because BI bisects $\angle B$. Hence $\angle MBI = \frac{1}{2}\angle A + \frac{1}{2}\angle B$. For the other base angle, use the collinearity: $\angle MIB$ is the *exterior* angle of $\triangle ABI$ at I , so

$$\angle MIB = \angle IAB + \angle IBA = \frac{1}{2}\angle A + \frac{1}{2}\angle B.$$

Thus $\angle MBI = \angle MIB$; triangle MBI is isosceles, and $MI = MB = MC$. ■

Why it is a summit. Three ideas had to arrive together: the arc-midpoint (for $MB = MC$ and $\angle MBC = \frac{1}{2}\angle A$), the bisector (for $\angle CBI = \frac{1}{2}\angle B$), and the collinearity AIM (which turned $\angle MIB$ into an exterior angle). No one of them cracks the problem; chained, they do. And the result — M is equidistant from B, I , and C — is the single most-summoned lemma in olympiad geometry. Learn it, and half of every incentre problem swings open.

Ascent 2 — two circles, two tangents

Circles ω_1 and ω_2 meet at A and B . A line through A meets ω_1 again at C and ω_2 again at D . The tangent to ω_1 at C and the tangent to ω_2 at D meet at P . Prove that B, C, P, D lie on one circle.

The hunt. Two tangents should fire one reflex — the tangent–chord angle — and the goal, “four points concyclic,” is itself an *angle* statement. So translate each tangent into an inscribed angle and hope the two meet. The one hazard is configuration: which side of each chord we land on. That is exactly what *directed angles* (Chapter 17) were built to ignore, so we chase modulo 180° .

Translate the tangents. By the tangent–chord theorem in ω_1 , written with directed angles, $\angle(CP, CB) = \angle(AC, AB)$; and in ω_2 , $\angle(DP, DB) = \angle(AD, AB)$.

The hinge. Why should the two right-hand sides agree? Because C, A, D are collinear — the lines AC and AD are the *same line*. So $\angle(AC, AB) = \angle(AD, AB)$, and therefore

$$\angle(CP, CB) = \angle(DP, DB).$$

That is precisely the directed-angle test for B, C, P, D to lie on one circle. ■

Why it is a summit. The tangent–chord theorem fired twice, the collinearity of C, A, D welded the two halves together, and the directed-angle concyclicity test closed the door — three tools, one chain. The whole problem turns on the quiet observation that AC and AD are a single line. See that, and it falls; miss it, and you flounder. *That* is the difference the Summit is meant to teach.

Twenty more, for the road

A solver is made by mileage. Here are twenty problems, roughly in order of difficulty, drawing on the whole book. Hints for the starred ones are in Appendix C.

1. In $\triangle ABC$, $\angle A = 60^\circ$. Prove $BC^2 = AB^2 + AC^2 - AB \cdot AC$.
2. A chord of a circle has length equal to the radius. Find the angle it subtends at the centre and at the circumference.
3. Prove the angle bisector from A and the perpendicular bisector of BC meet on the circumcircle.*
4. Prove that the feet of the perpendiculars from a point on the circumcircle to the three sides are collinear (Simson line).*
5. In a cyclic quadrilateral with perpendicular diagonals, prove the sum of squares of two opposite sides equals that of the other two.
6. Prove that the four triangles into which the diagonals split a cyclic quadrilateral come in two similar pairs.
7. P is inside square $ABCD$ with $PA = 1, PB = 2, PC = 3$. Find $\angle APB$.*
8. Prove that the midpoints of the three diagonals of a complete quadrilateral are collinear (the Newton–Gauss line).*
9. In $\triangle ABC$, the incircle touches BC, CA, AB at X, Y, Z . Prove AX, BY, CZ concur (Gergonne).
10. Prove that the reflection of the orthocentre over a side lies on the circumcircle.
11. Two circles meet at A, B . Lines through A and through B create four points; prove a resulting pair of lines is parallel.
12. Prove the tangents at B and C to the circumcircle meet on the symmedian from A .*
13. A circle is tangent to sides AB and AC of $\triangle ABC$ at P, Q and internally tangent to the circumcircle (the A -mixtilinear incircle). Prove that the incentre I is the midpoint of PQ .**
14. Prove Ptolemy's *inequality* for four arbitrary points.
15. In $\triangle ABC$, prove $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$.*

16. Prove the nine-point circle is tangent to the incircle (Feuerbach).**
17. Given four mutually tangent circles, relate their curvatures (Descartes).**
18. Prove that the Euler lines of the four triangles in a cyclic quadrilateral configuration are concurrent.**
19. Find the locus of points from which a given segment subtends a fixed angle.
20. Prove that among all triangles inscribed in a fixed circle, the equilateral has the greatest area, perimeter, and sum of squared sides.*

KEY IDEAS & FORMULAS

The whole arsenal, on one page.

- **Angles:** angle chasing and directed angles; inscribed-angle theorem, cyclic quadrilaterals, the tangent–chord (alternate segment) angle.
- **Lengths & circles:** similar triangles; Law of Sines ($= 2R$) and Cosines; $\text{area} = \frac{1}{2}ab \sin C = \frac{abc}{4R} = rs$; Stewart; power of a point; Ptolemy; the radical axis.
- **Concurrency / collinearity:** Ceva, Menelaus (and their trig forms), poles & polars, the cross-ratio.
- **Transformations:** reflection, rotation, homothety, spiral similarity, inversion — change your view until the truth is obvious.
- **The bashes:** coordinates, vectors, complex numbers, barycentrics — certainty when cleverness stalls.
- **The method:** draw it large, name the goal, match it to a tool, hunt the key claim, write it up. The secret was never *more* tools — only *seeing which one*.

★ OLYMPIAD SPARK

(The capstone.) Return to the very first *Olympiad Spark* of this book — the folded square hiding a 3–4–5 triangle, back in the Prologue. With everything you now command — coordinates, similar triangles, the Pythagorean theorem, perhaps a reflection — write a complete, rigorous proof. Then sit for a moment with how far you have come: that problem was once impossible, and now it is an exercise.

Epilogue: The Garden in Bloom

Turn back, for a moment, to where we began.

A loop of knotted rope on the banks of the Nile. Twelve equal parts, pulled taut into a triangle of 3, 4, and 5 — and a perfect right angle, conjured from nothing but string. We did not know, then, *why* it worked. We only knew that it did, and that somewhere behind the trick lay a reason worth chasing.

Look how far that chase has carried you.

You began unable to define a point. You learned that mathematics is built not by defining everything, but by choosing where to *stop* — and from three undefined words you built the entire plane. You proved your first theorem, vertical angles, and felt the floor of certainty hold under your weight. You watched a triangle's three angles, scattered and unequal, slide along one auxiliary line and fall into a perfect straight angle. The miracle was a straight line in disguise; you saw through it.

Then the tools began to gather, one by one, like a team assembling.

Congruence and similarity taught you when two figures are truly the same, and when they are merely echoes at different scales. *Pythagoras* arrived — and you met him not once but four times, because a truth seen from many sides is a truth you *own*. The triangle gave up its four hidden centres, and a single line — Euler's — threaded three of them together as though they had been waiting all along.

You walked into the circle and found it was a machine for trading angles against lengths: the inscribed angle, frozen as a point slides along its arc; Ptolemy, binding the diagonals of a cyclic quadrilateral to its sides; the power of a point, one number a point quietly remembers about a circle; the tangent, whispering across the circle through the alternate segment.

You assembled the toolbox. The *angle chase*, workhorse of them all. *Trigonometry*, where the Law of Sines turned out to be measuring the very circle through the triangle. *Ceva and Menelaus*, twins separated by a single sign. *Coordinates, vectors, complex numbers* — whole languages in which a rotation is just multiplication and a centroid is just an average. And *transformations*: the reflection that straightens a path, the 60° rotation that folded 3, 4, 5 into a single right angle at last.

And then the summit. The nine-point circle, nine coincidences collapsing into one scaling. The radical centre, conjured by pure transitivity. The spiral

similarity, hiding at the crossing of two circles. Inversion, the great simplifier, turning circles into lines and making Ptolemy fall out of the humblest inequality there is.

AHA!

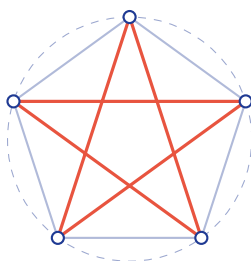
Here is the secret the whole book has been quietly teaching: **olympiad geometry is not about knowing more theorems than the next person.** It is about *seeing* — looking at a crowded, stubborn figure and recognising, in a heartbeat, which one idea will open it. That sight is not a gift you are born with. It is built, theorem by theorem, diagram by diagram, exactly as you have built it here.

The pentagram from the Prologue still hangs in the first pages, and the golden ratio still hides inside it. You could not reach it then. You can now. The folded square, the manhole cover that cannot fall through its own hole, the rope-stretchers' triangle — every wonder we opened a chapter with is, today, an exercise you could finish before the tea goes cold.

So here is the truth the garden was always growing toward: **you are the geometer now.** The next time you face a figure no one has shown you — on a contest, on a blackboard, in your own notebook at midnight — you will not freeze. You will draw it large and true. You will name what you want. You will reach, without hesitation, for the one tool among many that fits. And the figure will open, the way the rope-stretchers' right angle finally opened for you, because behind every beautiful coincidence there is a reason, and you have learned how to find it.

The road from a knotted cord to the International Mathematical Olympiad is long. You have just walked it. Whatever problem you meet next — harder, stranger, more beautiful — meet it the way you met every *Wonder* in this book: not with fear, but with the question that started everything.

Why?



Welcome to the garden. It was yours all along.

— *The Author*



Theorem & Formula Cheat-Sheet

A one-stop reference to every key result in the book. Use it to recall a fact fast; use the chapters to recall *why*.

Angles & triangles (Parts I–II)

- Angles on a line sum to 180° ; around a point, 360° ; vertical angles are equal.
- Across parallels: corresponding =, alternate =, co-interior sum 180° .
- Triangle angle sum 180° ; exterior angle = sum of the two remote interior angles.
- Congruence: **SSS, SAS, ASA, AAS, RHS**. Similarity: **AA**; areas scale as (ratio)².
- Isosceles: equal sides face equal angles (and conversely).
- Pythagoras $a^2 + b^2 = c^2$ and its converse.
- Pythagorean triples: $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, ...

Centres & the Euler line (Ch. 10, 24)

- Centroid G : medians, divides each $2 : 1$; $G = \frac{1}{3}(A + B + C)$.
- Circumcentre O (perp. bisectors), incentre I (angle bisectors, $r = \text{Area}/s$), orthocentre H (altitudes).
- **Euler line**: O, G, H collinear, $OG : GH = 1 : 2$; with nine-point centre N , $OG : GN : NH = 2 : 1 : 3$.
- **Nine-point circle**: $N = \text{midpoint } OH$, radius $\frac{1}{2}R$; through 3 midpoints, 3 altitude-feet, 3 Euler points.
- Euler: $OI^2 = R^2 - 2Rr$, hence $R \geq 2r$.

Area (Ch. 11)

$$[T] = \frac{1}{2}bh = \frac{1}{2}ab \sin C = \frac{abc}{4R} = rs = \sqrt{s(s-a)(s-b)(s-c)}.$$

Shoelace: $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$. Pick: $\text{Area} = I + \frac{B}{2} - 1$.

The circle (Part III)

- Inscribed angle = $\frac{1}{2}$ central; same-segment angles equal; semicircle angle = 90° .
- Cyclic quadrilateral: opposite angles sum 180° (and converse).
- **Ptolemy**: $AC \cdot BD = AB \cdot CD + AD \cdot BC$ (equality iff cyclic; $\leq$ otherwise).
- **Power of a point** = $OP^2 - r^2$; chords $PA \cdot PB = PC \cdot PD$; tangent $PT^2 = PA \cdot PB$.
- Tangent $\perp$ radius; two tangents from a point are equal; tangent-chord = alternate-segment inscribed angle.
- Chord angle: inside = $\frac{1}{2}(\text{arc} + \text{arc})$, outside = $\frac{1}{2}(\text{arc} - \text{arc})$.

The toolbox (Part IV)

- **Law of Sines** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$; **Law of Cosines** $c^2 = a^2 + b^2 - 2ab \cos C$.
- **Ceva** $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$ (concurrent); **Menelaus** = -1 , signed (collinear).
- **Stewart** $b^2m + c^2n = a(d^2 + mn)$; median $m_a^2 = \frac{1}{4}(2b^2 + 2c^2 - a^2)$; bisector $t_a^2 = bc - BD \cdot DC$.
- Distance $\sqrt{(\Delta x)^2 + (\Delta y)^2}$; perpendicular slopes multiply to -1 ; circle $(x - h)^2 + (y - k)^2 = r^2$.
- Complex: $\times e^{i\theta}$ rotates; equilateral $\iff a^2 + b^2 + c^2 = ab + bc + ca$. Vectors: $\vec{u} \cdot \vec{v} = 0 \iff \perp$.
- Homothety ratio k scales lengths by $|k|$, areas by k^2 ; spiral similarity multiplier $(C - D)/(A - B)$.

Olympiad engines (Part V)

- Radical centre: three radical axes concur; equal tangents from it.
- Poles & polars: La Hire (P on polar of $Q \iff Q$ on polar of P); cross-ratio is projection-invariant; harmonic = -1 .

- Spiral similarity centre = second intersection of circles $(PAC), (PBD)$ where $P = AB \cap CD$.
- **Inversion** $OP \cdot OP' = r^2$: lines $\leftrightarrow$ circles through O ; $P'Q' = PQ r^2 / (OP \cdot OQ)$; conformal.
- Barycentric (areal) $\alpha:\beta:\gamma = [PBC]:[PCA]:[PAB]$; centroid $(1:1:1)$, incentre $(a:b:c)$.

B

A Gallery of Constructions

Every construction below uses only the straightedge and compass of Chapter 6. Each comes with its one-line justification — a construction is only as good as the proof that it works.

Perpendicular bisector of AB .

Equal arcs from A and B meet at P, Q ; draw PQ . (P, Q are equidistant from A, B .)

Midpoint of AB .

The perpendicular bisector meets AB at its midpoint.

Bisector of an angle at A .

Arc from A cuts the sides at P, Q ; equal arcs from P, Q meet at R ; draw AR . ($\triangle APR \cong \triangle AQR$ by SSS.)

Perpendicular to ℓ from a point P off it.

Arc from P cuts ℓ at X, Y ; the perpendicular bisector of XY passes through P .

Perpendicular to ℓ at a point P on it.

Equal marks X, Y on either side of P , then the perpendicular bisector of XY .

Copy an angle onto a ray.

Reproduce the arc and chord that define it. (SSS.)

Equilateral triangle on AB (Euclid I.1).

Circles centred A (through B) and B (through A) meet at C . (*All sides = AB , radii.*)

$60^\circ, 30^\circ, 15^\circ$ angles.

Equilateral gives 60° ; bisect for 30° , again for 15° .

$90^\circ, 45^\circ$ angles.

Perpendicular gives 90° ; bisect for 45° .

Parallel to ℓ through P .

Copy the angle ℓ makes with a transversal, at P (equal corresponding angles).

Divide a segment into n equal parts.

Mark n equal steps along an auxiliary ray from one end; connect and draw parallels (Basic Proportionality, Ch. 8).

Centre of a given circle.

Two chords; their perpendicular bisectors meet at the centre.

Tangent from an external point P .

Circle on diameter OP meets the given circle at the contact points (Thales: $\angle OTP = 90^\circ$).

Circumcircle of $\triangle ABC$.

Two perpendicular bisectors give O ; draw the circle through A .

Incircle of $\triangle ABC$.

Two angle bisectors give I ; drop a perpendicular to a side for the radius.

 $\sqrt{n}$ from a unit segment.

Stack unit-height right triangles on growing hypotenuses — the spiral of Theodorus.

Known to be impossible with straightedge and compass alone (proved with algebra in the 1800s): trisecting a general angle, doubling the cube, squaring the circle, and constructing a regular 7- or 9-gon. Knowing the limits is part of the craft.



Hints & Solutions

Hints, not full write-ups — the joy is in finishing them yourself. A hint should be just enough to get unstuck.

Selected Olympiad Sparks

Ch. 4, the pentagram.

Each star-point angle is an inscribed angle; the five subtend, between them, the whole circle once. The five angles total $\frac{1}{2} \cdot 360^\circ = 180^\circ$.

Ch. 5, $\angle EAF = 45^\circ$ in a square.

Rotate $\triangle ABE$ by 90° about A so $AB \rightarrow AD$; the image of E, F , and D become collinear, and $EF = BE + DF$ drops out.

Ch. 9, P inside an equilateral, $PA = 3, PB = 4, PC = 5$.

Rotate 60° about a vertex to build a 3–4–5 right angle; the side is $\sqrt{25 + 12\sqrt{3}}$.

Ch. 14, sine addition via Ptolemy.

In a unit-diameter circle a chord on inscribed angle θ has length $\sin \theta$; apply Ptolemy to the cyclic quadrilateral.

Ch. 18, $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

Use $C = 180^\circ - (A + B)$ and the tangent addition formula.

Ch. 22 & 30, equilaterals on two sides.

One 60° rotation about A carries $F \rightarrow B$ and $C \rightarrow G$, so $CF \rightarrow BG$.

The “twenty more” (Chapter 30)

3.

Both the $\angle A$ -bisector and the perpendicular bisector of BC pass through the midpoint of arc BC not containing A .

4 (Simson).

P and two feet see a side at 90° , giving cyclic quadrilaterals; a short directed-angle chase forces the three feet collinear.

7.

Rotate 90° about B (so $A \mapsto C$); a $1-2\sqrt{2}-3$ right triangle appears and $\angle APB = 135^\circ$.

8 (Newton–Gauss).

Stage 1: a complete quadrilateral is four lines; its three diagonals join the three pairs of “opposite” intersection points. *Stage 2:* write each diagonal’s midpoint as the average of its two endpoints (position vectors). *Stage 3:* show one midpoint is an affine combination of the other two — a single linear relation — so the three are collinear. (Placing two of the lines on the axes also works.)

12 (symmedian).

The tangents at B, C meet at the pole of BC ; tangent–chord angles show it lies on the reflection of the median over the $\angle A$ -bisector.

13 (mixtilinear).

Its centre lies on the $\angle A$ -bisector, so the touch points P, Q on AB, AC are symmetric across that bisector and $PQ \perp$ the bisector; prove I lies on PQ at its midpoint. (Hard.)

15.

Use $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$ and $r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$.

16 (Feuerbach).

Stage 1: the incircle (centre I , radius r) and the nine-point circle (centre N , radius $\frac{1}{2}R$) are internally tangent exactly when $NI = \frac{1}{2}R - r$; that single equation is the whole target. *Stage 2:* place N as the midpoint of OH on the Euler line and compute NI^2 using Euler’s $OI^2 = R^2 - 2Rr$ and the known distances. *Stage 3:* simplify to $NI = \frac{1}{2}R - r$. (A purely synthetic proof by inversion at the contact point also exists, but is intricate — this is one of geometry’s gems.)

17 (Descartes).

Stage 1: invert at the point where two of the four circles touch; those two become *parallel lines*, and the remaining two become equal circles caught between them. *Stage 2:* in that strip the curvature relation is almost visible by inspection. *Stage 3:* invert back, tracking how each curvature transforms, to land on $(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$.

18.

The four Euler lines pass through the *anticentre* of the cyclic quadrilateral.

20.

Fix a chord and slide the third vertex (Ch. 11’s equal-area idea) to see each

adjustment toward equilateral increases the area; iterate or apply Jensen to $\sin$.

The capstone: the folded square, proved

The Prologue promised a proof “when you are a master.” Here it is. Place the unit square with $A = (0, 0)$, $B = (1, 0)$, $C = (1, 1)$, $D = (0, 1)$, and fold corner C onto $M = (\frac{1}{2}, 0)$, the midpoint of the non-adjacent side AB .

The crease is the perpendicular bisector of CM . A short computation places it as the line $y = -\frac{1}{2}x + \frac{7}{8}$, meeting the right edge BC at $E = (1, \frac{3}{8})$; reflecting edge CD across the crease, its image meets the left edge AD at $G = (0, \frac{2}{3})$.

The right-hand triangle. Because E lies on the crease, the fold gives $EM = EC$. Writing $BE = e$ (so $EC = 1 - e$), the right triangle BEM (right angle at B) has legs $BE = e$ and $BM = \frac{1}{2}$ and hypotenuse $EM = 1 - e$. Pythagoras gives $e^2 + \frac{1}{4} = (1 - e)^2$, hence $e = \frac{3}{8}$, and

$$BE : BM : EM = \frac{3}{8} : \frac{1}{2} : \frac{5}{8} = 3 : 4 : 5.$$

The other two. The fold sends the right angle of the square at C to a right angle at M , so the three triangles strung along the base AB are right-angled at A, M, B :

$$\triangle AMG : \frac{1}{2}, \frac{2}{3}, \frac{5}{6} = 3 : 4 : 5,$$

$$\triangle GME : \frac{5}{8}, \frac{5}{6}, \frac{25}{24} = 3 : 4 : 5,$$

$$\triangle BEM : \frac{3}{8}, \frac{1}{2}, \frac{5}{8} = 3 : 4 : 5.$$

Each is similar to the rope-stretchers' triangle. The motif that opened the book closes in four lines of algebra. ■

For the unstarred problems, the relevant chapter's main theorem is the intended tool — and the Cheat-Sheet (Appendix A) will point you to it. Resist looking; a problem you crack yourself is worth ten you are shown.

D

Further Reading & the Olympiad Path

You have finished the journey of this book. Here is how to keep climbing.

Books to grow with

Coxeter & Greitzer, *Geometry Revisited*.

The classic next step — elegant, deep, and humane. Read it slowly.

Evan Chen, *Euclidean Geometry in Mathematical Olympiads (EGMO)*.

The modern standard for olympiad geometry; directed angles, configurations, and a superb problem set.

Yaglom, *Geometric Transformations (I–IV)*.

Everything reflections, rotations, and inversion can do, taught through problems.

Honsberger, *Episodes in Nineteenth and Twentieth Century Euclidean Geometry*.

Beautiful results, gently told.

Johnson, *Advanced Euclidean Geometry*.

An encyclopaedic reference for the triangle's hidden riches.

Posamentier & Salkind, *Challenging Problems in Geometry*.

A large, friendly bank of problems with solutions.

The competition ladder

Geometry lives inside the wider olympiad movement. A typical path:

- **United States:** AMC 10/12 → AIME → USA(J)MO → MOP → IMO.
- **India:** IOQM → RMO → INMO → IMO Training Camp → IMO.
- **Most countries** run a similar funnel of national rounds feeding the **International Mathematical Olympiad**.

Whatever your country, the engine is the same: a national body selects, through escalating rounds, a team of six for the IMO each July.

How to practise

- **Solve past papers**, untimed at first, then timed. Olympiad archives are freely available online.
- **Write full solutions**. A problem is not finished until your write-up has no “clearly.” Graders reward rigour.
- **Keep a key-claim log** (Chapter 29): for each problem, record the one idea that cracked it. Patterns will emerge.
- **Read others’ solutions** after you have tried hard — communities like the Art of Problem Solving forums are a treasure.
- **Draw, always**. A large, accurate diagram remains your sharpest tool.

A last word

Talent helps, but it is wildly overrated next to patience, curiosity, and the habit of finishing what you start. Every olympian was once a beginner who could not define a point — and who kept asking *why*. You now carry the whole toolkit of Euclidean geometry, the discipline to deploy it, and the taste to find the elegant path. The rest is mileage, and mileage is just time spent doing something you have come to love.

Go find a hard, beautiful problem. Draw it large. And begin.

The garden is yours. Tend it well.

E

A Problem Compendium

A book is read; a solver is *built* — by mileage. What follows is a graded course of forty problems spanning the whole book, from one-line warm-ups to the hard canon of the subject. They are drawn from the classical olympiad tradition and from originals in its spirit; the famous named results are flagged so you can recognise old friends. Ratings are honest:

★ one idea, a few minutes. ★★ a solid contest problem. ★★★ olympiad.
★★★★ the hard canon.

Work each with the five-move loop of Chapter 29: draw it large, name the goal, match a tool, hunt the key claim, write it up. Cryptic one-line hints for every problem are gathered at the end — consult them only after a genuine struggle. Numerical answers are listed after the hints.

Tier I ★ — Warm-ups

1. The angles of a triangle are in the ratio 2 : 3 : 4. Find them.
2. Find the angle between the hands of a clock at 3:30.
3. A chord of length 16 lies 6 from the centre of a circle. Find the radius.
4. In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 5$, $BC = 12$. Find the circumradius.
5. Two parallel chords of a circle of radius 5, on opposite sides of the centre, have lengths 6 and 8. Find the distance between them.
6. An equilateral triangle has area $\sqrt{3}$. Find its side length.
7. Find the inradius of a 5–12–13 right triangle.
8. A regular hexagon has perimeter 24. Find its area.
9. In $\triangle ABC$ with $AB = AC = 5$ and $BC = 6$, find the length of the median to BC .
10. From a point 13 from the centre of a circle of radius 5, find the length of

the tangent.

Tier II ★★ — Contest

11. Prove the bisector of $\angle A$ divides BC in the ratio $AB : AC$.
12. In $\triangle ABC$ with $\angle A = 60^\circ$, prove $a^2 = b^2 + c^2 - bc$.
13. In cyclic quadrilateral $ABCD$ with $AB = AD$, prove that AC bisects $\angle BCD$.
14. **British Flag.** For any point P and rectangle $ABCD$, prove $PA^2 + PC^2 = PB^2 + PD^2$.
15. The incircle of $\triangle ABC$ touches BC at X . Prove $BX = s - b$, and that X and the A -excircle touch-point are reflections in the midpoint of BC .
16. Two circles meet at A, B ; a variable line through A meets them again at P, Q . Prove $\angle PBQ$ is independent of the line.
17. Prove the three medians cut a triangle into six triangles of equal area.
18. Prove $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$ for the medians.
19. A line through the incentre parallel to BC meets AB, AC at M, N . Prove $MN = BM + CN$.
20. Points A, B, C, D lie on a circle; $AC \perp BD$. Prove $AB^2 + CD^2 = BC^2 + DA^2$.

Tier III ★★★ — Olympiad

21. **The Butterfly.** M is the midpoint of a chord PQ ; two further chords AB and CD pass through M ; lines AD and BC meet PQ at X and Y . Prove $MX = MY$.
22. **Ptolemy.** For cyclic $ABCD$, prove $AC \cdot BD = AB \cdot CD + AD \cdot BC$.
23. **Van Schooten.** For P on arc BC of equilateral $\triangle ABC$, prove $PA = PB + PC$.
24. **Nine points.** Prove the three side-midpoints, three altitude-feet, and three Euler points of a triangle lie on one circle.
25. Prove the reflection of the orthocentre over a side of the triangle lies on the circumcircle.
26. **Newton–Gauss.** Prove the midpoints of the three diagonals of a complete quadrilateral are collinear.
27. Prove the tangents to the circumcircle at B and C meet on the symmedian from A .

28. The incentre is I ; ray AI meets the circumcircle again at M . Prove $MB = MC = MI$ (the *incentre-arc-midpoint* lemma).
29. Prove that among all triangles inscribed in a fixed circle, the equilateral maximises the area.
30. P lies inside a square $ABCD$ with $PA : PB : PC = 1 : 2 : 3$. Find $\angle APB$.

Tier IV ★★ ★ — The Hard Canon

31. **Feuerbach.** Prove the nine-point circle is internally tangent to the incircle and tangent to each excircle.
32. **Pascal.** For a hexagon inscribed in a circle, prove the three intersections of opposite sides are collinear.
33. **Brianchon.** For a hexagon circumscribed about a circle, prove the three main diagonals concur.
34. **Morley.** Prove the three intersection points of adjacent angle trisectors of any triangle form an equilateral triangle.
35. **Descartes.** For four mutually tangent circles of curvatures k_i , prove $(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$.
36. **Steiner porism.** If one Steiner chain closes between two circles, prove every chain does, after the same number of steps.
37. **Mixtilinear.** The circle tangent to AB , AC and internally to the circumcircle touches AB , AC at P , Q . Prove the incentre is the midpoint of PQ .
38. **Japanese theorem.** Prove the incentres of the four triangles cut from a cyclic quadrilateral by its diagonals form a rectangle.
39. **Casey.** State and use the generalisation of Ptolemy to four circles tangent to a fifth.
40. **The capstone.** Two circles meet at A , B ; a line through A meets them again at C , D ; the tangents at C and D meet at P . Prove B, C, P, D are concyclic (Chapter 30, Ascent 2) — then generalise.

Hints

Tier I. 1. $2x + 3x + 4x = 180$. 2. the hour hand has moved half-way past 3. 3. half-chord 8, then Pythagoras. 4. hypotenuse $\div 2$. 5. distances 4 and 3, opposite sides. 6. $\frac{\sqrt{3}}{4}s^2 = \sqrt{3}$. 7. $r = s - c$. 8. six unit-side equilateral triangles. 9. the median is the altitude here. 10. $\sqrt{13^2 - 5^2}$.

Tier II. 11. equal-height area ratio, or the sine rule. 12. Law of Cosines, $\cos 60^\circ = \frac{1}{2}$. 13. equal chords $\Rightarrow$ equal arcs $\Rightarrow$ equal inscribed angles. 14. place $P = (x, y)$ at a

corner-origin and expand. 15. equal tangents, $BX = s - b$, $BX' = s - c$, sum = a . 16. $\angle BPA$ stands on the fixed chord BA . 17. G divides each median $2 : 1$; pair the six pieces. 18. Stewart/Apollonius, summed. 19. the parallel makes BMI and CNI isosceles via the bisector. 20. right angles at the diagonal intersection.

Tier III. 21. drop perpendiculars from M to AB, CD , or use cross-ratio; power of M . 22. auxiliary point on AC , or invert at A . 23. Ptolemy on $BPCA$ with the equilateral side cancelling. 24. homothety $h(H, \frac{1}{2})$ sends the circumcircle to it. 25. $\angle BH'C = \angle BHC = 180^\circ - \angle A$. 26. vectors/coordinates; each midpoint satisfies one linear relation. 27. P is the pole of BC ; the second intersection makes a harmonic quadrilateral. 28. chase $\angle MBI = \angle MIB = \frac{1}{2}(\angle A + \angle B)$. 29. fix two vertices and slide the third to the arc-midpoint. 30. rotate 90° about B ; a $1, 2\sqrt{2}, 3$ right triangle appears, so $\angle APB = 135^\circ$.

Tier IV. 31. show $NI = \frac{1}{2}R - r$ (Apollonius in $\triangle OHI$). 32. project a conic to a circle; equal cross-ratios from two points. 33. dualise Pascal through pole/polar. 34. trigonometric identities on the trisected angles (or the standard auxiliary equilateral). 35. invert at a tangency to two parallel lines and a circle. 36. invert at a limiting point to make the circles concentric. 37. homothety at the contact T to the circumcircle, plus the arc-midpoint lemma (Prob. 28). 38. Fact 5 (incentre–arc-midpoint) on each of the four triangles. 39. replace each tangent length by a common-tangent length. 40. tangent–chord twice, then the collinearity of C, A, D welds the halves; directed angles.

Answers (numerical problems)

1. $40^\circ, 60^\circ, 80^\circ$. 2. 75° . 3. 10. 4. 6.5. 5. 7. 6. 2. 7. 2. 8. $24\sqrt{3}$. 9. 4.
10. 12. 30. 135° .